

# Unimodality of Forest Independence Polynomials

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## Abstract

We prove that the independence sequence of every finite forest is unimodal, resolving the tree unimodality conjecture of Alavi, Malde, Schwenk and Erdős. Our computer-assisted proof lowers the large-order threshold to 59 vertices and extends the finite-order argument through 60 vertices, so the overlapping ranges cover all orders. The proof combines structural counting inequalities with hard-core conditional-binomial reductions and certificates verified by exact integer and outward-rounded rational arithmetic.

**Keywords.** Independence polynomial; forest; unimodality; hard-core model; rational certificate; computer-assisted proof.

## 1 Introduction

For a finite simple graph  $F$ , its independence polynomial is

$$I_F(x) = \sum_{r=0}^{\alpha(F)} c_r(F) x^r, \quad c_r(F) = |\{S \subseteq V(F) : S \text{ independent, } |S| = r\}|.$$

A finite sequence is unimodal if it is nondecreasing up to some index and nonincreasing thereafter. We include the empty forest, with polynomial 1, and forests with isolated vertices or multiple components. The number of vertices is denoted by  $n$ .

The independence polynomial, introduced by Gutman and Harary [14], records how the number of independent vertex sets varies with their size. Its coefficient shape connects graph structure with the theory of combinatorial generating polynomials. For general graphs, there is no universal unimodality principle: Alavi, Malde, Schwenk and Erdős showed that the positive-index coefficients can realize any prescribed strict ordering [1]. In the same 1987 paper, they asked whether the independence sequence of every tree, or more generally every forest, is unimodal. The question asks whether the absence of cycles alone forces a global order on these counting coefficients, despite unrestricted vertex degrees and branching patterns.

For polynomials with nonnegative coefficients and no internal zeros, real-rootedness implies log-concavity, which in turn implies unimodality; see Stanley [27] for this classical framework. Heilmann and Lieb's matching-polynomial theorem gives real-rootedness for independence polynomials of line graphs [17]. For claw-free graphs, Hamidoune proved log-concavity [15],

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and Chudnovsky and Seymour subsequently proved real-rootedness [10]. These hypotheses do not cover arbitrary forests with branching vertices.

Progress on the conjecture has proceeded through several kinds of tree structure. Levit and Mandrescu proved unimodality for well-covered spiders and related well-covered trees [19]. Wang and Zhu used recurrences and polynomial factorizations to prove log-concavity for regular caterpillars and firecracker trees [28]. Bahls, Ethridge and Szabo treated non-regular caterpillars under explicit monotonicity and growth conditions on their pendant-edge counts [3]. Galvin and Hilyard established unimodality for periodic tree attachments and showed that attaching sufficiently many pendant edges at one vertex, or at every vertex, makes any given tree unimodal [13]. Mandrescu and Spivak proved log-concavity for all odd-order trees maximizing the number of maximal independent sets, as well as certain even-order batons [22].

Algebraic approaches give further families. Bencs developed real-rootedness methods for trees [6], while Zhu and Wang obtained real-rootedness and unimodality results for rooted products [30]. Zhu and Chen used recurrences and factorizations to establish log-concavity for additional specified tree families [29]. Bendjeddou and Hardiman used Lorentzian polynomials to prove log-concavity for forests obtained by replacing every edge of an arbitrary forest with their fixed caterpillar  $W_4$  [7]. Li, Li, Yang and Zhang proved log-concavity for all spiders using chromatic symmetric functions [21]. The edge-replacement and special-family hypotheses in these results do not cover an arbitrary forest itself.

For general trees and forests, partial monotonicity narrows the possible location of a mode. Levit and Mandrescu established a nonincreasing final segment for bipartite graphs [20]. Basit and Galvin extended tail bounds to general graphs and proved long increasing and decreasing segments for uniformly random labelled trees with probability tending to one [4]. Heilman obtained an increasing initial segment through  $[0.26543n]$  for uniformly random labelled trees with exponentially high probability [16]. These probabilistic theorems concern typical trees and do not settle the assertion for every tree. Fang, Lu, Nevo, Yao and Zheng proved unimodality for every sufficiently large forest, using a uniform hard-core central limit theorem and monotone initial and final segments [11]. Their theorem gives an absolute threshold without the explicit value required below. Our proof combines an explicit large-order argument with an overlapping finite-order argument.

The stronger log-concavity assertion fails for trees, as shown by Kadrawi and Levit [18] and subsequent constructions of Galvin [12], Ramos and Sun [24], and Bautista-Ramos, Guillén-Galván and Gómez-Salgado [5]. These results concern log-concavity and do not disprove unimodality. A different conjecture for well-covered graphs, proposed by Brown, Dilcher and Nowakowski [9], was disproved by Michael and Traves [23]. Bhattacharyya and Kahn also constructed bipartite graphs with non-unimodal independence sequences [8]. Thus bipartiteness alone cannot replace the forest hypothesis.

We prove the tree conjecture in the stronger form covering every finite forest. This extension is substantive: independence polynomials multiply over components, but the product of two arbitrary unimodal polynomials need not be unimodal. The result establishes the coefficient-shape property that survives for all forests even though log-concavity fails. It also closes the gap between qualitative large-order results and the remaining finite orders: a threshold of 59 for the large-order argument and coverage through 60 for the finite-order argument leave no uncovered order. The exact certificates make this final connection independently reproducible.

**Theorem 1.1.** *The independence sequence of every finite simple forest is unimodal.*

The proof has two parts.

**Proposition 1.2.** *Every forest on at most 60 vertices has a unimodal independence sequence.*

**Proposition 1.3.** *Every forest on at least 59 vertices has a unimodal independence sequence.*

*Proof of Theorem 1.1.* Apply Proposition 1.2 when  $n \leq 60$  and Proposition 1.3 when  $n \geq 61$ .  $\square$

The finite-order argument expresses independent-set counts relative to one suitably chosen maximum independent set. Counting, matching and deletion inequalities define a relaxation containing the counts of every forest. Direct comparisons, 670 rational separating inequalities and 1907 rational linear certificates cover its full parameter domain. The large-order argument uses a maximum-weight independent set for the actual hard-core marginals. Its complement has a provable matching lower bound, leading to a joint constraint on the free-binomial size and the complement occupancy. Positive expectations of fixed nonnegative mixtures of adjacent-coefficient kernels then exclude all necessary weak valleys.

The finite inequalities are part of a computer-assisted proof: they establish universally quantified scalar statements after the structural reductions. The argument does not infer the theorem from the absence of counterexamples in graph samples. The fixed proof data and accepting programs are supplied as ancillary material.

The paper is organized as follows. Section 2 derives the structural counting inequalities and exact linear certificates, proving unimodality through order 60. Section 3 locates the relevant coefficient ranks and reduces the large-order problem to a hard-core conditional-binomial mixture. Section 4 uses complement matchings to strengthen this reduction at orders 59 through 79. Section 5 combines rational mixture certificates with the inherited parameter coverage to prove unimodality from order 59 onward. Section 6 proves the activity-two mean bounds; Sections 7 and 8 establish the rooted-source estimates, mean refinements and finite-kernel bounds used by the large-order argument. Sections 9 and 10 specify the finite-order certificate rows and give an explicit certificate, and Section 11 describes the proof data and their reproduction.

## 2 The finite-order argument

This section proves Proposition 1.2. The structural inequalities place the independent-set counts of each forest in a real polyhedron. Rational certificates establish the required difference inequalities for every parameter triple in (35).

Throughout this section, binomial coefficients are defined by

$$\binom{b}{r} = 0 \quad \text{unless } b, r \text{ are integers with } 0 \leq r \leq b.$$

For polynomials with real coefficients,  $P \leq_{\text{coeff}} Q$  means that every coefficient of  $Q - P$  is nonnegative. We write  $x_+ = \max(x, 0)$  and use  $\mathbf{1}_A$  for the indicator of a condition  $A$ .

### 2.1 A decomposition relative to a maximum independent set

Let  $F$  be a forest on  $n$  vertices. Write  $a = \alpha(F)$  and let  $v$  be its matching number. Leaf deletion gives

$$a + v = n, \quad \delta := a - v \geq 0. \tag{1}$$

For completeness, if  $u$  is a leaf with neighbour  $w$ , a maximum independent set may be chosen to contain  $u$ : replace  $w$  by  $u$  if necessary. Consequently  $\alpha(F) = 1 + \alpha(F - \{u, w\})$ . A maximum matching may likewise be chosen to contain  $uw$ , by replacing an edge incident with  $w$  by  $uw$  when necessary. Its size is therefore  $1 + \nu(F - \{u, w\})$ . An isolated vertex increases the independence number by one and leaves the matching number unchanged. These observations prove the first identity in (1) by induction. The inequality follows from  $2v \leq n$ .

Fix a maximum independent set  $I$  and put  $Y = V(F) \setminus I$ , so that  $|I| = a$  and  $|Y| = v$ . For any fixed maximum matching, an independent set contains at most one endpoint of each matched edge, and there are  $\delta = n - 2v$  unmatched vertices. Since  $|I| = v + \delta$ , equality holds in all of these restrictions. Hence  $I$  contains every unmatched vertex and exactly one endpoint of every matched edge. In particular, the matching saturates  $Y$  and matches it into  $I$ .

For an independent set  $J \subseteq Y$ , define

$$j = |J|, \quad m(J) = |I \setminus N_I(J)|.$$

Let  $t_{j,m}$  count the independent subsets  $J$  of  $Y$  for which these parameters equal  $j, m$ . The set  $J \cup (I \setminus N_I(J))$  is independent, so  $j + m \leq a$ . By convention  $t_{0,a} = 1$ , and all other  $t_{0,m}$  vanish. Every independent set of  $F$  is uniquely obtained by first choosing its intersection  $J$  with  $Y$  and then choosing an arbitrary subset of  $I \setminus N_I(J)$ . Thus

$$I_F(z) = (1+z)^a + \sum_{j=1}^v \sum_{m=0}^{a-j} t_{j,m} z^j (1+z)^m. \quad (2)$$

In particular,

$$c_r = \binom{a}{r} + \sum_{j=1}^v \sum_{m=0}^{a-j} \binom{m}{r-j} t_{j,m}. \quad (3)$$

Here  $J$  ranges over all independent subsets of  $Y$ .

Define

$$T_j = \sum_m t_{j,m}, \quad M_j = \sum_m m t_{j,m}, \quad W_j = \binom{v}{j}. \quad (4)$$

The basic constraints are

$$t_{j,m} \geq 0, \quad T_1 = v, \quad T_j \leq W_j \quad (2 \leq j \leq v). \quad (5)$$

All certificate arguments below remain valid when the  $t_{j,m}$  are arbitrary nonnegative real numbers satisfying the displayed constraints. Integrality is used in deriving some structural inequalities for actual forests, but is not imposed in the subsequent linear relaxation.

## 2.2 Coefficient bounds and the first cross-layer inequalities

We begin with bounds on individual coefficients, which will later settle many parameter triples by direct comparison.

**Lemma 2.1.** *For every forest of order  $n$  and independence number  $a$ , with  $v = n - a$ ,*

$$\binom{n-r+1}{r} \leq c_r \quad (r \geq 0), \quad (6)$$

and

$$I_F(z) \leq_{\text{coeff}} (1+z)^{a-v} (1+2z)^v. \quad (7)$$

*Proof.* The number of independent  $r$ -sets of the path on  $n$  vertices is  $p_{n,r} = \binom{n-r+1}{r}$ . A leaf  $u$  with neighbour  $w$  gives

$$I_F = I_{F-u} + z I_{F-\{u,w\}},$$

and  $p_{n,r} = p_{n-1,r} + p_{n-2,r-1}$ . Induction therefore proves (6) when a leaf is present. If  $u$  is isolated,  $I_F = (1+z)I_{F-u}$ ; the induction bound and  $p_{n-1,r-1} \geq p_{n-2,r-1}$  give the same conclusion. The empty and one-vertex forests start the induction. Coefficients beyond the degree are handled by the binomial convention.

For the upper bound,  $m \leq a - j$  in (2), and the layer mass is at most  $\binom{v}{j}$ . Consequently

$$\begin{aligned} I_F(z) &\leq_{\text{coeff}} \sum_{j=0}^v \binom{v}{j} z^j (1+z)^{a-j} \\ &= (1+z)^{a-v} (1+2z)^v. \end{aligned}$$

□

The next two inequalities relate neighbouring layers, adding restrictions that separate coefficient bounds do not capture.

**Lemma 2.2.** *For  $2 \leq r \leq v$  and every integer  $h \geq 0$ ,*

$$\sum_m ((r-1)m + a - r + 1 - rh)_+ t_{r,m} \leq (v - r + 1) \sum_m (m - h)_+ t_{r-1,m}. \quad (8)$$

*Proof.* Fix an independent  $r$ -set  $J \subseteq Y$ , put  $m = m(J)$  and  $d = a - m$ , and consider the bipartite forest induced by  $J \cup N_I(J)$ . Each vertex of  $J$  has a neighbour in  $I$ , since otherwise  $I$  would not be maximum. If this forest has  $b \geq 1$  components and  $p$  vertices of degree one on its  $I$  side, its number of edges is  $r + d - b$ , while summing degrees on that side gives at least  $p + 2(d - p)$ . Therefore

$$p \geq d - r + b \geq d - r + 1.$$

For  $u \in J$ , let  $p_u$  be the number of its private neighbours in  $I$ , that is, neighbours of  $u$  adjacent to no other member of  $J$ . Then

$$m(J \setminus \{u\}) = m + p_u, \quad \sum_{u \in J} p_u \geq d - r + 1. \quad (9)$$

It follows that

$$\sum_{u \in J} (m + p_u - h)_+ \geq \left( r(m - h) + \sum_{u \in J} p_u \right)_+ \geq ((r-1)m + a - r + 1 - rh)_+.$$

Sum over  $J$ . Each independent  $(r-1)$ -set of  $Y$  is obtained by deleting one vertex from at most  $v - r + 1$  independent  $r$ -sets. The resulting reverse count proves (8). □

**Lemma 2.3.** *Put*

$$H_{r,l} = \sum_m \binom{m}{l} t_{r,m}, \quad H_{0,l} = \binom{a}{l}.$$

*Whenever the factor on the right is nonnegative,*

$$(r+1)H_{r+1,l} \leq (v - r - \max(0, l - \delta))H_{r,l}. \quad (10)$$

*Proof.* Fix a set  $L \subseteq I$  of size  $l$ . At most  $\delta$  vertices of  $I$  are unmatched by a fixed maximum matching. The matched vertices of  $L$  therefore exclude at least  $\max(0, l - \delta)$  distinct vertices of  $Y$ . Among the remaining vertices, an independent  $r$ -set has at most  $v - r - \max(0, l - \delta)$  extensions. Double-count the pairs consisting of such an  $r$ -set and one extending vertex, and then sum over  $L$ . An independent set  $J$  contributes  $\binom{m(J)}{l}$  choices of  $L$ , giving precisely (10). □

### 2.3 Choosing a maximum independent set with a sparse complement

The preceding formulas hold for every maximum independent set. The remaining inequalities are used with a single maximum independent set chosen as follows.

**Lemma 2.4.** *A forest has a maximum independent set  $I$  whose complement  $Y$  satisfies*

$$e(F[Y]) \leq E := \max(0, \delta - 1). \quad (11)$$

*If  $v = |Y| \geq 1$ , then also  $e(F[Y]) \leq v - 1$ .*

*Proof.* Fix a maximum matching and a bipartite colouring of each component of  $F$ . Contract each matched edge to one paired node, and retain each unmatched vertex as a terminal. The contracted graph is a forest. Every maximum independent set contains all terminals and selects exactly one endpoint from each matched pair. Label a paired node by the original bipartite colour of its selected endpoint, and label a terminal by its own colour.

Consider an edge of the contracted forest. Its corresponding original edge joins opposite bipartite colours. If the two node labels agree, the original edge has exactly one endpoint in  $I$ . If they disagree, it has either both endpoints in  $I$  or both in  $Y$ ; the former is excluded by independence. Thus edges between different labels correspond precisely to edges of  $F[Y]$ . No such edge is incident with a terminal.

Choose a maximum independent set minimizing  $e(F[Y])$ . Suppose a monochromatic connected block in the contracted forest has no terminal and has a boundary edge. Swap the selected and unselected endpoints in every matched pair belonging to this block. Internal edges still have exactly one selected endpoint. Every boundary edge, previously an edge with both endpoints in  $Y$ , now has exactly one selected endpoint. Matched edges still contribute one selected endpoint each. Hence the new set is independent, has the same size, and has strictly fewer edges in its complement, a contradiction.

In an original component containing  $\delta_c > 0$  terminals, every monochromatic block consequently contains a terminal. There are at most  $\delta_c$  such blocks, and contracting them gives a tree, so there are at most  $\delta_c - 1$  edges between blocks. In a component without terminals there can be only one monochromatic block, hence no such edges. Summing over components proves (11). The final assertion follows because  $F[Y]$  is itself a forest.  $\square$

If  $v = 0$ , then  $F$  is edgeless and  $I_F(z) = (1 + z)^n$  is unimodal. Henceforth assume  $v \geq 1$  and fix the set supplied by Lemma 2.4. Define

$$D_0 = \min(E, v - 1), \quad e = e(F[Y]) = \binom{v}{2} - T_2. \quad (12)$$

The last identity is exact: a two-element subset of  $Y$  fails to be independent precisely when it is an edge. All constraints below refer to this same  $I$ , this same  $Y$ , and this same value of  $e$ .

### 2.4 Two-sided information from deletion of a vertex

The sparse complement also bounds extension counts from below, allowing us to use the deletion relation in both directions.

For an independent  $r$ -set  $J \subseteq Y$ , retain the notation  $m = m(J)$ ,  $d = a - m$ , and put  $L = d - r + 1$ . Since  $r + m \leq a$ , we have  $L \geq 1$ . Moreover, applying maximality of the independence number to  $J \setminus \{u\}$  gives  $(r - 1) + (m + p_u) \leq a$ . Together with (9) and the fact that private neighbours are distinct, this yields

$$0 \leq p_u \leq L, \quad L \leq \sum_{u \in J} p_u \leq d. \quad (13)$$

For  $2 \leq r \leq v$  and an integer  $h \geq 0$ , define

$$b_{r,h}(m) = \begin{cases} r, & m \geq h, \\ 1, & m < h \text{ and } rh \leq a + (r-1)m, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

**Lemma 2.5.** *For every  $2 \leq r \leq v$  and integer  $h \geq 0$ ,*

$$\sum_m b_{r,h}(m)t_{r,m} \leq (v-r+1) \sum_{m \geq h} t_{r-1,m}. \quad (15)$$

*Proof.* If  $m \geq h$ , all  $r$  deletions from  $J$  give an available-vertex count at least  $h$ . Otherwise let  $t = h - m \geq 1$ . The condition in the second line of (14) is  $rt \leq d$ . If every deletion had  $p_u < t$ , integrality would imply

$$\sum p_u \leq r(t-1) \leq d-r < L,$$

contrary to (13). Thus at least one deletion has the required count. Summing this lower bound over  $J$  and using at most  $v-r+1$  extensions for each parent set proves the inequality.  $\square$

For an independent  $(r-1)$ -set  $A \subseteq Y$ , the number of unselected vertices forbidden by its edges is at most the total number  $e$  of edges of  $F[Y]$ . Hence  $A$  has at least

$$s_r = \max(0, v-r+1-E) \quad (16)$$

extensions to an independent  $r$ -set.

To bound deletion sums in the other direction, write

$$d = qL + \rho, \quad 0 \leq \rho < L.$$

The identity  $rL - d = (r-1)(d-r) \geq 0$  gives  $q \leq r$ ; if  $q = r$ , then  $\rho = 0$ . Define

$$U_{r,h}(m) = q(m+L-h)_+ + \begin{cases} (m+\rho-h)_+ + (r-q-1)(m-h)_+, & q < r, \\ 0, & q = r, \end{cases} \quad (17)$$

and

$$V_{r,h}(m) = \begin{cases} r, & h \leq m, \\ \min(r, \lfloor d/(h-m) \rfloor), & m < h \leq m+L, \\ 0, & h > m+L. \end{cases} \quad (18)$$

**Lemma 2.6.** *For  $2 \leq r \leq v$  and integer  $h \geq 0$ ,*

$$s_r \sum_m (m-h)_+ t_{r-1,m} \leq \sum_m U_{r,h}(m)t_{r,m}, \quad (19)$$

$$s_r \sum_{m \geq h} t_{r-1,m} \leq \sum_m V_{r,h}(m)t_{r,m}. \quad (20)$$

*Proof.* For fixed  $J$ , consider the feasible release vector  $(p_u)_{u \in J}$  in (13). The function  $p \mapsto (m+p-h)_+$  is convex and nondecreasing. Under the upper constraints  $0 \leq p_u \leq L$  and  $\sum p_u \leq d$ , its sum is maximized by concentrating mass:  $q$  coordinates equal  $L$ , at most one further coordinate equals  $\rho$ , and the others are zero. This follows, for example, by repeatedly transferring mass between two coordinates strictly inside  $[0, L]$  toward an endpoint, choosing a direction that does not decrease the convex sum. Monotonicity permits total mass  $d$ . The resulting maximum is (17).

For the indicator sum, all deletions qualify when  $h \leq m$ . If  $0 < h - m \leq L$ , each qualifying coordinate consumes at least  $h - m$  of the total mass at most  $d$ , giving the middle bound in (18). If  $h - m > L$ , none qualifies.

Now sum either deletion statistic over all independent  $r$ -sets. The upper bounds just obtained give the right sides of (19) and (20). Counting the same incidence relation from its  $(r - 1)$ -set side gives the left sides, because every such set has at least  $s_r$  extensions. When  $s_r = 0$ , both inequalities remain valid and are automatic consequences of nonnegativity.  $\square$

## 2.5 Available-vertex means and the complement's actual edge count

We supplement the deletion constraints with bounds on the total number of available vertices across each layer.

**Lemma 2.7.** *For  $2 \leq r \leq v$ ,*

$$\sum_m (a - m)t_{r,m} \leq \binom{v-1}{r-1} \sum_m (a - m)t_{1,m}. \quad (21)$$

*Proof.* For each independent  $r$ -set  $J \subseteq Y$ , its covered vertices in  $I$  form a union of the neighbourhoods of its members. Its cardinality is at most the sum of those neighbourhood sizes. A fixed vertex of  $Y$  belongs to at most  $\binom{v-1}{r-1}$  such sets. Summing the union bound therefore gives the stated inequality.  $\square$

The next bound retains the actual edge count of the complement, so the information it supplies remains linked across all layers.

For  $2 \leq r \leq v$ , put

$$\beta_r = \binom{v-2}{r-1} - (a-2)\binom{v-2}{r-2}, \quad C_r^0 = a\binom{v-2}{r} + (\delta+1)\binom{v-2}{r-1}. \quad (22)$$

**Lemma 2.8.** *With  $e = e(F[Y])$  as in (12),*

$$M_r \geq C_r^0 + \beta_r e \quad (2 \leq r \leq v). \quad (23)$$

*Proof.* For  $i \in I$ , let  $d_i = |N(i) \cap Y|$ . For integer  $0 \leq d \leq v$ , the function  $d \mapsto \binom{v-d}{r}$  is discretely convex: its forward differences are  $-\binom{v-d-1}{r-1}$  and are nondecreasing. The line through its values at  $d = 1, 2$  is therefore a lower support line on the integer domain. Explicitly,

$$\binom{v-d_i}{r} \geq \binom{v-2}{r} + (2-d_i)\binom{v-2}{r-1}.$$

The number of edges between  $I$  and  $Y$  is at most  $n - 1 - e$ . Consequently

$$\begin{aligned} \sum_{i \in I} \binom{v-d_i}{r} &\geq a\binom{v-2}{r} + (2a - n + 1 + e)\binom{v-2}{r-1} \\ &= C_r^0 + e\binom{v-2}{r-1}. \end{aligned}$$

The sum on the left counts pairs consisting of any  $r$ -subset of  $Y$  and an available vertex of  $I$ , without yet requiring the subset to be independent.

For an edge  $uw$  of  $F[Y]$ , both  $u$  and  $w$  have nonempty neighbourhoods in  $I$ , and those neighbourhoods are disjoint, since a common neighbour would produce a triangle. Every  $r$ -set containing  $u, w$  therefore has at most  $a - 2$  available vertices of  $I$ . There are  $\binom{v-2}{r-2}$  candidate sets containing this edge. A union bound over the  $e$  edges shows that deleting all nonindependent subsets from the preceding count loses at most  $e(a - 2)\binom{v-2}{r-2}$ . Subtracting this amount proves the lemma.  $\square$

The coefficient  $\beta_r$  may be negative. Accordingly we do not replace  $e$  independently by an endpoint of its allowed interval for each  $r$ . We retain the common variable and then substitute the exact identity  $e = \binom{v}{2} - T_2$ ; this is the source of the linear row used in the certificates.

## 2.6 Coefficient bounds for a forest with a fixed edge count

Because the layer masses are themselves independent-set counts in the complement, its edge count also constrains these masses directly.

**Lemma 2.9.** *For a forest  $G$  on  $v \geq 1$  vertices and with  $e$  edges,*

$$i_r(G) \geq \binom{v}{r} - e \binom{v-2}{r-2}, \quad (24)$$

where  $i_r(G)$  counts its independent  $r$ -sets, and

$$I_G(z) \leq_{\text{coeff}} (1+z)^{v-1} + z(1+z)^{v-e-1}. \quad (25)$$

*Proof.* The first bound deletes from the  $\binom{v}{r}$  subsets all those containing an edge. Each edge occurs in  $\binom{v-2}{r-2}$  subsets, so the union bound proves the claim, including cases when its right side is negative.

For the upper bound, write  $B(v, e) = (1+z)^{v-1} + z(1+z)^{v-e-1}$ . If  $e = 0$ , equality holds; orders at most two are immediate. For  $v \geq 3$  and  $e > 0$ , choose a leaf  $u$  with neighbour  $w$  of degree  $d$ . The two smaller forests have parameters  $(v-1, e-1)$  and  $(v-2, e-d)$ . Induction and leaf deletion give

$$I_G(z) \leq_{\text{coeff}} B(v-1, e-1) + zB(v-2, e-d).$$

A direct subtraction gives

$$B(v, e) - B(v-1, e-1) - zB(v-2, e-d) = z^2((1+z)^{v-3} - (1+z)^{v-e+d-3}).$$

The remaining forest on  $v-2 \geq 1$  vertices has  $e-d \leq v-3$  edges, so the second exponent is nonnegative. Since  $e \geq d$ , it does not exceed the first exponent. The displayed difference has nonnegative coefficients, completing the induction.  $\square$

Apply the lemma to  $G = F[Y]$ , whose independent-set coefficients are  $T_r$ . From (25),

$$T_r \leq \binom{v-1}{r} + \binom{v-e-1}{r-1}. \quad (26)$$

For  $2 \leq r \leq v$ , the function on integer  $0 \leq e \leq v-1$  given by

$$g_r(e) = \binom{v-1}{r-1} - \binom{v-e-1}{r-1}$$

is concave, satisfies  $g_r(0) = 0$ , and has successive increments  $\binom{v-e-2}{r-2}$ , which are nonincreasing. If  $D_0 \geq 1$ , define

$$G_r = \binom{v-1}{r-1} - \binom{v-D_0-1}{r-1}. \quad (27)$$

The chord bound gives  $g_r(e) \geq eG_r/D_0$  for  $0 \leq e \leq D_0$ . Substituting (12) in the lower bound and in this linearized upper bound yields

$$-T_r + \binom{v-2}{r-2} T_2 \leq -\binom{v}{r} + \binom{v-2}{r-2} \binom{v}{2}, \quad (28)$$

$$D_0 T_r - G_r T_2 \leq D_0 \binom{v}{r} - G_r \binom{v}{2}. \quad (29)$$

When  $D_0 = 0$ , the implemented replacement for (29) is the ordinary bound  $T_r \leq \binom{v}{r}$ . This replacement is weaker than all information available in that case, but remains valid and is sufficient for the finite certificates.

## 2.7 Reduction to successive differences on a finite parameter domain

We now translate the structural constraints into sign conditions on successive coefficient differences and identify the indices that still require certificates.

Set

$$\Delta_k = c_k - c_{k-1}, \quad B_{m,j}(k) = \binom{m}{k-j} - \binom{m}{k-j-1}. \quad (30)$$

The exact coefficient identity gives

$$\Delta_k = B_{a,0}(k) + \sum_{j,m} B_{m,j}(k) t_{j,m}. \quad (31)$$

The following decreasing-tail bound for bipartite graphs is due to Levit and Mandrescu [20]. We give a counting proof in the forest case for completeness.

**Lemma 2.10.** *For a forest of independence number  $a$ ,*

$$(k+1)c_{k+1} \leq 2(a-k)c_k. \quad (32)$$

*In particular, with*

$$K(a) = \left\lfloor \frac{2a+1}{3} \right\rfloor, \quad (33)$$

*the sequence  $c_{K(a)}, c_{K(a)+1}, \dots, c_a$  is nonincreasing.*

*Proof.* Given an independent  $k$ -set  $S$ , let  $A(S)$  be the vertices that can be added to it. The induced forest on  $A(S)$  has independence number at most  $a-k$ , since any of its independent sets can be joined to  $S$ . A bipartite graph has an independent set containing at least half its vertices, so  $|A(S)| \leq 2(a-k)$ . Count pairs  $(S, u)$  with  $u \in A(S)$ . Every independent  $(k+1)$ -set occurs  $k+1$  times, proving (32). The factor  $2(a-k)/(k+1)$  is at most one when  $k \geq \lceil (2a-1)/3 \rceil = K(a)$ .  $\square$

For each  $1 \leq k < K(a)$  it is sufficient to prove one of the statements

$$\Delta_k > 0, \quad \Delta_{k+1} \leq 0, \quad \Delta_k \leq 0 \implies \Delta_{k+1} \leq 0. \quad (34)$$

Indeed, once a nonpositive difference occurs, these alternatives prevent a later positive difference before the tail. The first alternative cannot apply at an index where the difference is already nonpositive, while the other two give the required next sign. Lemma 2.10 supplies all remaining signs. Also  $c_0 = 1$  and  $c_1 = n$ .

The empty forest and forests with no edges have independence polynomial 1 or  $(1+z)^n$ , respectively, and are handled directly. For forests with at least one edge and order at most 60, all required triples lie in the explicitly regenerated domain

$$\mathcal{P}_{60} = \left\{ (n, a, k) : \begin{array}{l} 2 \leq n \leq 60, \quad \lceil n/2 \rceil \leq a \leq n-1, \\ 1 \leq k < \lfloor (2a+1)/3 \rfloor \end{array} \right\}. \quad (35)$$

It contains 17 100 triples. Empty ranges for small values of  $a$  require no certificate. These triples are scalar parameter choices, not representatives of graph isomorphism classes.

## 2.8 Exact certificate identities and the complete finite check

The remaining task is to certify one of these sign conditions for each parameter triple, using direct bounds where they suffice and linear combinations of the structural inequalities elsewhere.

**Direct criteria.** Let

$$q_{a,v}(r) = \sum_{j=0}^{\min(v,r)} \binom{v}{j} 2^j \binom{a-v}{r-j}, \quad (36)$$

the coefficient of  $z^r$  in the upper-bound polynomial of Lemma 2.1. The comparisons

$$\binom{n-k+1}{k} > q_{a,v}(k-1) \quad \text{or} \quad q_{a,v}(k+1) \leq \binom{n-k+1}{k}$$

prove, respectively,  $\Delta_k > 0$  or  $\Delta_{k+1} \leq 0$ .

The basic layer masses give further direct bounds. For each  $j$  define

$$L_j(r) = \begin{cases} \min_{0 \leq m \leq a-1} B_{m,1}(r), & j = 1, \\ \min(0, \min_{0 \leq m \leq a-j} B_{m,j}(r)), & j \geq 2, \end{cases}$$

$$U_j(r) = \begin{cases} \max_{0 \leq m \leq a-1} B_{m,1}(r), & j = 1, \\ \max(0, \max_{0 \leq m \leq a-j} B_{m,j}(r)), & j \geq 2. \end{cases}$$

Then

$$B_{a,0}(r) + \sum_{j=1}^v W_j L_j(r) \leq \Delta_r \leq B_{a,0}(r) + \sum_{j=1}^v W_j U_j(r). \quad (37)$$

The absence of the extra zero in the first layer is justified by  $T_1 = v$  exactly. A strictly positive lower bound at  $r = k$  or a nonpositive upper bound at  $r = k + 1$  proves the corresponding alternative in (34).

For a rational number  $s \geq 0$ , the same upper-bound rule can be applied to  $B_{m,j}(k+1) - sB_{m,j}(k)$ . A nonpositive resulting bound proves

$$\Delta_{k+1} \leq s\Delta_k, \quad (38)$$

which proves the implication in (34). These are the one-parameter rational comparison certificates called **separator** in the data.

**General linear certificates.** We use Farkas-type certificates: exact nonnegative combinations of valid linear inequalities, together with equality rows and explicit residual bounds; see Schrijver [25] for the underlying linear-inequality theory. Normalize the variables by  $x_{j,m} = t_{j,m}/W_j$ . Their layer masses are at most one, and the first layer mass is exactly one. For every valid integer inequality

$$\sum_{j,m} a_{j,m}^\rho t_{j,m} \leq b^\rho,$$

define its scale by

$$N_\rho = \max \left( 1, |b^\rho|, \max_{j,m} |W_j a_{j,m}^\rho| \right), \quad (39)$$

and write the corresponding normalized row as  $A^\rho x \leq b_\rho$ , where  $A_{j,m}^\rho = W_j a_{j,m}^\rho / N_\rho$  and  $b_\rho = b^\rho / N_\rho$ . Normalize the target affine function in the same way, obtaining

$$Q(x) = q_0 + \sum_{j,m} q_{j,m} x_{j,m}.$$

If  $\lambda_\rho \geq 0$  and  $\eta \in \mathbb{R}$ , set

$$R_{j,m} = q_{j,m} - \sum_{\rho} \lambda_{\rho} A_{j,m}^{\rho} - \eta \mathbf{1}_{j=1}.$$

The certificate principle is the following elementary bound:

$$Q(x) \leq q_0 + \sum_{\rho} \lambda_{\rho} b_{\rho} + \eta + \sum_{j=1}^v \max(0, \max_m R_{j,m}). \quad (40)$$

To see this, subtract the nonnegative combination of valid rows and use the first-layer equality for the  $\eta$  term. On each remaining layer, nonnegativity and mass at most one bound its residual contribution by the nonnegative maximum of its residual coefficients. No assertion about linear-programming optimality is needed.

There are three linear certificate types. A **positive** certificate uses the target  $-\Delta_k$  and makes the right side of (40) strictly negative. A **negative** certificate uses  $\Delta_{k+1}$  and makes the bound nonpositive. A **conditional** certificate has the latter target and is permitted to add the row  $\Delta_k \leq 0$ . It therefore proves exactly the implication in (34). That assumption cannot be discarded while retaining an unconditional negative target claim.

**Exact arithmetic and exhaustive parameter coverage.** Each stored linear certificate gives a positive integer denominator  $D_c$ , integer row-weight numerators, and an integer numerator for the unrestricted equality weight. The principal checker `verify.py` independently constructs the unweighted integer rows, computes a common multiple of their normalization denominators, and checks (40) using integers. It regenerates every triple in (35); it rejects missing, duplicate, or extraneous triples, invalid row parameters, negative inequality multipliers, and conditional assumptions with the wrong index. Direct and one-parameter comparisons are likewise performed exactly.

A second arithmetic implementation, `verify_grid.py`, uses the weighted rows from `kernel.py`. It rounds target coefficients and right sides upward and row coefficients downward on an integer grid. Nonnegative variables and nonnegative row multipliers make this an outward, conservative bound. This second implementation checks all 1 907 linear certificates; complete parameter coverage and the 670 one-parameter comparisons are checked by the principal verifier. Both implementations use the Python standard library, without a graph catalogue, an optimizer, or floating-point feasibility tolerances.

The resulting complete parameter partition is given in Table 1.

| Certificate or direct criterion                             | Number of triples |
|-------------------------------------------------------------|-------------------|
| Path lower bound exceeds the preceding matching upper bound | 6 724             |
| Direct positive lower bound for $\Delta_k$                  | 5 709             |
| Direct nonpositive upper bound for $\Delta_{k+1}$           | 2 090             |
| One-parameter rational comparison                           | 670               |
| <b>positive</b> linear certificate                          | 603               |
| <b>negative</b> linear certificate                          | 158               |
| <b>conditional</b> linear certificate                       | 1 146             |
| Total                                                       | 17 100            |

Table 1: Complete parameter coverage for Proposition 1.2.

*Proof of Proposition 1.2.* Forests without edges were dealt with above. For a forest with an edge, choose  $I$  by Lemma 2.4 and form its exact counts  $t_{j,m}$ . The preceding lemmas show that these counts satisfy every structural inequality used by the certificates. Its parameters lie in the stated ranges, and the exact finite check proves (34) for every required index. The first

nonpositive difference, if one occurs before  $K(a)$ , is followed only by nonpositive differences through index  $K(a)$ . The tail is nonincreasing by Lemma 2.10. Hence the whole coefficient sequence is unimodal.  $\square$

### 3 The monotone window and the hard-core reduction

We now prove Proposition 1.3. The argument uses the hard-core distribution and exact conditional-binomial inequalities.

The initial segment below is established by Fang et al. in [11, Proposition 8.2]. We give a direct injection proof and then prove the terminal bound used here by comparison with a matching.

**Lemma 3.1.** *Let  $F$  be an  $n$ -vertex forest with independence number  $a$ . Put*

$$L = \lceil n/4 \rceil, \quad U = \lfloor (n + 2a)/6 \rfloor + 1.$$

*Then  $c_0 \leq \dots \leq c_L$  and  $c_U \geq c_{U+1} \geq \dots \geq c_a$ .*

*Proof.* Root every component. For an independent  $k$ -set  $S$ , consider triples  $(S, v, u)$  with  $v \in S$  and  $u$  a child of  $v$ . If  $u$  has no child in  $S$ , send the triple to  $((S \setminus \{v\}) \cup \{u\}, u)$ . Otherwise choose the first child  $w$  of  $u$  belonging to  $S$  and send it to  $(S, w)$ . Outputs are independent  $k$ -sets with a marked nonroot vertex. In the first case the grandparent of the mark, if it exists, is absent; in the second it is the selected vertex  $v$ . The mark and its ancestors therefore recover the input, proving injectivity. If  $D(S) = \sum_{v \in S} \deg(v)$  and  $\rho_k = \sum_S |S \cap \{\text{roots}\}|$ , this gives

$$\sum_S D(S) - kc_k + \rho_k \leq kc_k - \rho_k.$$

Counting extensions now yields  $(k+1)c_{k+1} \geq (n-3k)c_k$ . For every  $k < L$ ,  $n-3k \geq k+1$ , proving the initial segment.

Fix a maximum matching of size  $v = n - a$ . Retaining only its edges gives a disjoint union of  $v$  pairs and  $a - v$  singletons, with polynomial

$$Q(x) = (1 + 2x)^v (1 + x)^{a-v} = \sum q_k x^k.$$

Uniform independent sets in two consecutive ranks of this reference graph admit an inclusion coupling. To see this, more generally give the blocks weights  $w_1, \dots, w_r > 0$ . Select a  $k$ -subset of blocks proportionally to the product of its weights. If  $e_j$  denotes the elementary symmetric sum of the first  $r-1$  weights, the probability of selecting the last block is  $w_r e_{k-1} / (e_k + w_r e_{k-1})$ . It increases with  $k$ , because the coefficient sequence of  $\prod_{i < r} (1 + w_i x)$  is log-concave. Couple these last-block indicators monotonically. If they agree, couple the remaining ranks by induction; if they differ, use the same remaining block set. Within common blocks use the same uniform element. All conditional marginals are preserved, including boundary ranks.

Independence in  $F$  is a downward-closed event under this coupling, so  $c_{k+1}/q_{k+1} \leq c_k/q_k$ . Finally

$$(k+1)(q_{k+1} - q_k) = (n + 2a - 6k + 1)q_k - 2(a - k + 1)(q_k - q_{k-1}).$$

For  $k \geq U$  the first coefficient is nonpositive. If  $q_k \geq q_{k-1}$  this proves  $q_{k+1} \leq q_k$  directly; otherwise log-concavity proves it. Ratio domination transfers this to  $c_k$ .  $\square$

The monotone end segments leave only the intermediate ranks to control; the next mean bound lets us center the hard-core distribution at each such rank.

At activity  $\lambda > 0$ , sample an independent set with probability  $\lambda^{|S|}/I_F(\lambda)$ . Let  $K = |S|$ ,  $\mu = \mathbb{E}K$ , and  $V = \text{Var } K$ . The following scale-corrected mean estimate is proved in Section 6:

**Lemma 3.2.** *Every forest satisfies  $3\mu_F(2) \geq a + 29n/60$ .*

Since  $d\mu/d\log \lambda = V > 0$  for a nonempty graph, the mean is strictly increasing. Conditioning on either bipartition class gives  $V \geq \mu/[2(1 + \lambda)]$ , whence

$$\mu(9/4) \geq \sqrt{27/26} \mu(2) > (n + 2a)/6.$$

The last inequality follows from  $a \geq n/2$  and  $27 \cdot 59^2 > 26 \cdot 60^2$ . Since every vertex has marginal occupation at most  $\lambda/(1 + \lambda)$ , also  $\mu(1/3) \leq n/4$ . Thus every necessary rank  $L \leq k < U$  has a unique activity  $\lambda_k \in [1/3, 9/4]$  with  $\mu(\lambda_k) = k$ .

With the mean at the target rank, we now expose a conditionally binomial part of the independent-set size.

For such an activity, let  $\pi_v$  be the actual marginal at  $v$ , and choose an independent set  $B$  maximizing  $W = \sum_{v \in B} \pi_v$ . This is a maximum-weight set, not necessarily a maximum-cardinality set. Bipartiteness gives  $W \geq \mu/2$ . Write  $C = V(F) \setminus B$ . Conditional on the entire occupied configuration in  $C$ , let  $Y$  be its size and  $M$  the number of unblocked vertices in  $B$ . Then

$$K = Y + \text{Bin}(M, q), \quad q = \frac{\lambda}{1 + \lambda}, \quad m = \mathbb{E}M, \quad qm = W. \quad (41)$$

At  $\mu = k$ , put  $j = k - Y$  and  $\delta = j - qM$ . The same probability space gives

$$\mathbb{E}\delta = 0, \quad \mathbb{E}\delta^2 = V - q(1 - q)m. \quad (42)$$

No independence of  $M$  and  $Y$  is asserted.

## 4 The final matching-count extension: orders 59 through 79

For  $59 \leq n \leq 79$ , the following matching constraint strengthens the conditional-binomial reduction. The larger orders are covered by the analytic and finite-kernel arguments in Sections 7 and 8.

**Lemma 4.1.** *Let  $B$  be as in (41),  $h = |B|$ , and  $H = F[C]$ . Then*

$$h \geq \lceil n/3 \rceil, \quad \nu(H) \geq \max(0, n - 2h), \quad \alpha(H) \leq \lfloor n/2 \rfloor.$$

*Proof.* For a leaf  $u$  whose component is not a single edge, let  $v$  be its neighbor. The formulas

$$\pi_u = \frac{\lambda I_{F - \{u, v\}}(\lambda)}{I_F(\lambda)}, \quad \pi_v = \frac{\lambda I_{F - N[v]}(\lambda)}{I_F(\lambda)}$$

show  $\pi_u > \pi_v$ , since the second induced graph is strictly smaller than the first. Replacing  $v$  by  $u$  would improve a maximum-weight independent set; therefore  $v \notin B$ , and maximality implies  $u \in B$ . Isolated vertices are in  $B$ , and in each edge component exactly one endpoint is in  $B$ .

Let  $\kappa$  be the number of components of  $F$ ,  $d$  the number of its edge components,  $c = n - h$ ,  $e = |E(H)|$ , and  $s_0$  the number of isolated vertices in  $H$ . Each  $C$ -vertex has a neighbor in  $B$ . An isolated vertex of  $H$  has at least two such neighbors, except for the  $d$  endpoints in edge components. Hence  $e(B, C) \geq c + s_0 - d$ , and  $e + s_0 \leq h - \kappa + d$ . The number of nontrivial components of  $H$  is  $c - e - s_0 \geq n - 2h$ . Choosing one edge from each gives the matching bound. Comparing twice its size with  $c$  gives  $h \geq n/3$ . If  $h \leq n/2$ , then  $\alpha(H) \leq c - (n - 2h) = h$ ; otherwise  $\alpha(H) \leq c = n - h$ .  $\square$

The matching bound restricts the number of configurations in the complement, yielding the coefficient envelopes used in the next conditional estimate.

For  $c \geq 2r$  define

$$D(c, r; y) = \sum_t \binom{r}{t} 2^t \binom{c-2r}{y-t}.$$

Deleting all edges of  $H$  except a matching bounds its independent-set coefficients by those of  $(1+2x)^r(1+x)^{c-2r}$ . Therefore safe integer envelopes are

$$A_{n,y}^{(M)} = \max_{\substack{[n/3] \leq h \leq n \\ h \geq M}} D(n-h, \max(0, n-2h); y), \quad A_{n,y} = A_{n,y}^{(0)}. \quad (43)$$

An empty maximum is zero; binomial coefficients outside their natural ranges are zero.

**Lemma 4.2.** *For every integer  $y$ ,*

$$\mathbb{E}[\mathbf{1}_{\{Y=y\}}(1-q)^M] = \frac{i_y(H)\lambda^y}{I_F(\lambda)} \leq \frac{A_{n,y}\lambda^y}{P_n(\lambda)},$$

where  $P_n(x) = \sum_t \binom{n-t+1}{t} x^t$  is the path polynomial.

*Proof.* The probability of an independent configuration  $S \subseteq C$  is  $\lambda^{|S|}(1+\lambda)^{M(S)}/I_F(\lambda)$ . Multiplication by  $(1-q)^{M(S)}$  cancels the free-vertex factor. Sum over  $|S| = y$  and use (43) and the path coefficient lower bound proved in the finite-order part.  $\square$

Fix a rectangle  $q \in [q_a, q_b]$ ,  $m \in [m_l, m_h]$ , and  $59 \leq n \leq 79$ . Put  $a_0 = q_a/(1-q_a)$ ,  $b_0 = q_b/(1-q_b)$ . The certified density bound and integer rounding give  $k \geq k_{\min}$  and  $m \geq k_{\min}/(2q_b)$ . All real target distributions are included in

$$\mathcal{A} = \{(n, k) \in \mathbb{Z}^2 : 59 \leq n \leq 79, k_{\min} \leq k \leq \lfloor 2q_b m_h \rfloor, n/4 \leq k \leq n/2, rn \leq k \leq q_b n\}.$$

The value  $r$  here is a certified density constant, not the matching size in  $D$ . Define fixed upward rational bounds

$$B_j \geq \max_{(n,k) \in \mathcal{A}} \frac{A_{n,k-j} b_0^{k-j}}{P_n(a_0)}.$$

Terms with  $k-j$  outside the range of independent-set sizes are zero. Lemma 4.2 proves  $\mathbb{E}[\mathbf{1}_{\{k-Y=j\}}(1-q)^M] \leq B_j$  throughout the rectangle. All possible conditional states belong to the finite set

$$\mathcal{S} = \{(M, j) \in \mathbb{Z}^2 : 0 \leq M \leq 79, \exists (n, k) \in \mathcal{A} : A_{n,k-j}^{(M)} > 0\}.$$

Excluding a state uses a zero combinatorial count, never a small numerical probability.

## 5 Rational mixture certificates and the inherited chain

We now combine the moment bounds and the complement-count constraint into inequalities whose positive expectations exclude a weak valley.

Let  $b_M(j; q)$  be the binomial point probability, extended by zero outside its support. Define

$$f_L = (1-q)b_M(j; q) - qb_M(j-1; q), \quad f_R = qb_M(j; q) - (1-q)b_M(j+1; q),$$

$$f_C = 2b_M(j; q) - b_M(j-1; q) - b_M(j+1; q).$$

Every certificate uses fixed nonnegative weights  $w_L, w_R, w_C$ , not all zero, and  $f = w_L f_L + w_R f_R + w_C f_C$ . Its scale  $S$  is positive. Source inequalities on the same distribution provide

$$\mathbb{E}\delta^2 \leq \alpha_0 m + \beta_0, \quad \text{Var } M \leq D_0 m, \quad \mathbb{E}(1 - q + qt_i)^M \leq e^{-\ell_i m}. \quad (44)$$

Here the finitely many  $t_i \in [0, 1]$  and  $\ell_i \geq 0$  are fixed rational parameters of the source bounds. Their proof is part of the chain, as described below; constants cannot be copied to a smaller threshold without checking the signs of the  $n$ -dependent terms.

The pointwise acceptance inequality is

$$Sf(M, j; q) \geq A_0 + B_0 M + C_0 \delta - D_\delta \delta^2 - D_M M^2 - \sum_i T_i (1 - q + qt_i)^M - U_j (1 - q)^M, \quad (45)$$

for every state in  $\mathcal{S}$  and every  $q$  in the rectangle, with  $D_\delta, D_M, T_i, U_j \geq 0$ . The other coefficients may have either sign. Taking expectations gives

$$\begin{aligned} S\mathbb{E}f \geq H(m) &:= A_0 + B_0 m - D_\delta(\alpha_0 m + \beta_0) - D_M(m^2 + D_0 m) \\ &\quad - \sum_i T_i e^{-\ell_i m} - \sum_j U_j B_j. \end{aligned} \quad (46)$$

Since  $H''(m) = -2D_M - \sum_i T_i \ell_i^2 e^{-\ell_i m} \leq 0$ , strictly positive endpoint budgets certify all real  $m \in [m_l, m_h]$ .

To apply the budget at the selected activity, we must certify the pointwise inequality throughout each activity interval.

For a fixed state let  $R(q)$  be the left side of (45) minus the right side. Write  $B_L(M)$ ,  $B_R(M)$ , and  $B_C(M)$  for the supplied uniform upper bounds on the second  $q$ -derivatives of the three kernels, valid for every integer  $j$  and all  $q \in [q_a, q_b]$ . Explicit bounds, including the cases  $M < 3$  and all integers  $j$ , are proved in Section 8. These bounds give

$$\begin{aligned} R''(q) &\leq Z_{M,j} := S(w_L B_L(M) + w_R B_R(M) + w_C B_C(M)) + 2D_\delta M^2 \\ &\quad + \sum_i T_i M(M-1)(1-t_i)^2(1-q_a + q_a t_i)^{M-2} \\ &\quad + U_j M(M-1)(1-q_a)^{M-2}. \end{aligned}$$

The terms containing  $M(M-1)$  are zero when  $M < 2$ . With  $h_q = q_b - q_a$ , it suffices to check  $R(q_a), R(q_b) \geq h_q^2 Z_{M,j}/8$ . The interpolation remainder then proves  $R(q) \geq 0$  on the entire interval. Rational enclosures for elementary functions and probabilities are directed; a precision scale is not a floating-point tolerance.

The source bounds used above are justified as follows. Section 7 derives the rooted-message identities, nonnegative-source variance estimates and heterogeneous activity integrals used in (44). Section 8 gives the refined flow and logarithmic sources, the density induction and the finite-kernel acceptance rules. These arguments apply to all forests and to the entire stated activity and response domains.

For the Laplace bound, the set  $B$  is fixed while only its activities are lowered. The exact identity is

$$\mathbb{E}e^{-s_\lambda(u)M} = \mathbb{E}e^{-uK_B}, \quad s_\lambda(u) = \log \frac{1 + \lambda}{1 + \lambda e^{-u}}.$$

The variance bounds used along this path are valid for heterogeneous activities. The density input at activity 1 uses the path-mean theorem of Andriantiana, Razanajatovo Misanantenaina and Wagner [2]; the remaining density inputs follow from the full-order induction in Section 8.

With the source bounds and continuous-activity checks in place, it remains to verify coverage of the required parameters. The 231 new certificates cover 154 activity strata,

subordinate to 43 parent activity intervals. Together with 188, 243, and 558 older finite kernels, respectively from stages 99, 130, and 170, they form 1220 finite rectangles. Their upper  $m$  boundary joins the stage-350 unbounded- $m$  estimates. All 236 stage-80 kernels are excluded for  $n < 80$ , because some of them use signed costs requiring  $n \geq 80$ . This distinction is checked by the source and coverage code.

For the new certificates, the stored and replayed integer endpoint-state count is 1,185,408, and the smallest normalized expectation budget is

$$\frac{424830401847072141613691}{10^{28}} > 0.$$

The corresponding scale is 23; thus this normalized number must be divided by 23 to interpret it as a lower bound on the unscaled mixed-kernel expectation. It is not a common lower bound on a coefficient curvature.

The positive mixture budget now translates into a comparison of adjacent coefficients, completing the link from parameter coverage to unimodality.

Suppose at a necessary rank  $k$  that  $c_k \leq c_{k-1}$  and  $c_k \leq c_{k+1}$ . Writing  $p_j = c_j \lambda^j / I_F(\lambda)$ , the first two inequalities give  $\mathbb{E}f_L \leq 0$  and  $\mathbb{E}f_R \leq 0$ . They also imply

$$p_{k-1} + p_{k+1} \geq (\lambda^{-1} + \lambda)p_k \geq 2p_k,$$

so  $\mathbb{E}f_C \leq 0$ . This contradicts the strictly positive budget in (46). Every necessary rank therefore satisfies  $c_k > \min(c_{k-1}, c_{k+1})$ . If a strict descent is followed later by a strict ascent, choose the first subsequent index  $k$  with  $c_{k+1} \geq c_k$ . Then  $c_{k-1} > c_k$  and  $c_{k+1} \geq c_k$ . The monotone end segments in Lemma 3.1 place such an index in  $[L, U-1]$ , giving a prohibited weak local minimum, including the possibility of a flat valley. Consequently all orders  $59 \leq n \leq 79$  are covered once the source and coverage certificates are accepted.

Table 2: Frozen large-order stages. Each row proves the new finite interval and inherits the larger orders; the base stage includes an unbounded-order argument.

| Stage | New orders            | Proof layer in the fixed data  |
|-------|-----------------------|--------------------------------|
| 900   | $n \geq 900$          | base source and kernel proof   |
| 500   | $500 \leq n \leq 899$ | stage 500 proof and replay     |
| 350   | $350 \leq n \leq 499$ | stage 350 proof and replay     |
| 170   | $170 \leq n \leq 349$ | stage 170 proof and replay     |
| 130   | $130 \leq n \leq 169$ | stage 130 proof and replay     |
| 99    | $99 \leq n \leq 129$  | stage 99 proof and replay      |
| 80    | $80 \leq n \leq 98$   | stage 80 proof and replay      |
| 59    | $59 \leq n \leq 79$   | final matching-count extension |

The analytic and finite-kernel sections prove the stated interfaces, and the exact certificates establish their parameter inequalities. Taking the base and each extension in the table proves Proposition 1.3. Section 11 specifies the fixed data and the full proof-only verification procedure.

**Computational assistance.** AI-assisted tools were used in mathematical review, exact computational reproduction, literature research, translation and manuscript preparation. The authors are responsible for the mathematical claims and the submitted text. The proof is not formalized in a proof assistant.

## 6 The activity-two mean inequalities

Andriantiana, Razanajatovo Misanantenaina and Wagner [2] studied mean independent-set size and proved path extremality at activity 1. The activity-two estimates required here are proved below. The finite certificates appearing below concern complete induction cases. No conclusion is inferred from testing a collection of forests.

For a finite graph  $F$ , put

$$Z_F(\lambda) = \sum_{S \text{ independent}} \lambda^{|S|}, \quad \mu_F(\lambda) = \frac{\lambda Z'_F(\lambda)}{Z_F(\lambda)}.$$

Write  $n(F) = |V(F)|$ ,  $\alpha(F)$  for the independence number, and  $c(F)$  for the number of nonempty connected components. Unless an activity is displayed, partition functions and logarithmic derivatives in this section are evaluated at  $\lambda = 2$ . Define

$$e(F) = 3\mu_F(2) - n(F), \quad G(F) = 3\mu_F(2) - \alpha(F) - \frac{29}{60}n(F).$$

Both quantities are additive under disjoint union and vanish on the empty forest.

**Theorem 6.1.** *Every finite forest satisfies*

$$\mu_F(2) \geq \frac{n(F)}{3} + \frac{2c(F)}{15}, \quad (47)$$

$$3\mu_F(2) \geq \alpha(F) + \frac{29}{60}n(F). \quad (48)$$

*Equality in (47) holds precisely when every nonempty component is  $K_2$ , including the empty forest.*

### 6.1 The component correction

For a nonempty graph  $H$ , a vertex  $v$ , and  $J = H - v$ , the vertex recurrence gives

$$1 \leq r := \frac{Z_H}{Z_J} = 1 + 2 \frac{Z_{H-N[v]}}{Z_J} \leq 3. \quad (49)$$

The upper bound follows because every independent set of  $H - N[v]$  is an independent set of  $J$ .

Attach two new vertices  $a, b$  with edges  $va, ab$ , and call the resulting graph  $F$ . The polynomial identity

$$Z_F(\lambda) = (1 + \lambda)Z_H(\lambda) + \lambda Z_J(\lambda)$$

implies

$$e(F) = \frac{3Z_H e(H) + 2Z_J e(J)}{3Z_H + 2Z_J}. \quad (50)$$

Indeed,  $n(F) = n(H) + 2 = n(J) + 3$ , and the two multiplier remainders are  $3\lambda(1+\lambda)' - 2(1+\lambda) = 0$  and  $3\lambda(\lambda)' - 3\lambda = 0$  at  $\lambda = 2$ .

Instead attach  $m \geq 2$  new leaves at  $v$ . Now

$$Z_F(\lambda) = Z_H(\lambda) + ((1 + \lambda)^m - 1)Z_J(\lambda).$$

With  $A = 3^m - 1$ , differentiation gives

$$e(F) = \frac{Z_H e(H) + A Z_J e(J)}{Z_H + A Z_J} + \frac{Z_J}{Z_H + A Z_J} ((m-1)3^m + m + 1 - mr). \quad (51)$$

Here  $n(F) = n(H) + m = n(J) + m + 1$ . The differentiated multiplier contributes  $(m - 1)3^m + m + 1$ , and the  $Z_H$  term contributes  $-mr$ . By (49), the last numerator is at least

$$(m - 1)3^m - 2m + 1 \geq 7m - 8 \geq 6. \quad (52)$$

We prove  $e(T) \geq 2/5$  for every nonempty tree by strong induction. The bases are

| $T$   | $Z_T(2)$ | $\mu_T(2)$ | $e(T)$   |
|-------|----------|------------|----------|
| $P_1$ | 3        | $2/3$      | 1        |
| $P_2$ | 5        | $4/5$      | $2/5$    |
| $P_3$ | 11       | $14/11$    | $9/11$ . |

For a tree of order at least four, root at a leaf and take a deepest vertex  $v$  having children. It is not the root, and all its children are leaves. If it has one child,  $v$  and that child form a pendant two-vertex path. Deleting them leaves a tree  $H$  of order at least two; deleting its attachment vertex leaves a nonempty forest  $J$ . Induction and additivity give  $e(H), e(J) \geq 2/5$ , so (50) applies. Equality would require both  $H$  and  $J$  to be single edges, which is impossible because their orders differ by one. If  $v$  has at least two children, delete these leaves while retaining  $v$ . Both the resulting host  $H$  and  $J = H - v$  are nonempty. Equations (51) and (52) give  $e(T) > 2/5$ . This proves the tree statement and its equality case. Summing over components proves (47).

We will also use the exact path formulas

$$P_j(2) := Z_{P_j}(2) = \frac{2^{j+2} - (-1)^j}{3}, \quad e(P_j) = \frac{2}{3} - \frac{(j+2)(-1)^j}{2^{j+2} - (-1)^j}. \quad (53)$$

They follow from  $P_0(\lambda) = 1$ ,  $P_1(\lambda) = 1 + \lambda$  and  $P_j = P_{j-1} + \lambda P_{j-2}$ , together with its logarithmic derivative. In particular, the density constant  $1/3$  in (47) cannot be increased uniformly over all forests.

## 6.2 Pruning a minimal negative tree

For the remainder of the proof suppose, for a contradiction, that  $G$  is negative on some forest. By additivity choose a negative tree  $F$  of minimum order. Every strictly smaller forest has nonnegative  $G$ . A branch vertex means a vertex of degree at least three. A pendant leg has length equal to its number of edges from its branch vertex to its leaf.

**Lemma 6.2.** *Every leaf of a minimum negative tree belongs to every maximum independent set. Every pendant leg has length at most thirteen, and all pendant legs at the same branch vertex have the same parity.*

*Proof.* Let  $\ell$  be a leaf with neighbour  $v$ , and put  $H = F - \ell$  and  $J = F - \{\ell, v\}$ . If some maximum independent set omits  $\ell$ , then  $\alpha(F) = \alpha(H) = \alpha(J) + 1$ . Differentiating  $Z_F = Z_H + \lambda Z_J$  gives

$$G(F) = \frac{rG(H) + 2G(J)}{r+2} + \frac{124 - 29r}{60(r+2)}, \quad r = Z_H/Z_J \in [1, 3]. \quad (54)$$

The last summand is decreasing in  $r$  and at least  $37/300 > 0$ , contrary to minimality.

If a pendant leg has length at least fourteen, delete its last fourteen vertices, obtaining a nonempty host  $H$  with attachment vertex  $v$ , and put  $J = H - v$ . Let  $\delta = \alpha(H) - \alpha(J) \in \{0, 1\}$ . Reattaching these vertices gives

$$Z_F = P_{13}Z_H + \lambda P_{12}Z_J, \quad \alpha(F) = \alpha(H) + 7.$$

At activity two,  $A = P_{13}(2) = 10923$  and  $B = 2P_{12}(2) = 10922$ . The derivative identity becomes

$$G(F) = \frac{rAG(H) + BG(J)}{rA + B} + \Psi_\delta(r), \quad \Psi_\delta(r) \geq \frac{45415 - 10873r}{10(10923r + 10922)} \geq \frac{6398}{218455} > 0. \quad (55)$$

The worse state is  $\delta = 1$ ; its two unnormalized multiplier remainders are  $-10873/10$  and  $9083/2$ . These follow by substituting the path recurrence and its derivative. The displayed linear-fractional function is decreasing on  $[1, 3]$ . This again contradicts minimality.

Finally, on an even leg the terminal leaf can belong to every maximum independent set only if the branch vertex is always selected. On an odd leg it can do so only if the branch vertex is always omitted. To see this, fix the state of the branch vertex and use the two alternating maximum choices on the remaining even-order path. In the opposite state there is a choice omitting the terminal leaf with unchanged cardinality. The two necessary states cannot coexist at one branch vertex.  $\square$

### 6.3 Short-leg clusters and spiders

Attach  $m \geq 2$  paths of positive lengths  $\ell_1, \dots, \ell_m$  at a vertex  $v$  of a nonempty host  $H$ . Put  $J = H - v$ ,  $r = Z_H/Z_J \in [1, 3]$ ,  $\delta = \alpha(H) - \alpha(J)$ , and

$$L = \sum_i \ell_i, \quad A(\lambda) = \prod_i P_{\ell_i}(\lambda), \quad B(\lambda) = \prod_i P_{\ell_i-1}(\lambda), \quad X_0 = A(2)/B(2) > 1.$$

Conditioning at  $v$  gives the polynomial identity

$$Z_F = BZ_H + (A - B)Z_J. \quad (56)$$

Here and below a path of length  $\ell$  attached at  $v$  contributes  $\ell$  new vertices.

For even lengths define

$$a_\ell = 3\mu_{P_\ell}(2) - \frac{59\ell}{60}, \quad b_\ell = 3\mu_{P_{\ell-1}}(2) - \frac{59\ell}{60}, \quad h = \delta + \frac{29}{60}.$$

Since  $\alpha(F) = \alpha(H) + L/2$ , differentiating (56) yields

$$G(F) = \frac{rG(H) + (X_0 - 1)G(J)}{X_0 + r - 1} + \frac{X_0(\sum_i a_{\ell_i} - h) + (r - 1)\sum_i b_{\ell_i} + h}{X_0 + r - 1}. \quad (57)$$

For odd lengths subtract  $1/2$  from both definitions of  $a_\ell, b_\ell$  and take  $h = 29/60$ . With at least two odd legs, a maximum independent set omits  $v$ , and  $\alpha(F) = \alpha(J) + (L + m)/2$ . Thus

$$G(F) = \frac{rG(H) + (X_0 - 1)G(J)}{X_0 + r - 1} + \frac{X_0(\sum_i a_{\ell_i} - h) + (r - 1)\sum_i b_{\ell_i} + h + r\delta}{X_0 + r - 1}. \quad (58)$$

All denominators are positive. The two host margins are retained; a cluster is excluded when its last numerator is positive for all permitted  $r, \delta$ .

For odd  $\ell \in \{1, 3, 5, 7, 9, 11, 13\}$ , the path formulas give

$$P_\ell(2)/P_{\ell-1}(2) > 2, \quad a_\ell \geq 343/1140, \quad b_\ell \geq -89/60.$$

Consequently, for  $m \geq 4$  the last numerator of (58) is bounded below by

$$f_m = 2^m \left( \frac{343m}{1140} - \frac{29}{60} \right) - \frac{89m}{30} + \frac{29}{60}.$$

Here  $f_4 = 53/380 > 0$ , and

$$f_{m+1} - f_m = 2^m \left( \frac{343(m+2)}{1140} - \frac{29}{60} \right) - \frac{89}{30}$$

is increasing for  $m \geq 4$ , with initial value  $691/38 > 0$ .

For even  $\ell \in \{2, 4, 6, 8, 10, 12\}$ , the corresponding bounds are

$$P_\ell(2)/P_{\ell-1}(2) \geq 5/3, \quad a_\ell \geq 13/30, \quad b_\ell > -11/60.$$

The worse state in (57) is  $\delta = 1$ . For  $m \geq 4$  its last numerator is bounded below by

$$g_m = \left(\frac{5}{3}\right)^m \left( \frac{13m}{30} - \frac{89}{60} \right) - \frac{11m}{30} + \frac{89}{60}.$$

We have  $g_4 = 788/405 > 0$ , and

$$g_{m+1} - g_m = \left(\frac{5}{3}\right)^m \left[ \frac{2}{3} \left( \frac{13m}{30} - \frac{89}{60} \right) + \frac{5}{3} \frac{13}{30} \right] - \frac{11}{30}$$

is increasing from its value  $47327/7290 > 0$  at  $m = 4$ . Thus no upper bound on the number of legs is imposed.

For  $m = 2, 3$ , the last numerator of (57) or (58) is affine in  $r$  once the lengths and  $\delta$  are fixed. Positivity is therefore decided by its values at  $r = 1, 3$ . The complete finite parameter domain consists of the unordered pairs and triples from the six even lengths or seven odd lengths, and the two states  $\delta = 0, 1$ . Its size is

$$2 \left[ \binom{7}{2} + \binom{8}{3} + \binom{8}{2} + \binom{9}{3} \right] = 378.$$

The exact endpoint certificates leave the following cases:

- even lengths: only  $(2, 2)$  with  $\delta = 1$ ;
- odd lengths with  $\delta = 1$ : none;
- odd lengths with  $\delta = 0$ : all 28 unordered pairs, and the following 16 unordered triples:  
 $(3, 5, 5), (3, 5, 7), (3, 5, 9), (3, 7, 7),$   
 $(5, 5, 5), (5, 5, 7), (5, 5, 9), (5, 5, 11),$   
 $(5, 7, 7), (5, 7, 9), (5, 7, 11), (5, 9, 9),$   
 $(7, 7, 7), (7, 7, 9), (7, 7, 11), (7, 9, 9).$

The proof obligation here is the 756 rational endpoint inequalities specified by (57)–(58), with path values given by (53). The complete table is supplied with `adjusted_mean_leg_cluster.json`; `adjusted_mean_leg_cluster.py` reconstructs this parameter domain and the inequalities. All cases other than the listed 45 have strictly positive endpoint minima. The listed cases are retained as possible induction configurations, not asserted to occur in a negative tree.

**Lemma 6.3.** *Every nonempty tree with at most one branch vertex has  $G(T) > 0$ .*

*Proof.* For a path, (53) gives  $e(P_n) \geq 2/5$  when  $n \geq 2$  is even and  $e(P_n) > 2/3$  when  $n$  is odd. Since

$$G(P_n) = e(P_n) - \frac{n \bmod 2}{2} + \frac{n}{60},$$

all nonempty paths have positive margin.

Choose a nonpositive spider of minimum order if one exists. The reductions of Lemma 6.2 use only smaller spiders or path forests, so its legs have common parity and length at most

thirteen. At centre degree at least five, retain one leg as the host and apply the preceding  $m \geq 4$  cluster argument to the remaining legs. At degree four with even legs, the three-leg endpoint certificates exclude the spider. At degree three with even legs, any leg of length other than two allows a two-leg cluster other than  $(2, 2)$ , which is also excluded.

The remaining bases are exactly the 84 unordered triples and 210 unordered quadruples from  $\{1, 3, 5, 7, 9, 11, 13\}$ , and the even triple  $(2, 2, 2)$ . These bases have a direct finite algebraic specification. For a leg multiset  $\ell$  of size  $d$ , put

$$Z_\ell(\lambda) = \prod_i P_{\ell_i}(\lambda) + \lambda \prod_i P_{\ell_i-1}(\lambda), \quad N = 1 + \sum_i \ell_i.$$

For odd legs,  $\alpha = (\sum_i \ell_i + d)/2$ ; for  $(2, 2, 2)$ ,  $\alpha = 4$ . The supplied certificate verifies the integer inequalities

$$180 [\lambda Z'_\ell(\lambda)]_{\lambda=2} - (60\alpha + 29N)Z_\ell(2) > 0 \quad (59)$$

on exactly these 295 indices, using the path recurrence and product rule. This is a complete set of induction bases derived above, rather than a test of spiders up to an arbitrary order. The data and reconstruction are again `adjusted_mean_leg_cluster.json` and `adjusted_mean_leg_cluster.py`. Thus no nonpositive spider exists.  $\square$

#### 6.4 General rooted attachments and terminal blocks

Let  $T$  be a new tree rooted at  $w$ , attach it by the edge  $vw$  to a nonempty host  $H$ , and put  $J = H - v$ . Define

$$A(\lambda) = Z_{T-w}(\lambda), \quad B(\lambda) = Z_T(\lambda) - Z_{T-w}(\lambda), \quad M_A = [\lambda A']_2, \quad M_B = [\lambda B']_2.$$

Both  $A, B$  have nonnegative coefficients and

$$Z_F = AZ_H + BZ_J. \quad (60)$$

Write  $N = n(T)$ ,  $\delta_T = \alpha(T) - \alpha(T - w)$  and  $\delta = \alpha(H) - \alpha(J)$ . Each state lies in  $\{0, 1\}$ , and

$$\alpha(F) = \alpha(H) + \alpha(T) - \delta\delta_T. \quad (61)$$

Indeed, if both roots belong to every maximum independent set of their respective trees, the joining edge costs exactly one vertex; otherwise a maximum set on at least one side can omit its root without loss.

Set

$$X = rG(H), \quad Y = G(J), \quad r = Z_H/Z_J, \quad c = \alpha(T) - \delta\delta_T + \frac{29N}{60}.$$

Taking the logarithmic derivative in (60) and using (61) gives the affine identity

$$\begin{aligned} E_T(X, Y, r) &:= 60(Ar + B)G(F) \\ &= 60AX + 60BY + (180M_A - 60cA)r \\ &\quad + 180M_B - (60c + 29 + 60\delta)B. \end{aligned} \quad (62)$$

The host derivative is entirely contained in  $X, Y$ .

Compress the paths between branch vertices of a tree to obtain its branch skeleton. A terminal branch vertex in a minimum negative tree has, away from the rest of this skeleton, one of the 45 leg clusters above. Let  $T_{\ell,b}$  denote the entire terminal block, where the bridge from the host to that branch vertex has  $b \geq 1$  edges and all its  $b$  vertices other than the host are new. Thus  $N = b + \sum_i \ell_i$ , and the root is the first new bridge vertex. The necessary state at the host is

$$\delta = 1 - \delta_T. \quad (63)$$

For the even cluster  $(2, 2)$  the state at the terminal attachment is one; for the retained odd clusters it is zero. Adding a new root before a rooted tree changes its independence-number state from  $\varepsilon$  to  $1 - \varepsilon$ , since the new independence number is the larger of  $\alpha(T)$  and  $1 + \alpha(T - w)$ . Propagating these states along the bridge gives (63).

It is also the worse state in (62): the difference of the two constant remainders is

$$E_T^{\delta=1}(0, 0, r) - E_T^{\delta=0}(0, 0, r) = 60[\delta_T(Ar + B) - B].$$

This is  $60Ar > 0$  for  $\delta_T = 1$  and  $-60B < 0$  for  $\delta_T = 0$ .

## 6.5 All bridge lengths and the two surviving terminal types

Fix a retained leg cluster and extend its bridge. The root polynomials satisfy

$$A_{b+1} = A_b + B_b, \quad B_{b+1} = \lambda A_b.$$

For the formal polynomial  $P_b = rA_b + B_b$ , with the external variable  $r$  held fixed, this gives

$$P_{b+4} = (1 + 2\lambda)P_{b+2} - \lambda^2 P_b. \quad (64)$$

Bridge lengths of the same parity have the same  $\delta_T$ , and  $\alpha(T_{b+2}) = \alpha(T_b) + 1$ . Therefore  $60c_{b+2} - 60c_b = 118$ . Differentiating (64), substituting  $\lambda = 2$ , and then using (62), gives

$$N_{b+4} = 5N_{b+2} - 4N_b + 130P_{b+2} - 496P_b, \quad N_b(r) = E_{T_b}(0, 0, r), \quad (65)$$

in the state (63).

Let  $x = B_b/A_b$ . Initially  $0 < x \leq 2$ ; adding a root maps  $x$  to  $2/(1+x)$ , and two steps map it to  $2(1+x)/(3+x)$ . Thus for  $b \geq 3$ ,  $2/3 \leq x \leq 6/5$ . Consequently

$$\frac{P_{b+2}}{P_b} = \frac{(3r+2) + (r+2)x}{r+x} \geq \frac{43}{11} \quad (1 \leq r \leq 3). \quad (66)$$

For fixed either variable this expression is linear-fractional with positive denominator, so its extrema on the rectangle occur at corners. The minimum occurs at  $(r, x) = (1, 6/5)$  and  $(3, 2/3)$ . It follows that the forcing term in (65) is at least  $(134/11)P_b > 0$ .

For each of the 45 leg clusters, the exact base certificate verifies

$$N_b(r) > 0, \quad N_{b+2}(r) - N_b(r) > 0 \quad (b = 25, 26; r = 1, 3). \quad (67)$$

There are 180 base indices, each carrying the two displayed inequalities. They are affine in  $r$ , so they hold throughout  $[1, 3]$ . Equation (65) says that the next increment is four times the previous increment plus a positive forcing term. Induction along both parity classes proves  $N_b(r) > 0$  for every  $b \geq 25$ . Thus the bridge length is not truncated at a tested maximum.

It remains to handle the finite domain of 45 leg clusters and  $1 \leq b \leq 24$ . Besides  $X, Y \geq 0$  and  $1 \leq r \leq 3$ , minimality supplies additional valid inequalities. For every  $1 \leq j < N$ , attach the endpoint-rooted path  $P_j$  to the same host vertex, obtaining a tree  $H_j$  smaller than  $F$ . By (62),

$$X \geq 0, \quad Y \geq 0, \quad r - 1 \geq 0, \quad 3 - r \geq 0, \quad E_{P_j}(X, Y, r) \geq 0 \quad (1 \leq j < N). \quad (68)$$

These are inequalities of actual smaller forests, not an assumption of the desired theorem at the order of  $F$ .

The complete 1080-case certificate has the following partition:

| acceptance condition                   | number of cases |
|----------------------------------------|-----------------|
| $E_T(0, 0, 1), E_T(0, 0, 3) \geq 0$    | 736             |
| nonnegative affine identity using (68) | 342             |
| retained terminal types                | 2.              |

For each of the 342 cases the proof object is an exact identity

$$E_T = \sum_{i=1}^q c_i H_i + c_0, \quad q \leq 3, \quad c_i, c_0 \geq 0, \quad (69)$$

where each  $H_i$  is one explicitly indexed row of (68). Acceptance checks the coefficients of  $X, Y, r, 1$  separately. For example, the (5, 5) cluster with bridge length two and  $\delta = 0$  has

$$E_T = 163860X + 221880Y - 185448r + 257054,$$

and

$$E_{P_1} = 60X + 120Y - 89r + 124, \quad E_{P_3} = 300X + 360Y - 315r + 384.$$

Its identity is

$$E_T = \frac{6744}{13}Y + \frac{13395}{26}E_{P_1} + \frac{57611}{130}E_{P_3} + \frac{1494748}{65} > 0.$$

The full identities, base values (67), and complete 1080-case index set are in `endbranch_path_context_certificate.json`. Their construction is reconstructed by `endbranch_path_context_certificate.py`; `audit_endbranch_path_context.py` independently verifies the affine identities. The proof uses their exact coefficients and nonnegativity, not any discovery-stage optimization.

The only retained cases have  $b = 1$ ,  $\delta = 1$ , and legs (3, 3) or (3, 5). Their affine targets are

$$\begin{aligned} E_{33} &= 7260X + 3000Y + 1837r - 3200, \\ E_{35} &= 28380X + 12600Y + 5367r - 10860. \end{aligned} \quad (70)$$

Accordingly every terminal branch vertex of a minimum negative tree is the root of  $T_{33}$  or  $T_{35}$ , directly adjacent to its next branch vertex. These rooted trees have orders seven and nine, respectively, and

$$\alpha(T) - \alpha(T - \text{root}) = 0. \quad (71)$$

After deleting such a terminal block, its host has state one at the attachment vertex. If the tree had exactly two branch vertices, deleting one block would leave the other block as a host of state zero, contrary to this requirement. Lemma 6.3 already excludes at most one branch vertex. A minimum negative tree must therefore have at least three branch vertices.

## 6.6 The final two-level configuration

Root the branch skeleton at one of its leaves and choose a deepest vertex  $u$  that still has skeleton children. Since the skeleton has at least three vertices,  $u$  is not its root. It has an upward direction, and all its skeleton children are leaves. The terminal reduction shows that all these children are adjacent to  $u$  and carry  $T_{33}$  or  $T_{35}$ . There is at least one such child.

The vertex  $u$  belongs to every maximum independent set of  $F$ . To prove this, delete any one terminal block  $T$ , leaving  $H_T$ . Its root state at  $u$  is one, while (71) gives the block state zero. Hence  $\alpha(F) = \alpha(H_T) + \alpha(T)$ , but omitting  $u$  allows at most  $\alpha(H_T) - 1 + \alpha(T)$  vertices.

All ordinary pendant legs at  $u$  are consequently even. If an odd leg were present, selecting  $u$  would either omit its leaf immediately (length one), or leave an even-order path with an equally large choice omitting its terminal leaf. This contradicts Lemma 6.2. Since the leg lengths are at most thirteen, the downward rooted branches belong to

$$\mathcal{T} = \{T_{33}, T_{35}, P_2, P_4, P_6, P_8, P_{10}, P_{12}\}, \quad (72)$$

with at least one terminal block. All eight root states are zero.

Delete these downward branches, retaining  $u$  and the upward direction, and call the resulting tree  $H$ ; put  $J = H - u$ . Then  $u$  is a leaf of  $H$ , and  $H$  has at least two vertices. Since all deleted root states are zero,

$$\alpha(F) = \alpha(H) + \sum_i \alpha(T_i).$$

If a maximum independent set of  $H$  omitted  $u$ , it could be combined with maximum sets in all branches, contradicting the preceding forced selection of  $u$ . Thus

$$\alpha(H) - \alpha(J) = 1. \quad (73)$$

If  $v$  is the neighbour of  $u$  in  $H$ , then

$$r = \frac{Z_H}{Z_J} = 1 + 2 \frac{Z_{J-v}}{Z_J} \in [5/3, 3], \quad (74)$$

using (49) for  $J$  and  $v$ .

Let  $m = \deg_F(u) - 1 \geq 2$  be the number of deleted branches. At least one has seven or nine vertices and each other branch has at least two, so their total order  $N$  is at least nine. Attaching a two-vertex path at  $u$  in  $H$  produces a strictly smaller tree  $H_2$ . Minimality therefore gives

$$X = rG(H) \geq 0, \quad Y = G(J) \geq 0, \quad G(H_2) \geq 0. \quad (75)$$

## 6.7 Closing all branch counts

For each branch  $T_i$ , write

$$Z_i = Z_{T_i}, \quad A_i = Z_{T_i - \text{root}}, \quad M_i = [\lambda Z'_i]_2, \quad M_{A_i} = [\lambda A'_i]_2, \quad n_i = n(T_i), \quad \alpha_i = \alpha(T_i).$$

Define

$$A = \prod_i A_i, \quad P = \prod_i Z_i, \quad B = P - A, \\ M_A = [\lambda A']_2, \quad M_B = [\lambda B']_2, \quad c_{60} = 60 \sum_i \alpha_i + 29N.$$

Conditioning at  $u$  gives  $Z_F = AZ_H + BZ_J$ . Since the branch states are zero and the host state is one, differentiation yields

$$E := 60(Ar + B)G(F) = 60AX + 60BY + (180M_A - c_{60}A)r \\ + 180M_B - (c_{60} + 89)B. \quad (76)$$

For the two-vertex path in (75), the same identity specializes to

$$E_2 := 60(3r + 2)G(H_2) = 180X + 120Y + 6r - 54 \geq 0. \quad (77)$$

Put

$$R_i = Z_i/A_i, \quad a_i = G(T_i), \quad b_i = \frac{3M_{A_i}}{A_i} - \alpha_i - \frac{29n_i}{60}, \quad R = \prod_i R_i, \quad h = \frac{89}{60}.$$

Because the root state is zero,  $b_i = G(T_i - \text{root}) - 29/60$ . The requisite exact values are

| $T_i$    | $R_i$     | $a_i$       | $b_i$        |
|----------|-----------|-------------|--------------|
| $T_{33}$ | 171/121   | 343/1140    | 167/660      |
| $T_{35}$ | 683/473   | 4399/13660  | 1789/9460    |
| $P_2$    | 5/3       | 13/30       | 1/30         |
| $P_4$    | 21/11     | 67/105      | -19/165      |
| $P_6$    | 85/43     | 25/34       | -77/430      |
| $P_8$    | 341/171   | 4042/5115   | -52/285      |
| $P_{10}$ | 1365/683  | 2267/2730   | -661/4098    |
| $P_{12}$ | 5461/2731 | 23641/27305 | -1799/13655. |

For the paths these entries follow from (53). For  $T_{ab}$  they follow from  $Z_{T_{ab}} = P_a P_b + \lambda P_{a-1} P_{b-1}$  and  $A_{T_{ab}} = P_a P_b$ , with  $(a, b) = (3, 3), (3, 5)$ , and the product rule. In particular,

$$R_i \geq 7/5, \quad a_i \geq 343/1140, \quad b_i \geq -11/60. \quad (78)$$

Applying the product rule once more to (76) gives

$$\frac{E - 60AX - 60BY}{60A} = R \left( \sum_i a_i - h \right) + (r - 1) \sum_i b_i + h. \quad (79)$$

For  $m \geq 6$ , the coefficient  $343m/1140 - h$  is positive. Equations (74), (78) and (79) bound the right side below by

$$f_m = \left( \frac{7}{5} \right)^m \left( \frac{343m}{1140} - \frac{89}{60} \right) - \frac{11m}{30} + \frac{89}{60}.$$

We have  $f_6 = 5068593/2968750 > 0$ , and

$$f_{m+1} - f_m = \left( \frac{7}{5} \right)^m \left[ \frac{2}{5} \left( \frac{343m}{1140} - \frac{89}{60} \right) + \frac{7}{5} \frac{343}{1140} \right] - \frac{11}{30}.$$

The bracket is  $11/20$  at  $m = 6$  and increases with  $m$ ; hence every increment is at least  $11/20 - 11/30 = 11/60 > 0$ . Therefore  $E > 0$  for all  $m \geq 6$ .

For  $2 \leq m \leq 5$ , the multisets of the eight branch types containing at least one of  $T_{33}, T_{35}$  number

$$\binom{m+7}{7} - \binom{m+5}{5} = 15, 64, 204, 540,$$

respectively, for a total of 823. All use the same exact elimination:

$$E - \frac{A}{3}E_2 = A(60R - 100)Y + AJ(r), \quad (80)$$

where

$$J(r) = 60R \left( \sum_i a_i - \frac{89}{60} \right) + 60(r - 1) \sum_i b_i + 107 - 2r. \quad (81)$$

Since  $m \geq 2$ ,  $R \geq (7/5)^2 > 5/3$ , so the coefficient of  $Y$  is positive. The remaining function is affine in  $r$ . The finite certificate verifies the four coefficients of (80) and the two endpoint inequalities for each of these 823 indices, with

$$\min J(r) = \frac{3523}{473} > 0 \quad \text{at the multiset } (T_{35}, P_2), \quad r = 5/3. \quad (82)$$

Thus (75) and (77) imply  $E > 0$  throughout every case. For example,

$$\begin{aligned} E_{T_{33}, P_2} &= 21780X + 29520Y + 6237r - 12360 \\ &= 121E_2 + 15000Y + 5511(r - 5/3) + 3359, \\ E_{T_{35}, P_2} &= 85140X + 119760Y + 18939r - 41808 \\ &= 473E_2 + 63000Y + 16101(r - 5/3) + 10569. \end{aligned}$$

The complete data are `adjusted_mean_global_closure.json`; `adjusted_mean_global_closure.py` reconstructs the eight-type data and all 823 identities. An independent reconstruction is supplied by `audit_terminal_fan_second_stage.py`. The finite statements are rational consequences of the displayed table and (80)–(81); no numerical optimization enters acceptance.

Every minimum negative tree was reduced in Section 6.6 to precisely this configuration. The cases  $2 \leq m \leq 5$  and  $m \geq 6$  exhaust all possibilities, and both give  $E > 0$ , hence  $G(F) > 0$ , a contradiction. This proves (48) for every forest and completes the proof of Theorem 6.1.

## 6.8 Transfer to the activity window and proof data

For completeness, the activity transfer requires only a bipartite variance identity. At activity  $\lambda > 0$ , let  $K$  be the size of the random independent set and  $q = \lambda/(1 + \lambda)$ . Conditioning on either colour class makes the unblocked vertices of the other class independent Bernoulli variables with parameter  $q$ . If their expected occupation counts are  $\mu_A, \mu_B$ , the total variance formula gives

$$\text{Var } K \geq (1 - q)\mu_A, \quad \text{Var } K \geq (1 - q)\mu_B, \quad \text{hence} \quad \text{Var } K \geq \frac{\mu_A + \mu_B}{2(1 + \lambda)}.$$

The two conditional variances are not added. Since  $d\mu/d\log \lambda = \text{Var } K$ , the function  $\mu(\lambda)/\sqrt{\lambda/(1 + \lambda)}$  is nondecreasing. Thus

$$\mu_F(9/4) \geq \sqrt{27/26} \mu_F(2).$$

For a nonempty forest,  $\alpha(F) \geq n/2$ . Applying (48), the strict inequality  $59\sqrt{27/26} > 60$  yields

$$\mu_F(9/4) \geq \sqrt{27/26} \left( \frac{\alpha(F)}{3} + \frac{29n}{180} \right) > \frac{n + 2\alpha(F)}{6}.$$

Indeed, the difference is increasing in  $\alpha(F)$  and its value at  $\alpha(F) = n/2$  is  $n(59\sqrt{27/26} - 60)/180 > 0$ ; squaring positive quantities reduces this to  $27 \cdot 59^2 > 26 \cdot 60^2$ .

The proof data referenced in this section are located in the `inputs/inherited_mean/research_code/` directory of the stage-900 source release preserved in the ancillary material. The index in Section 11 identifies the original mathematical source documents and their exact archive hierarchy. In particular, the active finite obligations are: the complete short-leg endpoint and spider-base domains; the long-bridge bases and 342 nonnegative affine identities; and the 823 final two-level identities. Auxiliary graph-instance diagnostics in the historical files are not premises of this proof.

## 7 Rooted sources and conditional-binomial estimates

This section gives the analytic estimates used for large values of the mean number of free vertices. All forests below are finite, and their degrees and numbers of components are unrestricted. The letter  $n$  denotes the number of vertices;  $M$  denotes a random number of free vertices; and  $m = \mathbb{E}M$  is a real number. In particular, an estimate for all  $m \geq m_*$  is not a check at a single integer order. The fixed scalar certificates referred to below are part of the ancillary material. Their purpose is to establish the stated scalar inequalities on their entire domains. The structural reductions and the analytic estimates that give those inequalities their meaning are provided here.

### 7.1 Messages, marginals and arbitrary nonnegative sources

The hard-core partition function and its exact tree recursion are standard; Scott and Sokal [26, Section 3.4] give the multivariate recursion whose hard-core specialization yields the message and partition-function identities below. Root each component and give vertex  $v$  activity  $\lambda_v > 0$ . Let  $R_v$  be its occupied-to-unoccupied partition-function ratio in its descendant subtree, and put  $p_v = R_v/(1 + R_v)$ . Then

$$R_v = \lambda_v \prod_{w \in C(v)} (1 + R_w)^{-1}, \quad Z = \prod_v (1 + R_v), \quad \pi_v = p_v(1 - \pi_{\text{par}(v)}). \quad (83)$$

Here  $\pi_v$  is the marginal in the original forest and the parent marginal of a root is zero. Write  $r_v = \pi_v/p_v$ . For  $0 < \lambda_v \leq b$  we have

$$0 < p_v \leq q_b := \frac{b}{1+b}, \quad \frac{1}{1+b(1-p_v)} \leq r_v \leq 1. \quad (84)$$

Indeed, after deleting the edge to its parent, the parent-side cavity odds are at most  $b$ ; the marginal of that parent in the original forest is at most  $b/(1+b+R_v)$ .

Let  $a_v \geq 0$  and replace the activities by  $\lambda_v e^{ta_v}$ . The logarithmic message derivatives at  $t = 0$  satisfy

$$d_v = a_v - \sum_{w \in C(v)} p_w d_w. \quad (85)$$

Responses may have either sign. If  $U_{vw} = p_w$  when  $w$  is a child of  $v$  and is zero otherwise, then

$$(I + U)d = a, \quad \pi^\top(I + U) = p^\top.$$

For  $f_v = (1 - p_v)d_v^2$ , differentiating again gives  $(I + U)d' = -Uf$ . Twice differentiating  $\log Z$  consequently proves

$$\text{Var}\left(\sum_v a_v \mathbf{1}_{v \in S}\right) = \sum_v \pi_v (1 - p_v) d_v^2, \quad \sum_v p_v d_v = \sum_v \pi_v a_v. \quad (86)$$

For completeness, the second derivative of  $\log Z$  is  $p^\top f + p^\top d' = p^\top f - \pi^\top U f = \pi^\top f$ . Thus the identity is valid for heterogeneous activities and sources, and does not require a common activity during differentiation. Zero activities follow by continuity, or by deleting vertices whose activities are zero before applying these formulas.

Here are three summable constraints for the responses. Define

$$T_b(p) = \log \frac{b(1-p)}{p}, \quad \Lambda_b(p) = \frac{1}{p T_b(p)}, \quad h(p) = 1 - \frac{p}{2},$$

and, for  $t \geq 0$ , set

$$H_b(t) = j b + e^{t-j \log(1+b)} - 1, \quad j = \left\lfloor \frac{t}{\log(1+b)} \right\rfloor. \quad (87)$$

The child messages obey

$$\sum_{w \in C(v)} \frac{p_w}{h(p_w)} \leq T_b(p_v), \quad \sum_{w \in C(v)} R_w \leq H_b(T_b(p_v)). \quad (88)$$

The first inequality follows by summing  $2R/(2+R) \leq \log(1+R)$ : the derivative of the difference is  $R^2/((1+R)(2+R)^2) \geq 0$ . For the second, put  $t_w = \log(1+R_w)$ . The convex function  $e^t - 1$  on  $[0, \log(1+b)]$  has its maximum sum, subject to a bound on  $\sum t_w$ , when all but at most one of the nonzero arguments are endpoints. This proves (87) for any number of children.

Weighted Cauchy–Schwarz, (85), and disjointness of different parents' child sets give

$$\sum_v p_v \{ \Lambda_b(p_v) (a_v - d_v)_+^2 - h(p_v) (d_v)_+^2 \} \leq 0, \quad (89)$$

$$\sum_v p_v \{ \Lambda_b(p_v) (d_v - a_v)_+^2 - h(p_v) (d_v)_-^2 \} \leq 0, \quad (90)$$

$$\sum_v p_v \left\{ \frac{r_v (a_v - d_v)^2}{H_b(T_b(p_v))} - (1 - r_v)(1 - p_v) d_v^2 \right\} \leq 0. \quad (91)$$

To verify the last inequality locally, use  $(\sum_w p_w d_w)^2 \leq (\sum_w R_w) \sum_w p_w (1 - p_w) d_w^2$ , multiply by  $\pi_v$ , and use  $1 - r_w = \pi_v$  for every child  $w$  of  $v$ . Root terms in the negative parts only strengthen the inequalities. Whenever  $T_b(p_v) = 0$ , there are no children and  $d_v = a_v$ ; the corresponding zero-numerator capacity term means zero.

Combining (86), (91) and (84) yields the useful upper bound

$$V_a \leq \sum_v p_v \{(1 - p_v) d_v^2 - \eta_b(p_v)(a_v - d_v)^2\}, \quad \eta_b(p) = \frac{1}{[1 + b(1 - p)] H_b(T_b(p))}. \quad (92)$$

Here and below  $V_a$  denotes the variance in (86).

**Lemma 7.1.** *Suppose  $A, u_+, u_- \geq 0$  and, for every  $0 < p < q_b$  and  $z \in \mathbb{R}$ ,*

$$0 \leq \frac{A}{p} + \eta_b(p)(1 - z)^2 - (1 - p)z^2 + u_+ \{\Lambda_b(p)(1 - z)_+^2 - h(p)z_+^2\} + u_- \{\Lambda_b(p)(z - 1)_+^2 - h(p)z_-^2\}. \quad (93)$$

*Assume also the corresponding leaf inequality at  $p = q_b, z = 1$ . Then  $V_a \leq A \sum_v a_v^2$  for every nonnegative source and every forest with activities at most  $b$ .*

*Proof.* For  $a_v > 0$ , substitute  $z = d_v/a_v$ , multiply by  $p_v a_v^2$  and sum. The capacity terms have nonpositive sums and (92) proves the claim. For  $a_v = 0$ , use the homogeneous limit of the same quadratic inequality, or replace  $a$  by  $a + \varepsilon \mathbf{1}$  and let  $\varepsilon \downarrow 0$ .  $\square$

One explicit finite acceptance rule for (93) is as follows. On a cell  $p \in [l, h_0]$ , choose lower bounds  $\eta_b(p) \geq E$  and  $\Lambda_b(p) \geq L$ . Put

$$A_0 = E + u_+ L, \quad B_+ = 1 - l + u_+(1 - l/2), \quad B_- = 1 - l + u_-(1 - l/2), \\ c_0 = A/h_0 - B_+, \quad c_2 = E + u_- L - B_+.$$

The conditions

$$A_0 \geq B_-, \quad c_0 > 0, \quad c_2 > 0, \quad c_0 c_2 > B_+^2 \quad (94)$$

suffice on the three regions  $z \leq 0$ ,  $0 \leq z \leq 1$ , and  $z \geq 1$ . For example, on the last region set  $z = 1 + x$ ; the lower quadratic is  $c_0 - 2B_+x + c_2x^2$ . On the middle region the lower expression is  $A/h_0 + A_0(1 - z)^2 - B_+z^2$ , which is at least  $A/h_0 - B_+ > 0$  on that interval. On the negative half-axis all coefficients after putting  $z = -x$  are nonnegative. The function  $\eta_b$  is increasing. Moreover the maximum of  $pT_b(p)$  is the unique  $K_* > 0$  satisfying  $K_* e^{1+K_*} = b$ ; a rational  $K$  with  $Ke^{1+K} > b$  therefore gives  $\Lambda_b \geq 1/K$ . Endpoint refinement uses  $(pT_b(p))' = T_b(p) - 1/(1 - p)$ . At a rational endpoint, the integer in (87) can be determined by comparing rational powers of  $1 + b$  with  $b(1 - p)/p$ . The  $p = 0$  endpoint is treated by the limiting coefficient bounds; it is not asserted to be an actual positive-activity message.

## 7.2 Selecting the smoothing set and retaining true marginals

In this subsection activities have a common value  $\lambda \in [a, b]$ . Write  $q_b = b/(1 + b)$  and

$$f_b(x) = \max \left\{ \frac{(1 + q_b)x}{2q_b}, \frac{(1 + q_b)x}{2q_b^2} + \frac{q_b - 1}{2q_b} \right\}.$$

For every edge  $uv$ , its two cavity odds are at most  $b$ , whence  $\pi_u + q_b \pi_v \leq q_b$  and  $\pi_v + q_b \pi_u \leq q_b$ . Also  $0 \leq f_b(\pi_v) \leq 1$ , since  $\pi_v \leq q_b$ , and consequently  $f_b(\pi_u) + f_b(\pi_v) \leq 1$ . To check this, split at the common breakpoint  $q_b/(1 + q_b)$ . If both arguments use the first branch, add the two edge inequalities; if just one uses the second branch, use the corresponding edge inequality. Both arguments can use the second branch only at their common breakpoint.

Every family  $0 \leq x_v \leq 1$  with  $x_u + x_v \leq 1$  on the edges of a forest is realizable as the marginals of a random independent set: select a root with probability  $x_v$ , and, conditional on its parent being absent, select a child  $v$  with probability  $x_v/(1 - x_{\text{par}(v)})$ . If the denominator is zero the child has marginal zero. Independent auxiliary randomness on different branches completes the construction. It follows that a maximum  $\pi$ -weight independent set  $B$  satisfies

$$W := \sum_{v \in B} \pi_v \geq \frac{\mu}{2}, \quad W \geq J_b := \sum_v \pi_v f_b(\pi_v). \quad (95)$$

Set

$$\phi_{a,b}(p) = \frac{q_b}{\log(1+b)} \left[ \log \frac{a(1-p)}{p} \right]_+.$$

The ratio  $R/[(1+R)\log(1+R)]$  decreases for  $R > 0$ . The message recursion therefore implies  $\sum_{w \in C(v)} p_w \geq \phi_{a,b}(p_v)$ . The parent-marginal identity gives

$$\sum_v p_v \{r_v(1 + \phi_{a,b}(p_v)) - 1\} \leq 0. \quad (96)$$

Indeed  $\sum_v p_v(1 - r_v) = \sum_v \pi_v \sum_{w \in C(v)} p_w$ . For the constant source  $a_v = 1$ , combining this with  $\sum_v p_v(d_v - r_v) = 0$  also gives  $\sum_v p_v(d_v - 1 + r_v \phi_{a,b}(p_v)) \leq 0$ .

The selection certificates prove, for all  $0 < p < q_b$ ,  $[1 + b(1 - p)]^{-1} \leq r \leq 1$  and real  $z$ ,

$$\begin{aligned} 0 \leq & C_J r f_b(pr) + C_\mu r - r(1 - p)z^2 \\ & + u_+ \{ \Lambda_b(p)(1 - z)_+^2 - h(p)z_+^2 \} + u_- \{ \Lambda_b(p)(z - 1)_+^2 - h(p)z_-^2 \} \\ & + w \{ z - 1 + r \phi_{a,b}(p) \} + t(z - r) \\ & + s \left\{ \frac{r(1 - z)^2}{H_b(T_b(p))} - (1 - r)(1 - p)z^2 \right\}. \end{aligned} \quad (97)$$

All multipliers except  $t$  are nonnegative. After substituting  $z = d_v$  and summing against  $p_v$ , the preceding identities give

$$V \leq C_J J_b + C_\mu \mu \leq C_W W, \quad C_W = C_J + 2C_\mu. \quad (98)$$

The definition of  $f_b$  stays attached to the certificate's original upper endpoint  $b$ , even if its activity interval is subsequently restricted.

The fixed selection family consists of 28 parameter sets. Their continuous-domain acceptance splits at the branch of  $f_b(pr)$  and at  $z = 0, 1$ . After replacing transcendental coefficients in safe directions on each  $p$  cell, the remaining expressions are quadratics in the response and polynomials of degree at most three in  $r$ . For a half-axis quadratic  $A + Bx + Cx^2$ ,  $x \geq 0$ , the rules  $A \geq 0$ ,  $C \geq 0$ , and either  $B \geq 0$  or  $4AC - B^2 \geq 0$  are sufficient. For a finite interval, endpoint and vertex tests, or nonnegative Bernstein coefficients, give full-interval inequalities. The cell coverage includes the limiting  $p \downarrow 0$  region and the separate leaf case  $p = q_b, z = 1$ . These are algebraic assertions about continuous  $p, r, z$ , not tests of a collection of forest messages.

### 7.3 Affine sources and independent-support sources

A second source family preserves a true marginal term. Suppose that  $A, C, u_+, u_-, s \geq 0$  and the following holds throughout the domain in (84), for every real  $z$ :

$$\begin{aligned} 0 \leq & \frac{A}{p} + Cr - r(1 - p)z^2 \\ & + u_+ \{ \Lambda_b(p)(1 - z)_+^2 - h(p)z_+^2 \} + u_- \{ \Lambda_b(p)(z - 1)_+^2 - h(p)z_-^2 \} \\ & + s \left\{ \frac{r(1 - z)^2}{H_b(T_b(p))} - (1 - r)(1 - p)z^2 \right\}. \end{aligned} \quad (99)$$

The homogeneous substitution used in Lemma 7.1 now proves

$$V_a \leq A \sum_v a_v^2 + C \sum_v \pi_v a_v^2 \quad (0 \leq \lambda_v \leq b). \quad (100)$$

No positive lower activity bound is imposed in this assertion.

For an independent set  $B$  there is an additional, different source bound. Let  $a_v = \mathbf{1}_{v \in B}$  and let  $b_v$  be the parent's source mark, with mark zero at a root. Exactly three pairs are possible:

$$(a_v, b_v) \in \{(0, 0), (0, 1), (1, 0)\}. \quad (101)$$

Assume here that  $a \leq \lambda_v \leq b$  with  $a > 0$ . Define  $G_a(p) = \log p - 2 \log(1 - p) - \log a$ . Summing the logarithmic message identity gives

$$\sum_v G_a(p_v) = \sum_v \log \lambda_v - \sum_{v \text{ root}} \log(1 - p_v) - n \log a \geq 0. \quad (102)$$

If the positive and negative capacity constraints are summed separately over parents of source mark 0 and 1, they become

$$\sum_v p_v \{u_{a_v} \Lambda_b(p_v)(a_v - d_v)_+^2 - u_{b_v} h(p_v)(d_v)_+^2\} \leq 0,$$

and the analogous inequality with  $u$ ,  $(a_v - d_v)_+$  and  $(d_v)_+$  replaced by  $v$ ,  $(d_v - a_v)_+$  and  $(d_v)_-$ , respectively. This follows because each nonroot vertex is a child of exactly one parent; including roots in the negative sum is safe.

The marked-source scalar family is

$$\begin{aligned} 0 \leq & r(A + Ca_0) - r(1 - p)z^2 - \ell G_a(p)/p \\ & + u_{a_0} \Lambda_b(p)(a_0 - z)_+^2 - u_{b_0} h(p)z_+^2 \\ & + v_{a_0} \Lambda_b(p)(z - a_0)_+^2 - v_{b_0} h(p)z_-^2 \\ & + w\{r(1 + \phi_{a,b}(p)) - 1\} + t(z - ra_0) \\ & + s \left\{ \frac{r(a_0 - z)^2}{H_b(T_b(p))} - (1 - r)(1 - p)z^2 \right\}, \end{aligned} \quad (103)$$

for all pairs in (101), all admissible  $p, r$ , and all real  $z$ . Every multiplier except  $t$  is nonnegative. Summation, using (102), (96) and (86), proves

$$\text{Var } K_B \leq A\mu + C\mu_B, \quad \mu_B = \mathbb{E}K_B. \quad (104)$$

This conclusion concerns an independent indicator source, rather than arbitrary nonnegative weights. It is used only on its stated activity interval; the Laplace-transform bounds below use (100), whose activity domain contains the entire downward tilt. The ten fixed marked-source parameter sets have no additional term proportional to  $n$ .

## 7.4 A single probability space for the moments and tails

Return to a common activity  $\lambda \in [a, b]$  and fix the maximizing set  $B$  from (95). Conditional on the full configuration on its complement, let  $Y$  be the occupied count there and  $M$  the number of unblocked vertices of  $B$ . Then

$$K = Y + \text{Bin}(M, q), \quad q = \frac{\lambda}{1 + \lambda}, \quad v = q(1 - q), \quad m = \mathbb{E}M = \frac{W}{q}. \quad (105)$$

The binomial here describes a conditional distribution. Neither  $M$  nor  $Y$  is presumed independent of the other. At  $\mu = k$ , put  $\delta = k - Y - qM$ . The laws of total expectation and variance give

$$\mathbb{E}\delta = 0, \quad \mathbb{E}\delta^2 = V - vm, \quad \text{Var } M = \frac{\text{Var } K_B - (1 - q)W}{q^2}. \quad (106)$$

If  $\mu \geq r_*n$ , then

$$m \geq \frac{r_*n}{2q_b}, \quad \frac{|B|}{m} \leq \frac{2q}{r_*} - 1. \quad (107)$$

For the second inequality, use  $|B| \leq n - (\mu - W)/q$ ,  $n \leq \mu/r_*$ ,  $\mu/W \leq 2$  and  $r_* \leq q$ . Thus (98) yields

$$\frac{\mathbb{E}\delta^2}{vm} \leq R := \frac{C_W}{1 - q_b} - 1. \quad (108)$$

The density bounds needed here have the following explicit form. Let  $r_0 = 276393/10^6 < (5 - \sqrt{5})/10$ . The activity-one mean theorem [2] gives  $\mu(1) \geq r_0n$ . Conditioning on one of the two color classes of a forest gives  $V \geq \mu/[2(1 + \lambda)]$ . Integration therefore gives, for  $a \geq 1$ ,

$$r_* = r_0 \sqrt{\frac{2a}{1 + a}}.$$

For  $a < 1$ , a uniform source bound  $V \leq A_0n$  valid between  $a$  and 1 gives  $\mu(a)/n \geq r_0 - A_0 \log(1/a) \geq r_0 - A_0(1 - a)/a$ . In a necessary rank of the monotone window one also has  $\mu = k \geq n/4$ , so the usable bound is

$$r_* = \max\{1/4, r_0 - A_0(1 - a)/a\}. \quad (109)$$

The  $1/4$  alternative is used at these ranks; it is not a mean-density claim at every activity below one. The fixed data use  $A_0 = \max(A_b, 43/100)$ , where  $A_b$  is the applicable uniform source constant, so that the full interval up to activity one is covered.

Three bounds for  $D \geq \text{Var } M/m$  are available. From the uniform source bound and (107), put  $C_s = A_b(2q_b/r_* - 1)$ . Then

$$\frac{\text{Var } M}{m} \leq \frac{C_s}{q_a^2} - \frac{1 - q_a}{q_a} \quad \text{if } q_b < 2C_s. \quad (110)$$

The notation  $D$  will always denote a chosen upper bound for the actual  $\text{Var } M/m$ . The condition makes  $1 + C_s/q^2 - 1/q$  decreasing over the interval. From (100) one obtains

$$\frac{\text{Var } M}{m} \leq 1 + \frac{2A/r_* + C - 1}{q} - \frac{A}{q^2}. \quad (111)$$

Its maximum occurs at an endpoint or, when present, at  $q = 2A/(2A/r_* + C - 1)$ . From (104) one obtains

$$\frac{\text{Var } M}{m} \leq 1 + \frac{2A + C - 1}{q}, \quad (112)$$

whose maximum is at an endpoint. Taking the least of applicable upper bounds preserves validity.

To control the lower tail, keep this same set  $B$  fixed and replace its activities by  $\lambda e^{-u}$ , leaving the complement activities unchanged. Write  $w(u) = \mathbb{E}_u K_B$  and  $\psi(u) = \log \mathbb{E} e^{-uK_B}$ . Then  $w' = -\text{Var}_u K_B$ ,  $\psi' = -w$ , and  $w(0) = qm$ . All tilted activities lie in  $[0, b]$ , so (100) applies for every  $u \geq 0$ . For  $C > 0$ , set  $F_C(u) = (1 - e^{-Cu})/C$ . Solving the differential inequality  $w' \geq -A|B| - Cw$  and integrating once more gives

$$\log \mathbb{E} e^{-uK_B} \leq -qmF_C(u) + \frac{A|B|}{C} \{u - F_C(u)\}. \quad (113)$$

The continuous limit at  $C = 0$  gives  $-qmu + A|B|u^2/2$ . The exact conditional-binomial identity is

$$\mathbb{E}e^{-uK_B} = \mathbb{E}e^{-s_q(u)M}, \quad s_q(u) = -\log(1 - q + qe^{-u}).$$

Markov's inequality and (107) prove

$$\mathbb{P}(M < \alpha m) \leq e^{-m\rho_\alpha}, \quad 0 < \alpha < 1, \quad (114)$$

where any nonnegative lower bound for

$$\min_{q \in \{q_a, q_b\}} \left[ qF_C(u) - \alpha s_q(u) - \frac{A}{C} \left( \frac{2q}{r_*} - 1 \right) (u - F_C(u)) \right] \quad (115)$$

is valid for each fixed  $u \geq 0$ . The minimum is at the endpoints because  $s_q(u)$  is convex as a function of  $q$ , so the bracket is concave. The fixed parameters use finitely many positive rational  $u$ ; the maximum of their valid rates is again a valid rate.

For comparison, the uniform source bound gives  $\psi(u) \leq -qmu + C_s mu^2/2$ . Since  $s_q(u) \leq qu$ , it gives  $\rho_\alpha \geq q_a^2(1 - \alpha)^2/(2C_s)$ . For  $0 \leq u \leq 1/2$ , the tilted single-site parameter is at least  $20a/(33 + 20a)$ , since  $e^{1/2} < 33/20$ . Let  $v_0$  be the minimum of  $t(1 - t)$  over the interval with this left endpoint and right endpoint  $q_b$ . The identity  $s_q''(u) = -q_u(1 - q_u)$  gives  $s_q(u) \leq qu - v_0 u^2/2$ . Hence, with  $A_1 = q_a(1 - \alpha)$ ,  $B_1 = C_s - \alpha v_0 > 0$  and  $u_1 = \min\{1/2, A_1/B_1\}$ , the additional valid rate is  $A_1 u_1 - B_1 u_1^2/2$ . All these estimates concern the original joint law in (105); a new maximizing set is not selected along the tilt.

## 7.5 Finite Bernoulli Fourier estimates

Let  $X$  be Bernoulli with parameter  $q$ , let  $\chi_q(t) = \mathbb{E}e^{it(X-q)}$ , and put  $v = q(1 - q)$ ,  $\eta = |1 - 2q|$  and  $g(t) = e^{-vt^2/2}$ . The elementary identity

$$|\chi_q(t)|^2 = 1 - 4v \sin^2(t/2) \implies |\chi_q(t)| \leq e^{-2vt^2/\pi^2} \quad (|t| \leq \pi) \quad (116)$$

will also handle small conditional variances.

Suppose  $0 < T < \pi$ ,  $v \geq 1/6$ , and

$$v(1 - T^2/12) \geq 1/6, \quad 1 - 5v - 10v^2 + (15/8)v^3 T^2 \leq 0. \quad (117)$$

Then, for  $|t| \leq T$ ,

$$|\chi_q(t)| \leq g(t), \quad 0 \leq g(t) - \Re \chi_q(t) \leq \frac{vt^4}{48}, \quad |\Im \chi_q(t)| \leq \frac{v\eta|t|^3}{6}. \quad (118)$$

For the modulus, differentiate its logarithm and use  $\sin t \geq t(1 - 2v + 2v \cos t)$ . This last inequality follows from  $\sin t/t \geq 1 - t^2/6$  and  $1 - \cos t \geq t^2/2 - t^4/24$ , together with the first condition in (117). The real part is at most  $g$  by the modulus bound. Lower Taylor bounds for the two cosines in  $\Re \chi_q$ , and the fourth-degree upper Taylor bound for  $e^{-vt^2/2}$ , give

$$g - \Re \chi_q \leq \frac{v(6v - 1)t^4}{24} + \frac{v(1 - 5v - 10v^2)t^6}{720} + \frac{v^4 t^8}{384} \leq \frac{vt^4}{48}.$$

Finally  $\Im \chi_q(t) = vt\{\text{sinc}((1 - q)t) - \text{sinc}(qt)\}$  and  $|\text{sinc}'(x)| \leq x/3$  on  $[0, \pi]$  prove the imaginary-part bound by integration.

**Lemma 7.2.** *If  $v \geq 9/40$ , all three inequalities in (118) hold for  $|t| \leq \pi$ .*

*Proof.* Put  $H(t) = (t - \sin t)/(2t(1 - \cos t))$  for  $0 < t < \pi$ . It increases from  $1/6$  to  $1/4$ . Indeed, the sign of its derivative is the sign of  $(t + \sin t)(1 - \cos t) - t^2 \sin t$ ; after  $x = t/2$ , this reduces to  $x \tan x + \sin^2 x \geq 2x^2$ . The inequalities  $\tan x \geq x + x^3/3$  and  $\sin^2 x \geq x^2 - x^4/3$  prove it. For  $F(t) = \log |\chi_q(t)| + vt^2/2$ , the derivative has sign  $H(t) - v$ . Thus its maximum is at an endpoint. Now  $F(0) = 0$ , and  $2F(\pi) = \log(1 - 4v) + \pi^2 v$  is decreasing for  $9/40 \leq v < 1/4$ . At  $9/40$  it is negative, using  $\pi^2 < 10$  and  $e^{9/4} < 10$ . The endpoint  $v = 1/4$  follows by continuity. This proves the modulus bound. In the displayed Taylor bound above, the coefficient of  $t^6$ , including its  $t^8$  correction, is nonpositive because

$$1 - 5(9/40) - 10(9/40)^2 + (15/8)(1/4)^3 10 = -433/1280 < 0.$$

The real and imaginary bounds follow as before.  $\square$

Let  $b_d(j) = \binom{d}{j} q^j (1-q)^{d-j}$ , extended by zero outside  $0 \leq j \leq d$ , and put  $S = dv$ ,  $D = j - dq$ ,  $\rho_d = d/(d-1)$ . For  $d > 1$ , the finite power-difference identity gives

$$|\chi_q(t)^d - g(t)^d| \leq dv \left( \frac{\eta|t|^3}{6} + \frac{t^4}{48} \right) e^{-(d-1)vt^2/2} \quad (119)$$

on the available interval. Fourier inversion, extended Gaussian integrals, and the two tails consequently give

$$\left| b_d(j) - \frac{e^{-D^2/(2S)}}{\sqrt{2\pi S}} \right| \leq E_{d,q}(T), \quad (120)$$

where

$$\begin{aligned} E_{d,q}(T) &= \frac{\eta\rho_d^2}{3\pi vd} + \frac{\rho_d^{5/2}}{16\sqrt{2\pi}v^{3/2}d^{3/2}} \\ &\quad + (1 - T/\pi)e^{-ST^2/2} + \frac{e^{-ST^2/2}}{\pi ST}. \end{aligned}$$

For  $T < \pi$ , the true binomial tail uses the monotonicity of  $|\chi_q(t)|$  on  $[0, \pi]$  and its bound at  $T$ . The last term is the continuous Gaussian tail. Under Lemma 7.2 take  $T = \pi$ , when the true-binomial tail vanishes but the continuous tail remains. The bound holds for every integer  $j$ , without an offset truncation.

Define

$$\mathcal{C}(S, D) = \frac{1 - D^2/S}{\sqrt{2\pi}S^{3/2}} e^{-D^2/(2S)}, \quad \Delta^2 b_d(j) = 2b_d(j) - b_d(j-1) - b_d(j+1).$$

Using  $2 - 2\cos t \leq t^2$  and  $0 \leq t^2 - (2 - 2\cos t) \leq t^4/12$  in the same integral proves

$$\begin{aligned} |\Delta^2 b_d(j) - \mathcal{C}(S, D)| &\leq \frac{4\eta\rho_d^3}{3\pi v^2 d^2} + \frac{(5/16)\rho_d^{7/2} + 1/4}{\sqrt{2\pi}S^{5/2}} \\ &\quad + 4(1 - T/\pi)e^{-ST^2/2} + \frac{1}{\pi} \left( \frac{T}{S} + \frac{1}{S^2 T} \right) e^{-ST^2/2}. \end{aligned} \quad (121)$$

These constants are the third through sixth absolute Gaussian moments in (119); the  $1/4$  term is the lattice-kernel error. Independently, for  $d > 0$ ,

$$|\Delta^2 b_d(j)| \leq \frac{\pi^3/8}{\sqrt{2\pi}(dv)^{3/2}}. \quad (122)$$

This follows directly from (116); in the full-period Gaussian range the factor  $\pi^3/8$  can be replaced by 1. For every  $d \geq 0$ , also  $\Delta^2 b_d(j) \geq -1$ , since the two subtracted probabilities have sum at most one.

## 7.6 Binomial first differences, including small variances

Fourier inversion and (116) give, for  $d > 0$  and every  $q$ ,

$$|b_d(j+1) - b_d(j)| \leq \frac{\pi}{4dv}. \quad (123)$$

Sharper bounds follow from an exact adjacent-mass identity. With  $D = d + 1$  and  $\sigma^2 = Dv$ ,

$$b_d(j) - b_d(j-1) = \frac{Dq-j}{Dv} b_D(j). \quad (124)$$

At a maximizing index for  $|Dq-j|b_D(j)$ , comparison with the adjacent index on the same side of  $Dq$  yields

$$|Dq-j| \leq \frac{1 + \sqrt{1+4\sigma^2}}{2} \leq \sigma + \frac{1}{2} + \frac{1}{8\sigma}. \quad (125)$$

The comparison follows by substituting  $b_D(j+1)/b_D(j) = (D-j)q/((j+1)(1-q))$  and solving the resulting quadratic. Endpoints are included in the comparison; the estimates below ensure that a maximizing index lies strictly inside  $[0, D]$ .

Let  $q \in [q_0, 1 - q_0]$ ,  $\eta \leq \eta_0$ , and  $\sigma \geq s$ . Set

$$X_s = 1 + \frac{1}{2s} + \frac{1}{8s^2}, \quad R_s = 1 - \frac{\eta_0 X_s}{s} - \frac{X_s^2}{4s^2}, \quad K_s = 1 + \frac{\eta_0 X_s}{3s}.$$

Require  $R_s > 0$  and  $s+1/2+1/(8s) < 4q_0s^2$ . If  $j/D = q+h$ , then (125) gives  $(q+h)(1-q-h) \geq vR_s$ . For the Bernoulli relative entropy,

$$\begin{aligned} D(q+h||q) &= h^2 \int_0^1 \frac{1-u}{(q+uh)(1-q-uh)} du \\ &\geq \frac{h^2}{2\{v + (1-2q)h/3 - h^2/6\}} \geq \frac{h^2}{2\{v + \eta_0|h|/3\}}. \end{aligned} \quad (126)$$

The first inequality is Jensen's inequality for the reciprocal under the probability density  $2(1-u)$ . The finite Stirling bounds

$$\sqrt{2\pi r}(r/e)^r e^{1/(12r+1)} < r! < \sqrt{2\pi r}(r/e)^r e^{1/(12r)} \quad (r \geq 1)$$

therefore imply

$$|Dq-j|b_D(j) \leq \sqrt{\frac{K_s}{2\pi e R_s} \frac{24s^2}{24s^2-1}}.$$

Indeed, put  $x = |Dq-j|/\sigma$ , maximize  $xe^{-x^2/(2K_s)}$ , and use  $D \geq 4\sigma^2$  and  $e^{1/(6D)} \leq (1 - 1/(24s^2))^{-1}$  for the squared Stirling factor. Combining with (124) gives constants  $c/(Dv)$ , and hence the weaker  $c/(dv)$ , in the following table. Every decimal here is an exact rational upper bound.

| $dv$ at least      | 9    | 25   | 49   | 100  |
|--------------------|------|------|------|------|
| $q \in [.3, .7]$   | .279 | .261 | .255 | .251 |
| $q \in [.45, .55]$ | .255 | .248 | .246 | .245 |

The four columns use  $s = 3, 5, 7, 10$ , respectively.

We also use the bound

$$|b_d(j+1) - b_d(j)| \leq \frac{31/100}{(d+1)v} \quad (d \geq 0, q \in [.3, .7], j \in \mathbb{Z}). \quad (127)$$

For  $d \geq 43$ ,  $dv > 9$ , and the preceding  $.279/(dv)$  bound implies (127), since  $.279 \cdot 37/36 < .31$ . For  $0 \leq d \leq 42$ , reflection reduces  $q$  to  $[1/2, 7/10]$ . There remain precisely the two polynomial inequalities

$$\frac{31}{100} \pm (d+1)q(1-q)\{b_d(j+1) - b_d(j)\} \geq 0, \quad j = -1, \dots, d.$$

Outside this index range both masses vanish. Thus  $2 \sum_{d=0}^{42} (d+2) = 1978$  polynomial inequalities are the complete finite part. Their rational Bernstein certificates are supplied in the `recovered_gradient` data. The binomial polynomials, the reflection and the bound for every  $d \geq 43$  specify the complete quantifiers of this finite reduction.

## 7.7 Negative moments and lower-tail integration

Suppose  $M \geq 0$ ,  $\mathbb{E}M = m > 0$  and  $\text{Var } M \leq Dm$ . For  $p > 0$  and  $0 < \alpha < 1$ , set

$$B_p(\alpha) = \frac{\alpha^{-p} - 1 + p(\alpha - 1)}{(1 - \alpha)^2}, \quad Q_{p,\alpha}(y) = 1 - p(y - 1) + B_p(\alpha)(y - 1)^2.$$

The third derivative of  $y^{-p}$  is negative, so Hermite interpolation at  $\alpha$  and at the double node 1 gives  $Q_{p,\alpha}(y) \geq y^{-p}$  for  $y \geq \alpha$ . Also

$$B_p(\alpha) = \int_0^1 (1-u)p(p+1)[1+u(\alpha-1)]^{-p-2} du \geq p(p+1)/2 > p^2/4.$$

Thus this quadratic is positive on the whole real line. Applying it on the whole probability space proves

$$\mathbb{E}[(M/m)^{-p}; M \geq \alpha m] \leq 1 + B_p(\alpha)D/m =: J_p^{(0)}(m). \quad (128)$$

In particular  $B_1(\alpha) = 1/\alpha$ ,  $B_2(1/3) = 15$  and  $B_{5/2}(1/3) = (81\sqrt{3} - 24)/4$ .

There is a useful refinement when (114) is available. Choose  $\alpha < \beta < 1$ . On  $[\alpha, \beta]$  the residual  $[y^{-p} - Q_{p,\beta}(y)]_+$  decreases with  $y$ : it equals  $[B_p(y) - B_p(\beta)]_+(1-y)^2$ , and both factors decrease. Choose a partition  $\alpha = t_0 < \dots < t_s = \beta$  and decreasing upper bounds  $U_i$  for this residual at  $t_i$ , with  $U_s = 0$ . Then, on  $y \geq \alpha$ , its residual is bounded by  $\sum_{i=0}^{s-1} (U_i - U_{i+1}) \mathbf{1}_{y < t_{i+1}}$ . Consequently

$$J_p(m) := \min \left\{ J_p^{(0)}(m), 1 + \frac{B_p(\beta)D}{m} + \sum_{i=0}^{s-1} (U_i - U_{i+1})e^{-m\rho t_{i+1}} \right\} \quad (129)$$

is a valid upper bound. The implemented refinement is used for  $p = 1, 3/2, 2, 5/2$ ; for  $p = 7/2$  the bound (128) suffices.

The same summation by parts prices small- $M$  layers. If  $0 < \gamma_0 < \dots < \gamma_s$ ,  $c_0 \geq \dots \geq c_{s-1} \geq 0$ , and  $F_i \geq \mathbb{P}(M/m < \gamma_i)$ , then

$$\sum_{i=0}^{s-1} c_i \mathbb{P}(\gamma_i \leq M/m < \gamma_{i+1}) \leq c_{s-1}F_s + \sum_{i=0}^{s-2} (c_i - c_{i+1})F_{i+1}. \quad (130)$$

The omitted term is  $-c_0 \mathbb{P}(M/m < \gamma_0) \leq 0$ . The bottom event is priced separately. No independence assumption about the events is involved.

## 7.8 A signed Gaussian curvature envelope

The reference Gaussian curvature can be negative, so an absolute error bound alone does not supply its main term. The following construction retains the correlation of  $M$  with the conditional mean. Fix  $0 < x_0 < 3/2$  and put  $c = (3 - 2x_0)e^{-x_0}$ ,  $g(x) = (1 - 2x)e^{-x}$  and

$$A(y) = \inf_{x \geq 0} \{y^{-3/2}g(x) + cyx\}, \quad y > 0. \quad (131)$$

If  $cy^{5/2} < 3$ , the unique minimizer lies in  $(0, 3/2)$  and solves

$$(3 - 2x)e^{-x} = cy^{5/2}. \quad (132)$$

Otherwise it is  $x = 0$ , and  $A(y) = y^{-3/2}$ . On the interior branch,

$$\begin{aligned} A(1) &= e^{-x_0}(1 + x_0 - 2x_0^2), \\ A'(1) &= e^{-x_0}(-3/2 + 6x_0 - 2x_0^2), \\ A''(y) &= y^{-7/2}e^{-x} \frac{5(4x^2 - 12x + 15)}{2(2x - 5)} < 0, \\ A'''(y) &= y^{-9/2}e^{-x} \frac{-5(16x^4 - 128x^3 + 360x^2 - 600x + 525)}{2(2x - 5)^3} > 0. \end{aligned}$$

The final numerator polynomial is positive: under  $x = 3/2 - z$  it is  $16z^4 + 32z^3 + 168z + 84$  for  $0 \leq z \leq 3/2$ .

For  $0 < \beta < 1$ , choose an upper bound  $B_\beta$  for

$$\frac{A(1) + A'(1)(\beta - 1) - A(\beta)}{(1 - \beta)^2},$$

and let  $Q_\beta(y) = A(1) + A'(1)(y - 1) - B_\beta(y - 1)^2$ . The positive third derivative gives

$$y^{-3/2}g(x) \geq Q_\beta(y) - cyx \quad (y \geq \beta, x \geq 0). \quad (133)$$

At the switch to  $A(y) = y^{-3/2}$ ,  $A$  and its first derivative match. Beyond it,  $A$  is convex whereas  $Q_\beta$  is concave, so the inequality continues to hold. Increasing  $B_\beta$  preserves this lower bound.

For  $0 < \alpha < \beta$ , the deficit  $[Q_\beta(y) - A(y)]_+$  on  $[\alpha, \beta]$  is decreasing. Indeed the exact coefficient

$$B(y) = - \int_0^1 (1 - u)A''(1 + u(y - 1)) du$$

decreases with  $y$ , and the deficit is  $[B(y) - B_\beta]_+(1 - y)^2$ . Decreasing upper bounds  $U_i$  on a partition as in (129) therefore give the pointwise extension

$$y^{-3/2}g(x) \geq Q_\beta(y) - cyx - \sum_i (U_i - U_{i+1})\mathbf{1}_{y < t_{i+1}} \quad (y \geq \alpha).$$

All bounds for the minimizer in (132) use the strict monotonicity of its left side; directed exponential bounds therefore provide certified bounds on  $A(\beta)$  and each  $U_i$ .

In (105), put

$$y = M/m, \quad x = \frac{\delta^2}{2vM} \quad (M > 0).$$

Then  $\mathbb{E}y = 1$ ,  $\text{Var } y \leq D/m$  and  $\mathbb{E}(yx) \leq R/2$ , with the product interpreted as  $\delta^2/(2vm)$  also on  $M = 0$  when taking its full-space bound. Let  $P_\alpha = \max_{0 \leq y \leq \alpha} [Q_\beta(y)]_+$ . The preceding inequality and (114) prove

$$\mathbb{E}[y^{-3/2}g(x); M \geq \alpha m] \geq G(m), \quad (134)$$

where

$$G(m) = A(1) - \frac{cR}{2} - \frac{B_\beta D}{m} - \sum_i (U_i - U_{i+1})e^{-m\rho_{t_{i+1}}} - P_\alpha e^{-m\rho_\alpha}. \quad (135)$$

Using full-space second moments is safe because the omitted  $-c y x$  term is nonpositive, and the possible positive polynomial on the bad event has been explicitly subtracted.

The older fixed choice  $x_0 = 2/5$ ,  $\alpha = \beta = 1/3$  gives  $c = (11/5)e^{-2/5}$ ,  $A(1) = (27/25)e^{-2/5}$  and  $A'(1) = (29/50)e^{-2/5}$ . It admits  $B_\beta = 4813/1000$  and  $P_\alpha = 0$ . An explicit bound for the required coefficient is

$$\frac{33\sqrt{3}}{4}e^{-4/3} + \frac{39}{25}e^{-2/5} < 4813/1000.$$

Here the minimizer at  $y = 1/3$  is at most  $4/3$ , as follows from  $e^{14/15} > 15\sqrt{3}/11$  in (132). The refined fixed family uses  $x_0 = 9/20$ , for which  $c = (21/10)e^{-9/20}$ ,  $A(1) = (209/200)e^{-9/20}$  and  $A'(1) = (159/200)e^{-9/20}$ , and chooses  $\beta$  and the residual partition separately in each row.

To obtain an actual binomial-curvature bound, average (121) over the good event. If  $\rho = (1 - 1/(\alpha m))^{-1}$ , its normalized cubic and quartic errors are bounded by

$$\frac{4\sqrt{2\pi}}{3\pi} \frac{\eta \rho^3 J_2(m)}{\sqrt{vm}}, \quad \frac{\{(5/16)\rho^{7/2} + 1/4\} J_{5/2}(m)}{vm}, \quad (136)$$

where the normalization is  $\sqrt{2\pi}(vm)^{3/2}$ . For  $M < \gamma_0 m$ , pay the lower bound  $-1$  from (122), namely  $\sqrt{2\pi}(vm)^{3/2}e^{-m\rho_{\gamma_0}}$ . On successive layers use the coefficients  $(\pi^3/8)\gamma_i^{-3/2}$ , or  $\gamma_i^{-3/2}$  in the full-period Gaussian range, and (130). The two frequency tails in (121) are bounded at  $M = \alpha m$  and retained. Thus a positive total lower bound proves  $2p_k > p_{k-1} + p_{k+1}$  for the true probabilities  $p_j = \mathbb{P}(K = j)$ . By AM-GM,  $p_k^2 > p_{k-1}p_{k+1}$ , and the common activity factors cancel to give  $c_k^2 > c_{k-1}c_{k+1}$ . This conclusion does not identify the arithmetic second differences of probabilities and coefficients when  $\lambda \neq 1$ .

## 7.9 The exact lattice variance and its residual

Put  $J(t) = (\sin(t/2)/(t/2))^2$ , with  $J(0) = 1$ . It is the characteristic function of the sum of two independent uniform variables on  $[-1/2, 1/2]$ , whose variance is  $1/6$ . Exactly,

$$2 - 2\cos t = t^2 J(t). \quad (137)$$

For  $|t| \leq \pi$ ,

$$0 \leq e^{-t^2/12} - J(t) \leq t^4/1440. \quad (138)$$

To prove the lower inequality, let  $x = t/2$  and differentiate  $G(x) = (1 - x^2/3)\sin x - x\cos x$ . Since  $G'(x) = x(\sin x - x\cos x)/3 \geq 0$ , integration gives  $\log(\sin x/x) \leq -x^2/6$ . For the upper inequality use

$$J(t) \geq 1 - t^2/12 + t^4/360 - t^6/20160, \\ e^{-t^2/12} \leq 1 - t^2/12 + t^4/288 - t^6/10368 + t^8/497664.$$

Their fourth-degree difference is  $t^4/1440$ , and the remaining terms have nonpositive sum because  $t^2 \leq \pi^2 < 10$  and  $1/20160 - 1/10368 + 10/497664 = -233/8709120 < 0$ .

Under Lemma 7.2, compare the true Fourier kernel  $t^2 J(t)\chi_q(t)^d$  with  $t^2 e^{-(dv+1/6)t^2/2}$ . Decompose their difference as

$$t^2 J(t)\{\chi_q(t)^d - e^{-dvt^2/2}\} + t^2\{J(t) - e^{-t^2/12}\}e^{-dvt^2/2}.$$

The first term keeps the full Bernoulli error. Integration of the second over the real line gives  $1/(96\sqrt{2\pi}(dv)^{7/2})$ , since  $15/1440 = 1/96$ . For  $d \geq 2$  we obtain

$$|\Delta^2 b_d(j) - \mathcal{C}(dv + 1/6, D)| \leq \frac{4\eta\rho_d^3}{3\pi v^2 d^2} + \frac{5\rho_d^{7/2}}{16\sqrt{2\pi}(dv)^{5/2}} + \frac{1}{96\sqrt{2\pi}(dv)^{7/2}} + \left\{ \frac{1}{dv} + \frac{1}{\pi^2(dv)^2} \right\} e^{-dv\pi^2/2}. \quad (139)$$

The final term bounds the continuous reference tail; replacing  $dv + 1/6$  there by  $dv$  enlarges it. Thus the lattice error has been replaced with a proved residual, not discarded.

For random  $M$ , put  $c_* = 1/(6v)$  and

$$\tilde{y} = \frac{M + c_*}{m + c_*}, \quad \tilde{x} = \frac{\delta^2}{2v(M + c_*)}.$$

These variables satisfy

$$\mathbb{E}\tilde{y} = 1, \quad \text{Var } \tilde{y} \leq D/m, \quad \mathbb{E}(\tilde{y}\tilde{x}) \leq R/2, \quad \{\tilde{y} < t\} \subseteq \{M < tm\} \quad (0 < t < 1).$$

The exponents in their lower-tail bounds remain  $m\rho_t$ . The good event is still  $M \geq \alpha m$ , not a newly defined event. On its complement,

$$\tilde{y} \leq \frac{\alpha m + c_*}{m + c_*} \leq \frac{\alpha m + c_{\max}}{m + c_{\max}}, \quad c_{\max} = 1/(6v_-),$$

where  $v_-$  is a lower bound for  $q(1 - q)$  on the activity interval. Each matched row checks that  $Q_\beta$  is nonpositive through this enlarged endpoint. Since  $A'(1) \geq 0$  in those rows,  $Q_\beta$  increases to the left of 1, so the endpoint check covers the whole bad interval. If  $G(m) > 0$ , the normalized matched main term is at least

$$\left( \frac{m}{m + c_{\max}} \right)^{3/2} G(m). \quad (140)$$

Its loss from the factor in front of  $G$  must be retained. In (136), replace the quartic error by

$$\frac{5\rho^{7/2}J_{5/2}(m)}{16vm} + \frac{J_{7/2}^{(0)}(m)}{96v^2m^2}. \quad (141)$$

The cubic error, small- $M$  error and continuous Gaussian tail are unchanged. These formulas prove the matched-curvature budget while allowing arbitrary dependence between  $M$  and  $Y$ .

## 7.10 Mean-preserving point mass and ordinary directions

Let  $y = M/m$  and  $x = \delta/\sqrt{vm}$ , and define  $F(y, x) = y^{-1/2}e^{-x^2/(2y)}$  for  $y > 0$ . Then  $\mathbb{E}x = 0$ ,  $\mathbb{E}x^2 \leq R$  and  $\text{Var } y \leq D/m$ . For any  $C \geq 0$ , the exponential tangent inequality gives

$$F(y, x) \geq e^{-C} \{y(1 + C - \tfrac{3}{2} \log y) - x^2/2\}. \quad (142)$$

This follows by writing  $F = y \exp(-\frac{3}{2} \log y - x^2/(2y))$ . Also  $y \log y - y + 1 \leq (y - 1)^2$ , so  $\mathbb{E}(y \log y) \leq D/m$ . With  $C = R/2 + 3D/(2m)$  the full-space expectation of the right side of (142) is at least  $e^{-C}$ . On  $0 \leq y < \alpha < 1$ , its positive part is at most  $3e^{-1/3}/2 < 43/40$ : first discard  $-x^2/2$ , and then use  $e^{-C}(1 + C - \frac{3}{2} \log y) \leq 1 - \frac{3}{2} \log y$  on  $y \leq 1$ . Thus, with  $\rho = (1 - 1/(\alpha m))^{-1}$ , averaging (120) proves a lower bound  $P(m)$  for  $\sqrt{m}p_k$ :

$$P(m) = \max \left\{ 0, \frac{e^{-R/2 - 3D/(2m)} - (43/40)e^{-m\rho\alpha}}{\sqrt{2\pi v}} - \mathcal{E}(m) \right\}, \quad (143)$$

where

$$\begin{aligned}\mathcal{E}(m) &= \frac{\eta\rho^2 J_1(m)}{3\pi v\sqrt{m}} + \frac{\rho^{5/2} J_{3/2}(m)}{16\sqrt{2\pi} v^{3/2} m} \\ &\quad + \sqrt{m}(1 - T/\pi)e^{-v\alpha m T^2/2} + \frac{e^{-v\alpha m T^2/2}}{\pi v\alpha T\sqrt{m}}.\end{aligned}\quad (144)$$

In an interval budget use  $v_+$  in the denominator of the main term once its bracket is positive, and  $v_-$  in the error terms. For the full-period bound use  $T = \pi$  as before.

To average the first difference, choose  $0 < \alpha_1 < \alpha < 1$  and valid gradient constants  $c_0, c_1$  on  $M \geq \alpha m$  and  $M \geq \alpha_1 m$ , respectively. Put  $A_0 = c_0/(v\alpha)$  and  $A_1 = c_1/(v\alpha_1)$ , and require  $0 < A_0 \leq A_1 < m$ . On the whole real axis the reciprocal majorant in (128) is positive, and on  $0 \leq y \leq \alpha$  it is at least  $1/\alpha$ . Consequently the pointwise gradient bound can use that quadratic on the whole space, adding  $(A_1 - A_0)/m$  below  $\alpha$  and  $1 - A_1/m$  below  $\alpha_1$ . Taking expectations gives

$$\sqrt{m}|p_{k+1} - p_k| \leq H(m) := \frac{c_0 J_1(m)}{v\sqrt{m}} + \frac{A_1 - A_0}{\sqrt{m}}e^{-m\rho_\alpha} + \frac{m - A_1}{\sqrt{m}}e^{-m\rho_{\alpha_1}}. \quad (145)$$

The same bound holds for the other adjacent difference. In (145) either inverse bound may be used. Indeed, the refined pointwise majorant is  $Q_{1,\beta}(y) + \sum_i (U_i - U_{i+1})\mathbf{1}_{y < t_{i+1}}$ . For  $y < \alpha$  its step term equals  $U_0$ , and the quadratic is decreasing up to  $\alpha$ . Its value is therefore at least  $Q_{1,\beta}(\alpha) + U_0 \geq 1/\alpha$ , as required for the same two additional layer payments.

Since  $p_j = c_j \lambda^j / I_F(\lambda)$ , a left activity interval  $[a, b]$  with  $b < 1$  has  $c_k > c_{k-1}$  whenever

$$(1 - b)P(m) - bH(m) > 0. \quad (146)$$

A right interval with  $a > 1$  has  $c_k > c_{k+1}$  whenever

$$(a - 1)P(m) - H(m) > 0. \quad (147)$$

These follow by bounding  $p_k - \lambda p_{k-1}$  on the left and  $\lambda p_k - p_{k+1}$  on the right. They are coefficient directions at the integer mean, rather than assertions that the probability mass function has its mode exactly at that mean.

### 7.11 A joint mass and signed-gradient envelope

A stronger directional bound estimates the Gaussian mass and its signed first derivative together. For  $y \geq 4/9$ ,  $x \in \mathbb{R}$  and  $|\theta| \leq \theta_0$ , the fixed envelope certificates prove

$$F(y, x)(1 - \theta x/y) \geq H - \frac{y-1}{2} - \theta x - A_x x^2 - B_y (y-1)^2, \quad H = 99/100. \quad (148)$$

The parameter triples  $(\theta_0, A_x, B_y)$  used are

|            |    |     |     |     |     |     |      |     |     |      |
|------------|----|-----|-----|-----|-----|-----|------|-----|-----|------|
| $\theta_0$ | .5 | .6  | .75 | .9  | 1   | 1.1 | 1.25 | 1.4 | 1.5 | 1.65 |
| $A_x$      | .6 | .64 | .69 | .75 | .79 | .84 | .9   | .96 | 1   | 1.07 |
| $B_y$      | 2  | 2.1 | 2.3 | 2.6 | 2.7 | 3   | 3.2  | 3.5 | 3.6 | 4    |

Convex interpolation between certified triples is also valid: both sides of (148) are affine in  $\theta, A_x, B_y$ . Interpolation with  $\theta = 0$ , followed by reflection  $x \mapsto -x$ , gives the entire interval  $|\theta| \leq \theta_0$ .

We describe the exact all-real-domain reduction underlying these certificates. Put

$$a_{j,k} = \frac{j!(2j-k)!}{(2j)!k!(j-k)!}, \quad P_j(z) = \sum_{k=0}^j (-1)^k a_{j,k} z^k, \quad Q_j(z) = \sum_{k=0}^j a_{j,k} z^k.$$

Repeated integration by parts gives the signed remainder

$$Q_j(z)e^{-z} - P_j(z) = \frac{(-1)^{j+1}z^{2j+1}}{(2j)!} \int_0^1 u^j(1-u)^j e^{-zu} du. \quad (149)$$

Since  $Q_j(z) > 0$  for  $z \geq 0$ , this proves  $P_3/Q_3 \leq e^{-z} \leq P_2/Q_2$  throughout  $[0, \infty)$ . Let  $y = s^2$ ,  $s \geq 2/3$ , and  $z = x^2/(2s^2)$ . Clearing the positive denominators gives

$$\begin{aligned} P_3^* &= 960s^6 - 240x^2s^4 + 24x^4s^2 - x^6, & Q_3^* &= 960s^6 + 240x^2s^4 + 24x^4s^2 + x^6, \\ P_2^* &= 48s^4 - 12x^2s^2 + x^4, & Q_2^* &= 48s^4 + 12x^2s^2 + x^4. \end{aligned}$$

For a fixed  $\theta > 0$ , use  $j = 3$  where  $s^2 - \theta x \geq 0$  and  $j = 2$  where  $s^2 - \theta x \leq 0$ . It is enough to prove

$$(s^2 - \theta x)P_j^* - s^3Q_j^*\{H - (s^2 - 1)/2 - \theta x - A_x x^2 - B_y(s^2 - 1)^2\} \geq 0. \quad (150)$$

The three substitutions

$$x = -u \ (u \geq 0), \quad x = ts^2/\theta \ (0 \leq t \leq 1), \quad x = s^2/\theta + u \ (u \geq 0)$$

cover the negative, middle and positive regions without gaps. For  $\theta = 0$  reflection permits  $x \geq 0$  and  $j = 3$ . Put  $s = 2/3 + w$ . Each unbounded variable  $u$  or  $w$  is mapped to  $r/(1-r)$ ,  $0 \leq r < 1$ , and multiply by the appropriate nonnegative power of  $1-r$ . Tensor-product Bernstein coefficients on the resulting unit square prove the whole compact rectangle, including its boundary at infinity by polynomial continuity. Splitting a rectangle by the de Casteljau identities preserves its polynomial exactly. Thus the stored subdivision tree and nonnegative rational leaf coefficients establish (148) on the unbounded domain, rather than on a bounded plot of that domain.

We next explain its use in the coefficient direction. Put  $\varepsilon = |1 - \lambda|$  and

$$A_\lambda = \begin{cases} \lambda/(1-\lambda), & \lambda < 1, \\ 1/(\lambda-1), & \lambda > 1, \end{cases} \quad h = (vm)^{-1/2}, \quad \theta = A_\lambda h.$$

Divide the true directional expression  $p_k - \lambda p_{k-1}$  or  $\lambda p_k - p_{k+1}$  by  $\varepsilon/\sqrt{2\pi vm}$ . Its Gaussian main term is a combination of  $F(y, x)$  and one adjacent shift of  $F$ . Taylor expansion produces  $F(y, x)(1 - \theta x/y)$ , with a possible reflection of  $x$ . Since  $\mathbb{E}x = 0$ , full-space averaging of (148) gives the main term

$$H - A_x R - B_y D/m. \quad (151)$$

On the bad event  $y < \alpha = 4/9$ , the polynomial's positive part is at most

$$P_{\text{dir}} = \left[ H + (1 - \alpha)/2 - B_y(1 - \alpha)^2 + \theta_0^2/(4A_x) \right]_+. \quad (152)$$

Here the maximum over  $x$  is obtained by completing its square, and the maximum over  $0 \leq y \leq \alpha$  is at  $\alpha$ , because  $1/2 - 2B_y(1 - \alpha) \leq 0$ . Subtract  $P_{\text{dir}}e^{-m\rho_\alpha}$  from (151).

The elementary bound  $F_{xx}(y, x) \leq 2e^{-3/2}y^{-3/2}$  controls the finite Taylor step. The resulting normalized cost is

$$A_\lambda e^{-3/2} \frac{J_{3/2}(m)}{vm}. \quad (153)$$

Small- $M$  terms of the true direction are also retained. Its mass part is nonnegative, so only  $A_\lambda$  times an actual binomial gradient is charged. Below  $\gamma_0 m$  the cost is  $A_\lambda \sqrt{2\pi vm} e^{-m\rho_0}$ . For  $\gamma_i m \leq M < \gamma_{i+1} m$ , use the gradient constant valid at  $\gamma_i m v$  and coefficient  $A_\lambda \sqrt{2\pi v} c_i/(v\gamma_i \sqrt{m})$ ; (130) combines the layers when these coefficients decrease.

It remains to bound the finite-binomial Fourier error for the joint direction. This is smaller than separately pricing two point errors. The exact multiplier satisfies

$$|1 - \lambda e^{it}|^2 = (1 - \lambda)^2 + 4\lambda \sin^2(t/2) \leq \varepsilon^2 + \lambda t^2. \quad (154)$$

Use the same nonnegative error density from (119) inside  $[-\pi, \pi]$  and the Gaussian reference density outside it. Let  $I_0(M)$  and  $I_2(M)$  denote its integrals with weights 1 and  $t^2$ , including the Fourier factor  $1/(2\pi)$ . Cauchy–Schwarz first in frequency and then in the original probability space shows that, if

$$U_0 \geq \sqrt{m} \mathbb{E}[I_0(M); M \geq \alpha m], \quad U_2 \geq m^{3/2} \mathbb{E}[I_2(M); M \geq \alpha m],$$

the normalized joint error is at most

$$\sqrt{2\pi v} \sqrt{U_0^2 + \frac{\lambda}{\varepsilon^2 m} U_0 U_2}. \quad (155)$$

The same bounds apply to the reversed multiplier on the other side. Explicit valid expressions in the full-period Gaussian range are

$$U_0 = \frac{\eta \rho^2 J_1(m)}{3\pi v \sqrt{m}} + \frac{\rho^{5/2} J_{3/2}(m)}{16\sqrt{2\pi} v^{3/2} m} + \frac{e^{-\pi^2 v \alpha m/2}}{\pi^2 v \alpha \sqrt{m}}, \quad (156)$$

$$U_2 = \frac{4\eta \rho^3 J_2(m)}{3\pi v^2 \sqrt{m}} + \frac{5\rho^{7/2} J_{5/2}(m)}{16\sqrt{2\pi} v^{5/2} m} + \left\{ \frac{\sqrt{m}}{v\alpha} + \frac{1}{\pi^2 v^2 \alpha^2 \sqrt{m}} \right\} e^{-\pi^2 v \alpha m/2}. \quad (157)$$

The use of a common error density is essential for (155); two unrelated signed errors cannot be substituted into that inequality.

Subtract (152) times its tail probability, (153), (155), and the small- $M$  layer cost from (151). A positive result proves the required coefficient direction. Across an activity interval, replace  $A_\lambda$  by  $b/(1-b)$  on the left or  $1/(a-1)$  on the right; likewise use  $b/(1-b)^2$  or  $a/(a-1)^2$  for  $\lambda/\varepsilon^2$ .

**A sharper finite-step bound.** There is a sharper replacement for (153). Since

$$-F_{xx}(y, x) = y^{-3/2}(1 - x^2/y)e^{-x^2/(2y)},$$

the construction of Section 7.8, applied on the same good event  $\mathcal{G} = \{M \geq \alpha m\}$ , gives  $\mathbb{E}[-F_{xx}(y, x); \mathcal{G}] \geq G_0(m)$ . The variables here have  $\mathbb{E}x = 0$ . Thus  $\mathbb{E}(x+s)^2 = \mathbb{E}x^2 + s^2$ , and the very same proof gives

$$\mathbb{E}[-F_{xx}(y, x+s); \mathcal{G}] \geq G_0(m) - cs^2/2. \quad (158)$$

The polynomial maximum on the bad event is unchanged by this translation, since the maximum over its offset is still attained when the squared offset vanishes. In particular, no independence of  $y$  and  $x$  is needed.

Integrating (158) twice over the actual step  $h = (vm)^{-1/2}$  gives

$$\mathbb{E}[F(y, x) - F(y, x+h) + hF_x(y, x); \mathcal{G}] \geq G_0(m)h^2/2 - ch^4/24. \quad (159)$$

If the row begins at  $m_*$ , take the fixed constant  $G_* := \min\{0, G_0(m_*)\}$ . The construction in (135) makes  $G_0(m)$  nondecreasing, so

$$A_\lambda \left\{ \frac{-G_*}{2vm} + \frac{c}{24v^2m^2} \right\} \quad (160)$$

is a valid replacement cost for (153) for all  $m \geq m_*$ . Positive curvature gains are not required. This replacement concerns only the conditional Gaussian reference's finite step; all the finite-binomial and small- $M$  costs remain.

## 7.12 Fixed rows and the unbounded mean domain

We specify how the preceding estimates give statements uniform in both activity and order. A row fixes a closed activity interval  $[a, b]$ , applicable source and selection certificates, density  $r_*$ , bounds  $D, R$ , cutoffs and tail-rate parameters, and an initial real mean

$$m_* = \frac{r_* N}{2q_b}. \quad (161)$$

The following choices cover an entire activity interval: take the minimum and maximum of  $v = q(1 - q)$  over  $q \in [q_a, q_b]$ , the maximum of  $|1 - 2q|$  at its endpoints, the maxima described after (111) and (112), and the endpoint minimum in (115). Keep every chosen multiplier, cutoff, envelope and interpolation weight fixed thereafter. The underlying forest is not restricted to any finite set of message values, vertex degrees or component types.

The activity and mean hypotheses for every row are checked at  $m_*$ . They include  $\alpha m_* > 1$ , the Fourier conditions (117) or  $v_- \geq 9/40$ , and each chosen gradient threshold. For the ordinary direction, require  $m_* > A_1$  and

$$2\rho_{\alpha_1}m_*(m_* - A_1) \geq m_* + A_1. \quad (162)$$

For a joint row, the fixed  $\theta_0$  must dominate  $A_\lambda/\sqrt{v-m_*}$ . For a matched-curvature row, require  $G(m_*) > 0$  and the enlarged bad-event polynomial condition after (139).

Here is the extension to all  $m \geq m_*$ . The factors  $\rho = (1 - 1/(\alpha m))^{-1}$  decrease, the two alternatives in (129) decrease, and hence their minimum decreases. The functions  $G(m)$  and the joint main term (151) increase. Every remaining error is a sum or product of nonnegative decreasing factors, except possibly factors of the form  $m^p e^{-\tau m}$  and the last term in (145). The first is decreasing whenever  $\tau m \geq p$ , a condition checked at  $m_*$  for  $p = 1/2$  or  $3/2$  as applicable. For the last term, direct differentiation shows that  $(m - A_1)m^{-1/2}e^{-\rho_{\alpha_1}m}$  decreases if and only if  $2\rho_{\alpha_1}m(m - A_1) \geq m + A_1$ . The function  $(m + A_1)/(m(m - A_1))$  decreases for  $m > A_1$ , so (162) suffices on the entire half-line.

For the lattice correction,  $[m/(m + c_{\max})]^{3/2}$  increases and  $G(m) > 0$  increases, so their product increases. The enlarged bad-event endpoint  $(\alpha m + c_{\max})/(m + c_{\max})$  decreases, retaining the verified polynomial sign. The extra term in (141) decreases. For the curvature-assisted direction,  $G_*$  is fixed and nonpositive, and (160) decreases. The actual  $\theta = \lambda_X/\sqrt{vm}$  decreases, so its fixed envelope continues to apply. These facts justify evaluating each row's sufficient expression at its lower mean endpoint; they do not replace an infinite range by a finite selection of means.

The fixed large-mean family has 43 rows: six left intervals covering  $[1/3, .88]$ , seventeen central intervals covering  $[.88, 1.14]$ , and twenty right intervals covering  $[1.14, 9/4]$ . Eleven of the direction rows use (146) or (147); fifteen use the joint bound and (160); all seventeen central rows use (140). The parameter records, their parent source indices, and the positive rational lower bounds are specified by `fixed_plan_350.json`, `coverage_350.json`, and `results/complete_350.json.gz` in the frozen ancillary certificate package. In particular, these records fix  $N = 350$ , and every row has a strictly positive accepted lower bound. The smallest central lower bound in those normalizations is

$$\frac{4578083662128341986864739127}{500000000000000000000000000000} > 0.$$

Different methods use different normalizations; this displayed number is not a common lower bound for all original coefficients.

All forests with  $n \geq 350$  satisfy the lower mean requirement (161). More generally, a row's analytic statement applies to any forest for which its actual  $m$  meets that same threshold, whether or not  $n > 350$ . This latter interface is what permits the finite conditional-state

arguments of Section 8 to use the large-mean rows above their respective cutoffs and cover the complementary bounded-mean region separately.

Finally, put  $R_j = c_{j+1}/c_j$ . On left ranks the direction says  $R_{k-1} > 1$ ; on central ranks it says  $R_{k-1} > R_k$ ; and on right ranks it says  $R_k < 1$ . Since the activity at the integer mean increases strictly with the rank, these three ranges occur in this order. A strict descent of coefficients cannot be followed by a strict ascent: inside the central block this is excluded by the decreasing ratio chain, and its endpoint ratios are constrained by the neighboring left and right blocks. If the central block contains no integer rank, at most one ratio lies between the forced left and forced right ratios. The initial and final monotone segments from Lemma 3.1 complete the unimodality conclusion for the ranks covered here.

### 7.13 The finite algebraic acceptance interface

The analytic proof has two types of fixed finite input. Source inputs assert (93), (97), (99) or (103) on their full scalar domains. Kernel inputs assert the 1978 gradient polynomial inequalities and the joint-envelope inequalities (150), and supply the constants for the 43 final rows. Each is accepted by rational inequalities after outward bounds on the explicit logarithms, exponentials and square roots. No floating-point search value is an accepted coefficient or bound.

For clarity, if  $P(t) = \sum_{k=0}^d a_k t^k$  on  $[0, 1]$ , its Bernstein coefficients of degree  $d$  are

$$b_j = \sum_{k=0}^j a_k \frac{\binom{j}{k}}{\binom{d}{k}}, \quad P(t) = \sum_{j=0}^d b_j \binom{d}{j} t^j (1-t)^{d-j}.$$

Their nonnegativity proves  $P \geq 0$  throughout the interval. The product basis proves the corresponding rectangle assertion. Affine substitution, exact denominator clearing and exact subdivision specify the cells and preserve their union. The accepting checks reconstruct these polynomials from the scalar formulas and check every cell, branch and endpoint, including the distinct leaf constraints. Rational interval bounds on transcendental coefficients are used only in directions that lower the asserted nonnegative expression or upper-bound a subtracted error. The precision and recorded margins are part of the finite proof data; they are not assumptions about an underlying floating-point model.

The earlier conditional-Gaussian and central-curvature arguments needed in the frozen input are thus represented here by (120), (128), (142), (133) and the gradient reduction in Section 7.6. Their weaker historical numerical budgets are not additional hypotheses. The combinatorial monotone window and the all-order activity-two mean estimate are separate inputs, proved in the main text and Section 6. The archive layout and reproduction interface are specified in Section 11. The finite certificate checking remains a computer-assisted part of the proof, rather than a proof-assistant formalization.

## 8 Mean refinements, source inequalities and finite kernels

This section gives the common proof behind the finite extensions  $350 \rightarrow 170 \rightarrow 130 \rightarrow 99 \rightarrow 80 \rightarrow 59$ . The quantities  $M, Y, m, q, \delta$  have the meanings in (41)–(42). All sources refer to the same forest and probability distribution. A density bound is denoted by  $r_*$  to distinguish it from a message retention ratio.

### 8.1 Moment bounds and integer target ranks

Suppose a source gives  $V \leq C_W W$  on an activity interval and  $q \in [q_a, q_b]$ . Then

$$\mathbb{E}\delta^2 \leq Rv_+m, \quad R = \frac{C_W}{1 - q_b} - 1, \quad v_+ = \max_{q \in [q_a, q_b]} q(1 - q).$$

The parameters used here have  $R \geq 0$ . A sharper valid choice is

$$\mathbb{E}\delta^2 \leq \max_{q=q_a, q_b} \{C_W q - q(1-q)\} m,$$

since the expression in braces is convex in  $q$ . If  $\text{Var } K_B \leq \kappa W$ , conditional variance gives

$$\text{Var } K_B = q(1-q)m + q^2 \text{Var } M, \quad \text{Var } M \leq \left(1 + \frac{\kappa - 1}{q}\right) m. \quad (\text{FK1})$$

The maximum of the last coefficient on a  $q$  interval is at an endpoint. Several independently valid upper bounds may be replaced by their minimum.

If  $\mu \geq r_* n$  and  $n \geq N$ , then the target rank  $\mu = k \in \mathbb{Z}$  satisfies

$$k \geq \lceil r_* N \rceil, \quad m \geq \frac{\lceil r_* N \rceil}{2q_b}. \quad (\text{FK2})$$

Only  $k$  is rounded; the mean  $m$  remains real. The complement marginals also give  $\mu - W \leq q(n - |B|)$ . Using  $n \leq \mu/r_*$ ,  $\mu \leq 2qm$  and  $r_* \leq q$ , we obtain

$$\frac{|B|}{m} \leq \frac{2q}{r_*} - 1. \quad (\text{FK3})$$

This is a bound on the fixed independent set, not a claim that  $M \leq n - |B|$ .

## 8.2 Independent support with two activity bounds

Let  $B$  be any fixed independent set. Write  $\sigma_v = \mathbf{1}_{\{v \in B\}}$ , and assume activities satisfy  $0 \leq x_v \leq b_1 = bt$  on  $B$  and  $0 \leq x_v \leq b_0 = b$  elsewhere, where  $0 < t \leq 1$ . For positive activities let

$$R_v = x_v \prod_{w \text{ child of } v} (1 - p_w), \quad p_v = \frac{R_v}{1 + R_v}, \quad z_v = \sigma_v - \sum_{w \text{ child of } v} p_w z_w, \quad r_v = \frac{\pi_v}{p_v}.$$

The message variance identity of Section 7 yields

$$\text{Var } K_B = \sum_v p_v r_v (1 - p_v) z_v^2. \quad (\text{FK4})$$

Independence of  $B$  restricts the current and parent marks  $(\sigma, \eta)$  to  $(0, 0), (0, 1), (1, 0)$ . Their full domain includes

$$0 < p \leq \frac{b_\sigma}{1 + b_\sigma}, \quad \frac{1}{1 + b_\eta(1 - p)} \leq r \leq 1. \quad (\text{FK5})$$

A root can be assigned a fictitious parent mark 0, with  $r = 1$ .

Put  $h(p) = 1 - p/2$ ,  $T_\sigma(p) = \log[b_\sigma(1 - p)/p]$  and  $\Lambda_\sigma(p) = 1/[pT_\sigma(p)]$ . The inequality  $p/h(p) \leq -\log(1 - p)$  and weighted Cauchy–Schwarz imply

$$\begin{aligned} p_v \Lambda_{\sigma_v}(p_v) (\sigma_v - z_v)_+^2 &\leq \sum_{w \text{ child of } v} p_w h(p_w) (z_w)_+^2, \\ p_v \Lambda_{\sigma_v}(p_v) (z_v - \sigma_v)_+^2 &\leq \sum_{w \text{ child of } v} p_w h(p_w) (z_w)_-^2. \end{aligned}$$

Thus nonnegative constants  $u_0, u_1, v_0, v_1$  prove  $\text{Var } K_B \leq C \mathbb{E} K_B$  if, throughout (FK5) and for every real  $z$ ,

$$\begin{aligned} 0 \leq & Cr\sigma - r(1 - p)z^2 + u_\sigma \Lambda_\sigma(\sigma - z)_+^2 - u_\eta h(p)z_+^2 \\ & + v_\sigma \Lambda_\sigma(z - \sigma)_+^2 - v_\eta h(p)z_-^2. \end{aligned} \quad (\text{FK6})$$

Indeed, multiply by  $p$  and sum over vertices. Each negative child potential has the coefficient belonging to that child's parent. The two summed Cauchy remainders are nonpositive; the root has only an additional nonpositive boundary contribution.

For a probability cell  $[p_l, p_h]$ , set  $d = 1 - p_l$ ,  $h_l = 1 - p_l/2$  and use a certified lower bound  $L \leq \Lambda_\sigma(p)$ . With  $x \geq 0$ , the three reconstructed quadratics are

$$\begin{aligned} z = -x : & \quad u_\sigma L \sigma^2 + Cr\sigma + 2u_\sigma L \sigma x + (u_\sigma L - v_\eta h_l - rd)x^2, \\ \sigma = 1, z = x \in [0, 1] : & \quad u_1 L + Cr - 2u_1 L x + (u_1 L - u_0 h_l - rd)x^2, \\ z = \sigma + x : & \quad -u_\eta h_l \sigma^2 + Cr\sigma - rd\sigma^2 \\ & \quad + (-2u_\eta h_l \sigma - 2rd\sigma)x + (v_\sigma L - u_\eta h_l - rd)x^2. \end{aligned}$$

For  $\sigma = 0$  the constant and linear coefficients vanish; only the homogeneous quadratic coefficient is required to be nonnegative. At  $p = b_\sigma/(1 + b_\sigma)$  the vertex has no children and  $z = \sigma$ , so the separate leaf condition is

$$-u_\eta h(p)\sigma^2 + r\{C\sigma - (1 - p)\sigma^2\} \geq 0.$$

Zero activities follow by continuity of the finite partition function. The 36 tilted-support certificates at stage 130 use precisely this template, with no discarded constant or vertex-count term.

### 8.3 Marked mean sources and covariance control

On a heterogeneous activity interval  $[a, b]$ , another source template gives

$$\text{Var } K_B \leq A\mu + Cw, \quad w = \mathbb{E}K_B, \quad A, C \geq 0. \quad (\text{FK7})$$

For completeness its scalar inequality, for current and parent marks  $(\sigma, \eta)$ , is

$$\begin{aligned} 0 \leq & \quad Ar + Cr\sigma - r(1 - p)z^2 + u_\sigma \Lambda_b(\sigma - z)_+^2 - u_\eta h(p)z_+^2 \\ & \quad + v_\sigma \Lambda_b(z - \sigma)_+^2 - v_\eta h(p)z_-^2 + w_0[-1 + r(1 + \phi)] + t_0(z - r\sigma) \\ & \quad + s_0 \left[ \frac{r(\sigma - z)^2}{H_b(T_b(p))} - (1 - r)(1 - p)z^2 \right] - \ell G_a(p)/p. \end{aligned} \quad (\text{FK8})$$

Here  $u_\sigma, v_\sigma, w_0, s_0, \ell \geq 0$ ,  $t_0$  is unrestricted, and

$$\phi = \frac{q_b}{\log(1 + b)} \left[ \log \frac{a(1 - p)}{p} \right]_+, \quad G_a(p) = \log \frac{p}{a(1 - p)^2}, \quad q_b = \frac{b}{1 + b}.$$

The capacity  $H_b$  is defined in Section 7. All extra terms in (FK8), after multiplication by  $p$  and summation, are zero or nonpositive. In particular,

$$\sum_v G_a(p_v) = \sum_v \log(x_v/a) - \sum_{\text{roots}} \log(1 - p_v) \geq 0,$$

the response identity gives  $\sum p_v z_v = \sum \pi_v \sigma_v$ , the energy term cancels along edges by Cauchy-Schwarz, and  $\phi_v \leq \sum_{w \text{ child of } v} p_w$  gives the edge-count term. Consequently (FK8) implies (FK7). Independent support uses the three mark pairs above. Full marking instead uses only  $(1, 1)$  and  $A = 0$ . It is a separate certificate domain, not an application of the independent-support domain to an inadmissible marking. The scalar domain is  $0 < p < q_b$ ,  $1/[1 + b(1 - p)] \leq r \leq 1$  and all real  $z$ , divided at  $z = 0, \sigma$  as in (FK6). At  $p = q_b$  the true response is  $z = \sigma$ ; the separate leaf condition, retaining every constant term, is

$$Ar + Cr\sigma - r(1 - p)\sigma^2 - u_\eta h(p)\sigma^2 + w_0(r - 1) + t_0\sigma(1 - r) - s_0(1 - r)(1 - p)\sigma^2 - \ell G_a(p)/p \geq 0.$$

Thus the singular child-capacity terms at this endpoint are replaced by their actual zero contributions, rather than evaluated separately.

Stage 99 has 31 independent-support and three full-marking sources. The latter give  $\text{Var } K \leq C\mu$  with

| $[a, b]$         | $C$                   |
|------------------|-----------------------|
| $[17/20, 1]$     | $352048071/500000000$ |
| $[19/20, 57/50]$ | $174100049/250000000$ |
| $[4/5, 57/50]$   | $372069839/500000000$ |

In the initial common-activity distribution,  $\mu \leq 2W$  turns (FK7) into  $\text{Var } K_B \leq (2A + C)W$ , so (FK1) applies.

Now multiply only the activities on the fixed set  $B$  by  $e^{-u}$ . Write  $w(u) = \mathbb{E}_u K_B$ ,  $\mu(u) = \mathbb{E}_u K$  and  $q_u = \lambda e^{-u}/(1 + \lambda e^{-u})$ . Exponential-family differentiation gives  $w' = -\text{Var}_u K_B$  and  $\mu' = -\text{Cov}_u(K, K_B)$ . Conditional on the complement configuration, put  $Q = \mathbb{E}_u[K_B \mid C] = q_u M$  and  $\varepsilon = K_B - Q$ . Since  $\mathbb{E}[\varepsilon \mid C] = 0$  and  $\mathbb{E}\varepsilon^2 = (1 - q_u)w$ , expansion yields

$$\begin{aligned} \text{Cov}_u(K, K_B) &= (1 - q_u)w + \text{Var}_u(Q) + \text{Cov}_u(Y, Q) \\ &= (1 - q_u)w + \text{Var}_u(Q + Y/2) - \frac{1}{4} \text{Var}_u Y. \end{aligned} \quad (\text{FK9})$$

Hence

$$\mu'(u) \leq \frac{1}{4} \text{Var}_u K_C - (1 - q_u)w(u), \quad (w/q_u)' = -q_u \text{Var}_u M \leq 0. \quad (\text{FK10})$$

In particular  $w(u)/m \leq q_u$ , where the normalizing  $m$  is fixed at  $u = 0$ . No monotonicity of  $\mu(u)$ , independence of  $M, Y$ , or continued optimality of  $B$  is needed.

#### 8.4 Certified integration of the tilted sources

If throughout a time step  $\text{Var } K_B \leq A|B| + Cw$ , then  $w' \geq -A|B| - Cw$ . For  $F_C(d) = (1 - e^{-Cd})/C$  this implies

$$\begin{aligned} w(u + d) &\geq w(u)e^{-Cd} - \frac{A|B|}{C}(1 - e^{-Cd}), \\ \int_u^{u+d} w(s) ds &\geq w(u)F_C(d) - \frac{A|B|}{C}(d - F_C(d)). \end{aligned}$$

The parameters used have  $C > 0$ . The independent-support sources in Section 8.2 become valid when  $b \leq b_s$  and  $be^{-u} \leq b_s t_s$ , and remain valid thereafter. For one fixed source over the full tilt and  $t = e^{-u}$ , (FK3) gives the simpler rate

$$\mathbb{E}(1 - q + qt)^M \leq e^{-\ell(q)m}, \quad \ell(q) = qF_C(u) - \frac{A}{C} \left( \frac{2q}{r_*} - 1 \right) (u - F_C(u)).$$

This expression is affine in  $q$ , so its minimum over a closed  $q$  interval is controlled at the two endpoints.

For the stronger stepwise rates put  $\gamma = 2q_b/r_*$  and  $\beta = \gamma - 1$ . Then  $n/m \leq \gamma$ ,  $|B|/m \leq \beta$  and, since  $M \leq |B|$ ,  $|C|/m \leq \gamma - 1$ . Initially  $w/m \geq q_a$  and  $\mu/m \leq 2q_b$ . Let  $\mathbf{M}$  be a valid general source constant for all activities at most  $b$ . Cauchy–Schwarz gives a global derivative bound  $|\mu'|/m \leq \eta := \mathbf{M}\sqrt{\gamma\beta}$ , and all vertices have occupation at most  $q_b$ , giving  $\mu/m \leq q_b\gamma$ .

Suppose a step starts with  $w/m \geq L \geq 0$  and  $\mu/m \leq U$ . A full-step upper bound is  $U_c = \min\{U + \eta d, q_b\gamma\}$ . Only sources whose entire activity domain contains the closed step are used. In particular a marked-mean source on  $[a_s, b_s]$  requires  $b_s \geq b$  and  $a_s \leq ae^{-(u+d)}$ . Given any full-step bound  $U_f$  on  $\mu/m$ , the source (FK7) implies

$$(w/m)' \geq -AU_f - C(w/m).$$

Thus each valid source supplies the two lower bounds

$$Le^{-Cd} - \frac{D}{C}(1 - e^{-Cd}), \quad LFC(d) - \frac{D}{C}(d - FC(d)), \quad (\text{FK11})$$

where  $D = AU_f$  for a mean source and  $D = A\beta$  for a support-size source. Taking the maximum of the valid endpoint bounds and zero gives a valid new  $L$ ; taking the maximum of the valid integral bounds and zero gives a valid step integral. These two maxima may use different sources, since each bounds the same respective real quantity.

The covariance refinement supplies a better  $U_f$ . First obtain a valid endpoint lower bound  $L_{\text{end}}$  from  $U_c$ . Since  $w$  is nonincreasing,  $w/m \geq L_{\text{end}}$  throughout the step. Valid source bounds for  $K_C$  give

$$\text{Var } K_C/m \leq V_C := \min\{\mathbf{M}(\gamma - 1), A(\gamma - 1) + C \max(0, U_c - L_{\text{end}})\},$$

with the minimum also taken over all admissible general affine sources valid for  $K_C$ ; independent-support sources alone do not suffice here. Together with a full-step bound  $q_u \leq q^{\text{up}}$ , (FK10) gives  $d_\mu = V_C/4 - (1 - q^{\text{up}})L_{\text{end}}$ . Cauchy bounds for  $\text{Var } K_B, \text{Var } K$  may further reduce the growth estimate. A safe full-step bound and a safe endpoint bound are formed separately, using respectively

$$U + \max(0, d_\mu)d, \quad U + d_\mu d.$$

Their minima with other proved upper bounds remain valid. The endpoint bound is used for the next step, never for earlier times in the current step. This retains a possible decrease without assuming one.

The fixed data use rational time steps  $1/20$  in stage 130 and  $1/100$  in the covariance-refined stages, with a final rational remainder. Directed logarithmic bounds choose a terminal time no larger than  $\log(1/t)$ . Omitting the remaining nonnegative integral of  $w$  weakens the estimate safely. The retained values of  $t$  are

$$\frac{19}{20}, \frac{9}{10}, \frac{4}{5}, \frac{7}{10}, \frac{3}{5}, \frac{1}{2}, \frac{9}{20}, \frac{2}{5}, \frac{7}{20}, \frac{3}{10}, \frac{1}{4}, \frac{1}{5}, \frac{3}{20}, \frac{1}{10}, \frac{1}{20}, \frac{1}{100}, \frac{1}{1000}.$$

The last three occur from stage 130 onward. Every positive integral lower bound  $\ell m$  proves

$$\mathbb{E}(1 - q + qt)^M = \mathbb{E}t^{K_B} \leq e^{-\ell m}. \quad (\text{FK12})$$

All source eligibility, endpoint and integral calculations are made with outward rational bounds.

## 8.5 Boundary and density bounds

The path-minimal mean theorem at activity 1 [2], together with the explicit path formula, gives a useful additive gain. Set  $\rho = (1 - 1/\sqrt{5})/2$ ,  $\rho_0 = 276393/10^6 < \rho$  and  $s = (1 - \sqrt{5})/(1 + \sqrt{5})$ . The Fibonacci expression and its logarithmic derivative give

$$\mu_{P_n}(1) = \rho n + \frac{3}{5} - \frac{1}{\sqrt{5}} - \frac{(n+2)s^{n+2}}{\sqrt{5}(1 - s^{n+2})}.$$

Use  $v = 223607/500000 > 1/\sqrt{5}$  and  $t = 191/500 > |s|$ . Odd  $n$  has a positive correction. For even  $n \geq 2$ , the negative correction is at most  $v(n+2)t^{n+2}/(1 - t^{n+2})$ . The expression  $Nt^N/(1 - t^N)$  decreases for  $N \geq 4$  because the successive ratio is at most  $5t/4 < 1$ . The rational inequalities

$$\frac{3}{5} - v - \frac{4vt^4}{1 - t^4} > \frac{11}{100}, \quad \frac{3}{5} - v - \frac{10vt^{10}}{1 - t^{10}} > \frac{19}{125}$$

and the direct one-vertex value give a gain greater than 11/100 for every nonempty tree, and greater than 19/125 for every tree of order at least 8. Additivity over components therefore yields

$$\mu_F(1) \geq \rho_0 n + \frac{19}{125} \quad (n \geq 8). \quad (\text{FK13})$$

If  $a < 1$ , a piecewise valid upper bound  $V \leq C(x)\mu$  gives

$$\mu(a) \geq \exp\left(-\int_a^1 C(x) d\log x\right) \mu(1).$$

The integration is partitioned at all applicable source endpoints. For  $a \geq 1$  the universal lower variance bound gives  $\mu(a) \geq \sqrt{2a/(1+a)} \mu(1)$ . On a finite order range  $N \leq n \leq n_{\max}$ , a resulting  $\mu \geq r_t n + h$ ,  $h \geq 0$ , implies  $\mu \geq (r_t + h/n_{\max})n$ . The integer lower bound may use both  $\lceil r_* N \rceil$  and  $\lceil r_t N + h \rceil$ .

The next argument supplies density bounds at selected rational activities. We adapt the branch-decomposition induction in [2, Section 4], with new activity-dependent constants and explicit acceptance conditions. For each of 106 rational activities  $x$ , the density data specify  $r, g_4, s_0, B_0, h$  and prove

$$\mu_F(x) \geq rn + h \quad (n \geq 8). \quad (\text{FK14})$$

Here  $r$  denotes the density coefficient within this subsection. Write  $s = \sqrt{1+4x}$ ,  $\rho = (1-1/s)/2$  and  $t = (s-1)/(s+1)$ . The path recurrence  $Z_n = Z_{n-1} + xZ_{n-2}$ ,  $Z_0 = 1$ ,  $Z_1 = 1+x$  gives

$$\mu_{P_n}(x) = \rho n + \frac{(s-1)^2}{2s^2} - \frac{(n+2)(-t)^{n+2}}{s(1-(-t)^{n+2})}. \quad (\text{FK15})$$

Assume  $r \leq \rho$  and  $4t^2/3 < 1$ , as verified for every retained activity. For even  $n \geq 4$ , the absolute negative error  $E_n$  satisfies

$$\frac{E_{n+2}}{E_n} = \frac{n+4}{n+2} t^2 \frac{1-t^{n+2}}{1-t^{n+4}} < \frac{4}{3} t^2 < 1.$$

Odd-order corrections are positive. Consequently  $\mu_{P_n} - rn \geq \mu_{P_4} - 4r$  for every  $n \geq 4$ , and the analogous  $P_8$  bound holds for every  $n \geq 8$ . The complete initial conditions are

$$\begin{aligned} h > 0, \quad 0 < s_0 \leq g_4 \leq B_0, \quad g_4 \leq \mu_{P_4} - 4r, \quad s_0 \leq \min_{1 \leq i \leq 3} (\mu_{P_i} - ir), \\ \mu_{P_3} - 3r \geq g_4, \quad 2s_0 \geq g_4, \quad h \leq \min(\mu_{P_8} - 8r, B_0, 2s_0), \quad 4s_0 - r \geq B_0. \end{aligned} \quad (\text{FK16})$$

In particular the  $P_4$  condition is the rational inequality  $(g_4 + 4r)(1 + 4x + 3x^2) \leq 4x + 6x^2$ . The displayed conditions explicitly include the  $P_4$  and positive  $h$  checks present in the original accepting program.

Induct simultaneously on tree order: every nonempty tree has boundary  $\mu - rn \geq s_0$ , every tree of order at least 4 has boundary at least  $g_4$ , and every nonpath tree has boundary at least  $B_0$ . Paths have already been handled. Root a nonpath tree at a vertex of degree  $d \geq 3$ . Its branches have exactly five types: one vertex, two vertices, a three-vertex path rooted at an end, a three-vertex path rooted at its middle, and order at least 4. Their deleted-root to full partition-function ratios are respectively

$$\frac{1}{1+x}, \quad \frac{1+x}{1+2x}, \quad \frac{1+2x}{1+3x+x^2}, \quad \frac{(1+x)^2}{1+3x+x^2}, \quad \left[ \frac{1}{1+x}, 1 \right].$$

The last interval follows from root occupation at most  $x/(1+x)$ . The branch and deleted-branch boundary vectors are

$$\begin{aligned} A &= (\mu_1 - r, \mu_2 - 2r, \mu_3 - 3r, \mu_3 - 3r, g_4), \\ D &= (0, \mu_1 - r, \mu_2 - 2r, 2\mu_1 - 2r, g_4). \end{aligned}$$

For a large branch the deleted graph has at least three vertices. If it is connected, use the three-vertex condition or the induction hypothesis; if disconnected, use  $2s_0 \geq g_4$ . For each nonnegative type-count vector  $(n_1, \dots, n_5)$  with  $\sum n_i = d$ , let  $u'_1, \dots, u'_d$  be the actual branch ratios. Root nonoccupation is  $R = 1/(1 + x \prod_{j=1}^d u'_j)$ . Its rational bounds follow by using the four fixed ratios and the displayed interval independently for each large branch. Root conditioning gives

$$\mu_T - r|T| \geq R \left( \sum_i n_i A_i - r \right) + (1 - R) \left( 1 - (d + 1)r + \sum_i n_i D_i \right). \quad (\text{FK17})$$

For  $d = 3, 4$ , all  $\binom{7}{4} + \binom{8}{4} = 105$  type-count vectors are included. The right side is affine in  $R$ ; both rational endpoint values are required to be at least  $B_0$ .

For  $d \geq 5$  delete one branch  $S$ , leaving  $T'$ . Let  $R'$  be the root nonoccupation probability in  $T'$ ; then  $R \geq R'$ , and conditioning gives the exact identity

$$\begin{aligned} \mu_T &= R\mu_S + (1 - R)\mu_{S-\text{root}} \\ &\quad + \frac{1 - R}{1 - R'}\mu_{T'} + \frac{R - R'}{1 - R'} \sum_{i \neq S} \mu_{S_i}. \end{aligned}$$

The last two coefficients are nonnegative and sum to one. The smaller tree  $T'$  is nonpath, so its boundary is at least  $B_0$ . The at least four remaining branches give boundary at least  $4s_0 - r \geq B_0$  relative to the same  $r|T'|$  baseline. The contribution of  $S$ , after subtracting  $r|S|$ , is nonnegative: for the four small types this is an affine endpoint condition on  $R \in [1/(1 + xu_i), 1]$ ; for the large type it follows from  $g_4 - xr/(1 + x) \geq 0$ . All these gains are included in the fixed density certificate. This proves the induction for every degree. For a forest of order at least 8, one component uses the path or nonpath conclusion; two or more components give total boundary at least  $2s_0 \geq h$ . Thus (FK14) holds.

For  $a \geq x$ , integrate  $V \geq \mu/[2(1 + \lambda)]$  to obtain

$$\mu_F(a) \geq \sqrt{\frac{a/(1 + a)}{x/(1 + x)}} \mu_F(x).$$

Directed rational lower bounds propagate both  $r$  and  $h$ . For  $n \leq 98$ ,  $h > 0$  permits  $rn + h \geq (r + h/98)n$ . Combining all applicable activities with the other density bounds and then applying (FK2) supplies the lower edge of each finite-kernel coverage region.

## 8.6 Message-dependent potentials, flow and logarithmic terms

For common activity  $\lambda \leq b$ , write  $q_b = b/(1 + b)$  and retain the rooted messages  $p$ , responses  $z$  and retention ratios  $r$  from Section 7. Then

$$\mu = \sum_v p_v z_v = \sum_v p_v r_v, \quad V = \sum_v p_v r_v (1 - p_v) z_v^2.$$

The parent message  $p_{\text{par}}$  lies in the coupled domain

$$\frac{1}{1 + b(1 - p)} \leq r \leq 1, \quad 1 - r \leq p_{\text{par}} \leq \frac{b(1 - p)}{1 + b(1 - p)}. \quad (\text{FK18})$$

Indeed  $1 - r$  is the actual parent marginal, which does not exceed its downward message, and the parent's message recurrence gives the upper bound. At a root take a fictitious parent message zero and  $r = 1$ . Its parent energy and flow terms vanish; its two negative parent potentials remain as additional nonpositive boundary contributions.

Use the exact Cauchy factor  $h(p) = p/[-\log(1-p)]$ , with  $h(0) = 1$ . Let  $u, v, s$  be nonnegative continuous piecewise-linear functions of the message. Multiplying the positive-response, negative-response and energy Cauchy inequalities by these functions at each parent gives three summed nonpositive remainders. At a child their negative terms must retain  $u(p_{\text{par}}), v(p_{\text{par}}), s(p_{\text{par}})$  with the same parent message. For any real-valued function  $\tau$ , the additional identity

$$\sum_v p_v [r_v \tau(p_v)(z_v - 1) + (1 - r_v) \tau(p_{\text{par}}) z_v] = 0 \quad (\text{FK19})$$

follows by regrouping the second sum by parents and using  $z_v - 1 + \sum_{w \text{ child of } v} p_w z_w = 0$ . Thus  $\tau$  need not be nonnegative.

The selection proxy from Section 7 is

$$f_q(x) = \max \left\{ \frac{1+q}{2q} x, \frac{1+q}{2q^2} x + \frac{q-1}{2q} \right\}, \quad J_b = \sum_v \pi_v f_{q_b}(\pi_v) \leq W.$$

With  $T_b, \Lambda_b, H_b, \phi, G_a$  as above, a common-activity scalar certificate has the following complete form:

$$\begin{aligned} 0 \leq & C r f_{q_b}(pr) + C_\mu r - r(1-p)z^2 + u(p)\Lambda_b(p)(1-z)_+^2 - u(p_{\text{par}})h(p)z_+^2 \\ & + v(p)\Lambda_b(p)(z-1)_+^2 - v(p_{\text{par}})h(p)z_-^2 + s(p) \frac{r(1-z)^2}{H_b(T_b(p))} \\ & - s(p_{\text{par}})(1-r)(1-p)z^2 + w(z-1+r\phi) + t_0(z-r) \\ & + r\tau(p)(z-1) + (1-r)\tau(p_{\text{par}})z + \frac{D_n - \ell G_a(p)}{p}. \end{aligned} \quad (\text{FK20})$$

Here  $C, C_\mu, w, \ell \geq 0$ ,  $t_0, D_n$  may be signed, and the last two lines may be absent in simpler certificates. Multiply by  $p$  and sum. The Cauchy terms are nonpositive, the flow and  $t_0$  terms sum to zero, and the  $w$  term is nonpositive by the same edge-count identity as in (FK8), since  $\sum pz = \sum pr$ . This proves

$$V \leq C J_b + C_\mu \mu + D_n n - \ell \sum_v G_a(p_v). \quad (\text{FK21})$$

The message recurrence also gives the exact product identity

$$\prod_v \frac{p_v}{a(1-p_v)^2} = (\lambda/a)^n \prod_{\text{roots}} (1-p_v)^{-1}.$$

Every nonempty tree can be rooted at a leaf. Its root odds are at least  $\lambda/(1+b)$ ; an isolated root has still larger odds. Choosing such a root in every component of a nonempty forest therefore yields

$$V \leq C J_b + C_\mu \mu + [D_n - \ell \log(\lambda/a)]n - \ell \log \left( 1 + \frac{\lambda}{1+b} \right). \quad (\text{FK22})$$

If  $w = 0$ , the parameter  $a$  serves only as a logarithmic reference and need not bound  $\lambda$  below. If  $w > 0$ ,  $\lambda \geq a$  is required for the edge-count term.

For complete probability-cell acceptance put  $F_a(p) = (D_n - \ell G_a(p))/p$ . Direct differentiation gives

$$F'_a(p) = -\frac{D_n + \ell[(1+p)/(1-p) - G_a(p)]}{p^2}.$$

The bracketed expression has its global minimum at  $p = \sqrt{2} - 1$ , with value  $1 + \sqrt{2} - \log[(1 + \sqrt{2})/(2a)]$ . Hence the retained condition

$$D_n + \ell \left[ 1 + \sqrt{2} - \log \frac{1 + \sqrt{2}}{2a} \right] \geq 0$$

proves that  $F_a$  decreases, including when  $D_n < 0$ .

**Full-domain scalar acceptance.** Each source partitions  $[0, q_b]$  into consecutive rational probability cells. On  $[p_l, p_h]$ , use lower bounds for the nonnegative  $u, v, s, \Lambda_b, 1/H_b$ , upper bounds  $h(p_l)$  and  $1 - p_l$  for negative quadratic terms, and the lower bound  $F_a(p_h)$ . For  $z \leq 1$  the multiplier  $r(z - 1)$  is nonpositive, so use an upper bound on  $\tau(p)$ ; for  $z \geq 1$  use its lower bound. These choices preserve a scalar lower bound. The selection term is bounded below by evaluating its increasing piecewise-affine proxy at  $p_l r$ .

The retention cells cover  $[1/(1 + b(1 - p_l)), 1]$ . Subdivide them also at the proxy switch and where  $1 - r$  crosses a potential knot. On each potential segment the resulting expression is affine in  $p_{\text{par}}$ . It therefore suffices to include the lower endpoint  $1 - r$ , the upper bound  $b(1 - p_l)/[1 + b(1 - p_l)]$ , and every intervening knot, all with the same parent argument for  $u, v, s, \tau$ . This covers the domain (FK18); the widened upper bound only makes the requirement stronger.

Substitute  $z = -x$ ,  $z = x$ ,  $z = 1 + x$  on  $z \leq 0$ ,  $0 \leq z \leq 1$ , and  $z \geq 1$ , respectively. Each expression has the form  $A(r) + B(r)x + C(r)x^2$ , with coefficient degree at most two in  $r$ . On a half-line, sufficient exact conditions are  $A > 0$ ,  $C \geq 0$  and either  $B \geq 0$  or  $4AC - B^2 > 0$ . On  $[0, 1]$ , positive endpoints together with concavity, monotonicity, positive Bernstein coefficients  $(A, A + B/2, A + B + C)$ , or a positive quadratic with  $4AC - B^2 > 0$  are sufficient. The accepting routines reconstruct these polynomials. Quadratic extrema are exact; higher-degree positivity uses rational Sturm sequences and positive endpoint values to exclude all roots in the retention interval. No response interval or half-line is truncated.

At  $p = q_b$  there are no children and the true response is  $z = 1$ . The zero child-capacity terms are handled separately, while the parent potential, parent flow and logarithmic terms remain. The retention interval is  $[1/(1 + q_b), 1]$  and lies in the upper proxy branch. Zero activities and probability endpoints are treated by their continuous limits. The stage-80 library has 37 potential sources, 14 flow sources and 14 logarithmic sources. Stage 59 adds four sources containing both flow and logarithmic terms; all use the same proved template.

**Corrections on a bounded order interval.** Let a kernel rectangle satisfy  $N \leq n \leq n_{\max}$ ,  $q \in [q_a, q_b]$  and  $m \in [m_l, m_h]$ . Set  $\lambda_a = q_a/(1 - q_a)$ , and take directed bounds

$$g \geq D_n - \ell \log(\lambda_a/a), \quad r_0 \leq \ell \log(1 + \lambda_a/(1 + b)).$$

The vertex term in (FK22) is at most

$$\beta_0 = \begin{cases} gN - r_0, & g \leq 0, \\ gn_{\max} - r_0, & g > 0. \end{cases}$$

Using  $\mu \leq 2qm$  and  $J_b \leq qm$  gives one admissible  $\mathbb{E}\delta^2 \leq \alpha m + \beta$  with

$$\alpha = \max_{q=q_a, q_b} \{(C + 2C_\mu)q - q(1 - q)\}, \quad \beta = \beta_0.$$

Alternatively, throughout this same rectangle,

$$k = \mu \leq k_{\max} := \min\{\lfloor 2q_b m_h \rfloor, \lfloor n_{\max} q_b \rfloor\}$$

gives

$$\alpha = \max_{q=q_a, q_b} \{Cq - q(1 - q)\}, \quad \beta = \beta_0 + C_\mu k_{\max}. \quad (\text{FK23})$$

All these are statements about the same real distribution. The input source and choice of bound are regenerated for every certificate. A negative correction  $gN$  must be recalculated when  $N$  is lowered. In particular none of the stage-80 kernels is used unchanged for  $n < 80$ .

## 8.7 Exact binomial kernels and continuous activity bounds

Besides the definitions in Section 5, the kernels satisfy the exact identities

$$\begin{aligned} f_L(M, j; q) &= \frac{M+1-2j}{M+1} b_{M+1}(j; q), \\ f_R(M, j; q) &= \frac{2j+1-M}{M+1} b_{M+1}(j+1; q), \\ f_C(M, j; q) &= \frac{(M+2)v - \delta^2 - (1-2q)\delta}{(M+2)(M+1)v^2} b_{M+2}(j+1; q), \quad v = q(1-q). \end{aligned}$$

They follow by taking common denominators in the adjacent binomial coefficients, including the boundary values  $j = -1, M+1$ .

For a fixed activity cell put  $v_- = \min_{q \in [q_a, q_b]} q(1-q)$ . The characteristic function satisfies  $|1 - q + qe^{it}| \leq e^{-c_* t^2/2}$  for  $|t| \leq \pi$ . One can take  $c_* = v_-$  when  $v_- \geq 9/40$ , using the full-period lemma in Section 7. Otherwise  $0 < c_* \leq 4v_-/\pi^2$  is valid by  $\sin(|t|/2) \geq |t|/\pi$  and  $|1 - q + qe^{it}|^2 = 1 - 4q(1-q)\sin^2(t/2)$ .

For  $M > 2$ , differentiating the Fourier expression of the central kernel twice in  $q$  gives the bound

$$|\partial_q^2 f_C| \leq \frac{3M(M-1)}{\sqrt{2\pi}[(M-2)c_*]^{5/2}} < \frac{6M(M-1)}{5[(M-2)c_*]^{5/2}}.$$

Indeed the central factor is  $2 - 2\cos t \leq t^2$ , the two derivatives contribute  $M(M-1)|e^{it} - 1|^2 \leq M(M-1)t^2$ , and extension of the nonnegative Gaussian integral to the real line gives the displayed fourth-moment integral. There is also the bound  $6M(M-1)$  from the coefficient vector  $(-1, 4, -6, 4, -1)$  applied to a probability distribution. Take the smaller bound; for  $M = 0, 1$  use zero and for  $M = 2$  use 12.

For a directional kernel put  $d = M+1$ . Its prefactor in the first two identities has absolute value at most one on the effective support. The Gaussian second-moment integral and the coarse second-difference bound give, for  $d > 2$ ,

$$|\partial_q^2 f_L|, |\partial_q^2 f_R| \leq \min \left\{ 2d(d-1), \frac{2d(d-1)}{5[(d-2)c_*]^{3/2}} \right\}.$$

For  $d = 1$  the bound is zero and for  $d = 2$  it is 4. The simplification uses  $1/\sqrt{2\pi} < 2/5$ , certified by  $2\pi > 25/4$ . All these bounds are uniform over every integer  $j$ .

For stages before 59 a certificate covers all  $0 \leq M \leq n_{\max}$  and  $j \in \mathbb{Z}$ , and has the form

$$Sf \geq A + BM + c\delta - d\delta^2 - eM^2 - \sum_i T_i(1 - q + qt_i)^M, \quad S > 0, \quad d, e, T_i \geq 0. \quad (\text{FK24})$$

Stage 59 adds the matching-count term in (45). If  $P$  is the pointwise residual, a nonnegative upper bound for  $P''$  is

$$Z_M = SB_M + 2dM^2 + \sum_i T_i M(M-1)(1-t_i)^2(1-q_a + q_a t_i)^{M-2}.$$

The sum is zero for  $M < 2$ . For a mixed kernel use the nonnegative weighted sum of the three derivative bounds; the matching penalty contributes its additional nonnegative second-derivative term as in the main text. Since

$$P(q) \geq \text{chord}(P)(q) - Z_M(q - q_a)(q_b - q)/2,$$

requiring  $P(q_a), P(q_b) \geq Z_M(q_b - q_a)^2/8$  proves the inequality on the entire continuous activity cell.

For the older kernels, the finite integer strip  $-2 \leq j \leq M + 2$  is checked explicitly. Outside it the kernel is zero and the residual at each  $q$  endpoint is a convex quadratic in  $\delta$ . Its minimum on each of  $\delta \leq -2 - qM$  and  $\delta \geq M + 2 - qM$  occurs at the boundary or at the quadratic vertex, if that vertex belongs to the half-line. If  $d = 0$ , the retained certificates require  $c = 0$ . This covers all remaining integer  $j$ , with the same uniform interpolation bound. Stage 59 instead uses the fully justified finite matching-count state domain from Section 4.

Taking expectation in (FK24), or in its matching version, uses (FK1), (FK12) and (FK23). The resulting function of  $m$  is affine minus nonnegative multiples of  $m^2$  and  $e^{-\ell_i m}$ . It is concave, so strictly positive values at the two real  $m$  endpoints prove positivity throughout the rectangle.

## 8.8 Exact arithmetic and coverage of all required parameters

At rational  $q = a/b$ , the binomial probability has the exact integer expression  $\binom{M}{j} a^j (b - a)^{M-j} / b^M$ . The accepting programs enclose the modal probability on the integer grid  $10^{70}$ , then recurse to either side with positive probability ratios at most one. Lower endpoints use floor division and upper endpoints use ceiling division. Induction therefore encloses every probability, including tiny tails. Laplace powers are enclosed by positive outward multiplication. The residuals are bounded below, derivative corrections and negative expectation costs above, and all comparisons use integer cross multiplication or exact rational arithmetic. The grid is an interval scale, not a feasibility tolerance.

The order ranges and finite proof domains are

| stage | new orders | maximum $M$ | new kernels |
|-------|------------|-------------|-------------|
| 170   | 170–349    | 349         | 558         |
| 130   | 130–169    | 169         | 243         |
| 99    | 99–129     | 129         | 188         |
| 80    | 80–98      | 98          | 236         |
| 59    | 59–79      | 79          | 231         |

For every parent activity interval, take the common refinement of all rational  $q$  endpoints. On each resulting closed stratum, collect only certificates covering the entire stratum, and form the union of their real  $m$  intervals. The lower endpoint is the independently proved density and integer-rank bound; the union must meet without gaps all the way to the corresponding stage-350 threshold  $m_{\text{old}}$ . The stage-350 analytic estimate holds for every  $m \geq m_{\text{old}}$ , with its full monotonicity conditions. Its constants and the threshold are compared exactly with the original 43 rows. Reusing that kernel estimate for a smaller forest is valid because its only order dependence was in establishing a lower bound on  $m$ .

The stage-170, 130 and 99 kernels similarly remain valid on their recorded  $m$  rectangles when  $M$  is restricted to a smaller upper bound: their source inequalities have no additional negative  $N$  correction. Stage 80 is treated separately by Section 8.6. For orders 59–79 only the 188, 243 and 558 older kernels are reused, together with the 231 new ones; their 1220 rectangles cover 154 strata and all 43 parents. For orders 80–98 there are 1225 rectangles, including the 236 stage-80 kernels. The other finite refinements have 130 strata at stage 170 and 148 at stages 130 and 99.

Every accepted rectangle supplies one fixed kernel, or one fixed nonnegative mixture, over the whole conditional distribution. Different rectangles may use different kernels. The weak-valley argument of Section 5 therefore applies at every required rank. The finite proof data establish these stated coverage properties with the exact acceptance rules just described.

## 9 Integer rows for the finite-order certificates

For clarity and reproducibility, the following specifies every row family used by the finite linear certificates. A row labelled  $\rho$  means  $\sum_{j,m} a_{j,m}^\rho t_{j,m} \leq b^\rho$ ; unspecified coefficients vanish. All indices satisfy  $1 \leq j \leq v$  and  $0 \leq m \leq a - j$ . The symbols in the rows are defined above, and  $r, h, l$  are integers.

**count**( $r$ ). For  $2 \leq r \leq v$ , use  $a_{j,m} = \mathbf{1}_{j=r}$  and  $b = \binom{v}{r}$ .

**path**( $r$ ). For  $2 \leq r \leq a$ , use  $a_{j,m} = -\binom{m}{r-j}$  and  $b = \binom{a}{r} - \binom{n-r+1}{r}$ .

**private**( $r, h$ ). For  $2 \leq r \leq v$  and  $0 \leq h < a$ , set

$$a_{j,m} = \mathbf{1}_{j=r}((r-1)m + a - r + 1 - rh)_+ - (v - r + 1)\mathbf{1}_{j=r-1}(m - h)_+, \quad b = 0.$$

**hall**( $r, l$ ). Take  $0 \leq l \leq a$ ,  $0 \leq r < \min(v, a - l)$ , and  $c = v - r - \max(0, l - \delta) \geq 0$ . Set

$$a_{j,m} = ((r+1)\mathbf{1}_{j=r+1} - c\mathbf{1}_{j=r})\binom{m}{l}, \quad b = \begin{cases} c\binom{a}{l}, & r = 0, \\ 0, & r > 0. \end{cases}$$

The implemented domain with  $l = 0$  uses  $1 \leq r < v$ ; the omitted  $(r, l) = (0, 0)$  case is the already imposed first-layer equality.

**tail**( $r, h$ ). For  $2 \leq r \leq v$  and  $0 \leq h \leq a - r + 1$ , set

$$a_{j,m} = \mathbf{1}_{j=r}b_{r,h}(m) - (v - r + 1)\mathbf{1}_{j=r-1}\mathbf{1}_{m \geq h}, \quad b = 0.$$

**release\_upper**( $r, h$ ). On the same range, set

$$a_{j,m} = s_r\mathbf{1}_{j=r-1}(m - h)_+ - \mathbf{1}_{j=r}U_{r,h}(m), \quad b = 0.$$

**tail\_upper**( $r, h$ ). On the same range, set

$$a_{j,m} = s_r\mathbf{1}_{j=r-1}\mathbf{1}_{m \geq h} - \mathbf{1}_{j=r}V_{r,h}(m), \quad b = 0.$$

**mean**( $r$ ). For  $2 \leq r \leq v$ , set

$$a_{j,m} = -m\mathbf{1}_{j=r} - \beta_r\mathbf{1}_{j=2}, \quad b = -C_r^0 - \beta_r\binom{v}{2}.$$

**union**( $r$ ). For  $2 \leq r \leq v$ , set

$$a_{j,m} = (a - m)\left(\mathbf{1}_{j=r} - \binom{v-1}{r-1}\mathbf{1}_{j=1}\right), \quad b = 0.$$

**edge\_lower**( $r$ ). For  $2 \leq r \leq v$ , set

$$a_{j,m} = -\mathbf{1}_{j=r} + \binom{v-2}{r-2}\mathbf{1}_{j=2}, \quad b = -\binom{v}{r} + \binom{v-2}{r-2}\binom{v}{2}.$$

**edge\_upper**( $r$ ). For  $2 \leq r \leq v$  and  $D_0 > 0$ , set

$$a_{j,m} = D_0\mathbf{1}_{j=r} - G_r\mathbf{1}_{j=2}, \quad b = D_0\binom{v}{r} - G_r\binom{v}{2}.$$

For  $D_0 = 0$ , use the **count**( $r$ ) row instead.

**assumption**( $k$ ). Only in a **conditional** certificate at the same index  $k$ , use

$$a_{j,m} = B_{m,j}(k), \quad b = -B_{a,0}(k).$$

These formulas specify a relaxation containing the counts of every forest after the prescribed choice of  $I$ . They do not assert that every point in the relaxation is realizable by a forest.

## 10 Finite proof data and an explicit certificate

The data contain 2 577 stored triples: 670 comparison parameters and 1 907 linear certificates. The latter contain 142 017 nonzero row-weight terms. There are at most 610 variables  $t_{j,m}$  for any parameter pair in this domain; their values are not enumerated. The largest common denominator used by the principal arithmetic check has 68 decimal digits. The minimum verified normalized linear margin is

$$\frac{953977}{42042000000} > 0,$$

attained at  $(n, a, k) = (34, 19, 10)$ . This is a certificate upper-bound margin after normalization, not a lower bound for an unnormalized coefficient curvature.

The exact data file `certificates.json` is identified by the SHA-256 digest

`a77f4fc43aec1b8eecab91c4418b6990f72dee69d802129c78efd67d2e1bcc5a.`

The archived proof data and checker source are mathematical supplements to the finite theorem; the execution logs alone are not substitutes for them.

### 10.1 An explicit affine comparison at the final parameter gap

For  $(n, a, k) = (60, 33, 19)$  the data include a certificate with 193 nonzero row multipliers, including one assumption row. Its denominator is  $D_c = 10^7$  and its equality-weight numerator is 1395. Retaining the assumption row algebraically on the left, rather than declaring it true, gives the unconditional affine comparison

$$\Delta_{20} \leq \frac{96885639}{100000000} \Delta_{19} - \frac{4010142596414187852788421881}{28302876966000000000}. \quad (163)$$

Since the last positive fraction is strictly greater than 141686748,

$$\Delta_{20} \leq \frac{96885639}{100000000} \Delta_{19} - 141686748. \quad (164)$$

The complete weights are in `example_60_33_19.json`, which agrees exactly with the corresponding entry of the master data file. The comparison has been checked by recomputing both normalization scales and the rational residual bound. It applies simultaneously to all forests with  $(n, a) = (60, 33)$ , rather than to a selected graph.

## 11 Proof data and reproduction

The ancillary directory `anc/` contains the accepting programs, fixed certificate data and a single entry point. It is not needed to typeset the paper. The computational checks use only Python’s standard library and have been run with Python 3.12.14. From the submission root, use a new or empty work directory outside `anc/`:

```
python -S anc/proof_reproduce.py -work /path/to/empty-work
```

Assertions must remain enabled: do not use Python’s `-O` or `-OO` options. The optional argument `-workers 1` reduces parallel memory use. On Windows, a short work path is convenient; the included runtime handles the inherited nested-release launch paths. The checks take tens of minutes and write their results only to the chosen work directory.

## 11.1 Fixed inputs and source locations

The directory `anc/repro/inputs/` contains precisely two frozen input archives, with the following SHA-256 identifiers:

```
forest_n60_extension_and_n100_gap.zip
3cbf5c0b32bbbce213e23db79655138aebd
9ceb01fc9f55df4ac52a91ed12759

forest_threshold_59_audited_handoff.zip
cbe5b28a4c0335ef099435356edaba0d440
e650f7fd7398eb3dab8e11d0fa7f1
```

Each displayed digest is the concatenation of its two lines. The first archive contains the finite-order row generators and certificates. The second recursively contains every release from stage 59 through stage 900, including the inherited activity-two mean data. No separate copy of an inherited archive is needed at the submission root.

All filenames in the proof sections refer to files inside the corresponding released stage. For example, the mean files are in `inputs/inherited_mean/research_code/` inside the stage-900 release; they are not scripts placed next to the historical readable documents. The machine-readable index `anc/repro/source_index.json` gives the complete nested-archive route and digest of each of the 32 historical proof documents. They can be extracted without executing any release code by

```
python -S anc/repro/extract_sources.py -output /path/to/empty-docs
```

The English arguments required for the theorem are included in this paper; those historical documents provide provenance. Historical notes, diagnostic examples and earlier thresholds inside the fixed inputs are not additional hypotheses of the theorem.

## 11.2 Acceptance and full coverage

The entry point authenticates both input archives before extraction, rejects unsafe or duplicate archive paths, and runs the finite-order checkers and the full chain

$$900 \longrightarrow 500 \longrightarrow 350 \longrightarrow 170 \longrightarrow 130 \longrightarrow 99 \longrightarrow 80 \longrightarrow 59.$$

At each stage it retains the original integrity checks, source reconstruction, exact budget and kernel acceptance, comparison of the mathematical fields with the frozen data, complete parameter coverage, and the analytic tail interface. An inherited stage is not replaced by the presence of a previously successful log.

There are three kinds of finite proof obligation. Integer counting rows and rational separating identities are accepted by exact arithmetic. Continuous scalar inequalities are accepted on a complete interval partition, using outward rational bounds or the explicit polynomial identities described in the preceding sections. Discrete kernel domains are reconstructed from their integer bounds or exact combinatorial support. In every case the mathematical reduction supplies the domain: a fixed precision or a finite list of test forests cannot supply that domain.

The entry point also checks the consistency of historical rational relaxation examples stored in the finite archive. These objects are not forests and are not premises of Theorem 1.1. They impose no additional case split in the proof.

On success, `proof_reproduction_summary.json` in the work directory records `all_mathematical_acceptance_checks_passed:true`, the input and execution-code digests, all eight stage summaries and the fresh return codes. Full logs and omission records remain in that directory. A failed subprocess, a missing stage, a source mismatch or an incomplete omission record causes failure of the entry point. The accompanying `anc/README.txt` specifies the output fields and commands.

### 11.3 Supplementary diagnostics and mathematical revisions

The historical releases also include graph-instance diagnostics. The supplied entry point omits their imports and calls, and omits named-host attachment and optional finite path-sampling checks in the inherited mean scripts. The complete symbolic short-leg domain, spider induction bases, terminal-block identities, two-level closure identities and all continuous and discrete certificate coverage remain active.

The complete omissions are recorded in `anc/repro/patches.json` and `anc/repro/diagnostic_omissions.diff`. Each of the thirteen affected source files is authenticated before a uniquely matching replacement is applied in memory; the resulting source is also checked against its declared digest. The archives and extracted source files are unchanged. Diagnostic fields are omitted from comparisons with historical summaries; all mathematical fields are still compared. Thus this is an explicitly documented proof-only execution of the released acceptance code.

The density induction in Section 8 states two conditions that were checked by the original code but absent from the displayed condition list in the historical stage-80 prose:  $h > 0$  and  $g_4 \leq \mu(P_4) - 4r$ . In particular, the sign of  $h$  is needed when transferring  $rn + h$  to a finite-order density. No certificate constant is altered by making these conditions explicit.

The accepting programs do not formalize the written structural and analytic reductions in a proof assistant. Their conclusion is exact acceptance of the specified certificates; the mathematical proof also uses the reductions and domain arguments given in the paper.

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